Linear Algebra I

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Based on the online textbook
A First Course in Linear Algebra
by Robert A. Beezer
and the lecture notes of Dr. Charles LI

Contents

1 System of Linear Equations **5**

1.1 Introduction 5

1.2 Elimination and equivalent system 7

2 Matrix and Gaussian Elimination **16**

2.1 Matrix representation for systems of equations 16

2.2 RREF and Gaussian elimination 21

2.3 Type of solution sets 31

3 Matrix Algebra **35**

3.1 Matrix addition, subtraction and scalar multiplication 35

3.2 Matrix multiplication and linear system 39

3.3 Transposes and symmetric matrices 47

3.4 Special matrices 51

3.5 Block matrices 57

4 Homogeneous System and Nonsingular Matrix **62**

4.1 Homogeneous System and Null Space 62

4.2 Solution Structure of Nonhomogeneous System 69

4.3 Nonsingular Matrix 71

5 Invertible Matrix **75**

5.1 Inverse of a Matrix 75

5.2 Equivalence between Nonsingular and Invertible 81

5.3 Computing the Inverse of a Matrix 86

6 Vector Space **89**

6.1 Definition and examples of vector spaces 89

6.2 Subspaces 101

6.3 Linear Combinations 111

6.4 Linear Dependence and Independence 117
6.5 Basis and Dimension 131
6.6 Null Space, Column Space, Row Space 140

7 Determinant **164**
7.1 Determinant of a Square Matrix 164
7.2 Properties of Determinants 167
7.3 More Examples 176
7.4 Adjoint and Cramer's Rule 187

8 Eigenvalues and Eigenvectors **191**
8.1 Eigenvalues and Eigenvectors 191
8.2 Similar Matrices (may be skipped if no enough time) 199
8.3 Diagonalization 202
8.4 Powers of Matrices 207

9 Inner Product Space **209**
9.1 Basic Properties of Inner Product 209
9.2 Orthogonal Sets 214
9.3 Gram-Schmidt Orthogonalization Process 221
9.4 Two Inequalities (Optional!) 225

Preface

This is a first course in linear algebra, covering the conventional topics including system of linear equations, matrix theory, vector space, determinant, eigenvalue and eigenvector, and inner product. We will follow this set of lecture notes which is mainly based on:

- the online textbook: *A First Course in Linear Algebra* , by Robert A. Beezer, downloaded at: `http://linear.ups.edu/download/fcla-3.50-print.pdf`
- the lecture notes of Dr. Charles LI, posted in the common site of MATH1030ABC in blackboard.

There are a large number of textbooks on linear algebra with different focus and different order of topics. Here are some good reference books:

Easy:

- *Introduction to linear algebra*, by Gilbert Strang.
- *Introduction to linear algebra*, by Serge Lang, accessed online at CUHK Library.

Normal:

1. *Linear algebra and its applications*, by Gilbert Strang.
2. *Linear algebra and applications*, by Steven J. Leon.

Rigorous, for math major:

1. *Linear algebra*, by S. H. Friedberg, A. J. Insel, L.E. Spence, accessed online at CUHK Library.
2. 高等代数，丘维声, accessed online at CUHK Library.
3. 高等代数与解析几何, 陈志杰.

Last advice: Do not rely on lecture notes! Go to find a textbook suiting you! Pay attention to those introductions, motivations, and explanations!

1 System of Linear Equations

1.1 Introduction

Definition 1.1.1 (System of Linear Equations). A linear equation in n unknowns is an equation of the form

$$a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = b$$

where $a_1, a_2, \cdots, a_n$ and b are real numbers and $x_1, x_2, \cdots, x_n$ are variables.
A **system of linear equations** is a collection of m equations in n unknowns of the form

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= b_2 \\
a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \cdots + a_{3n}x_n &= b_3 \\
&\vdots \\
a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= b_m
\end{aligned}$$

where the a_{ij}'s (called **coefficients** of the system) and b_i's are real numbers, and x_j's are variables. The above system is called an $m \times n$ system.

Definition 1.1.2 (Solution and Solution Set). By a **solution** to an $m \times n$ system, it means an n-tuple of numbers $(x_1, x_2, \cdots, x_n)$ that satisfies all the equations of the system.
The set of all solutions of a linear system is called the **solution set** of the system.

Example 1.1.1.

(a) 2×2 system:

$$\begin{aligned}
x_1 + 2x_2 &= 5 \\
2x_1 + 3x_2 &= 8
\end{aligned}$$

Check that $(1, 2)$ is a solution in the following way. Substitute $x_1 = 1, x_2 = 2$ into the system and verify that the equations are satisfied.

(b) 2×3 system:

$$\begin{aligned} x_1 - x_2 + x_3 &= 2 \\ 2x_1 + x_2 - x_3 &= 4 \end{aligned}$$

Check that $(2, 0, 0)$ is a solution, and moreover, $(2, a, a)$ is always a solution for any real number a. Therefore this system has infinitely many solutions.

(c) 3×2 system:

$$\begin{aligned} x_1 + x_2 &= 2 \\ x_1 - x_2 &= 1 \\ x_1 &= 4 \end{aligned}$$

This system has no solution. It follows from the third equation that x_1 must be 4. Substituting $x_1 = 4$ into the first and second equation, we see $x_2 = -2$ and $x_2 = 3$, which is a contradiction.

Definition 1.1.3 (Consistent System). A system of linear equations is **consistent** if it has at least one solution. Otherwise, the system is called **inconsistent**.

Thus, in the above example, system (c) is inconsistent, while systems (a) and (b) are both consistent.

1.2 Elimination and equivalent system

The main goal of this section is to solve systems of linear equations, and the key idea is **"elimination"**.

1.2.1 Idea of elimination

Example 1.2.1. Solve the following 2×2 system

$$3x + 4y = 2 \qquad ①$$

$$4x + 5y = 3 \qquad ②$$

$-\frac{4}{3} \times ① + ②$:

$$-\frac{1}{3}y = \frac{1}{3}$$

So $y = -1$. Substituting it into ①:

$$3x + 4(-1) = 2, \qquad x = 2.$$

So we obtain the solution $x = 2$, $y = -1$.

Remark 1.2.1.

1. The number $-\frac{4}{3}$ is chosen such that the variable x is eliminated in the second equation, thus y can be easily solved. This idea is called **"elimination"**.

2. After finding y, we substitute it into the first equation to solve x.

Remark 1.2.2. The above system can be also solved by direct **"substitution"**:

$$3x + 4y = 2 \tag{1.1}$$

$$4x + 5y = 3 \tag{1.2}$$

Use (1.2) to express y in terms of x:

$$y = \frac{3}{5} - \frac{4}{5}x. \tag{1.3}$$

Substitute this into (1.1):

$$3x + 4(\frac{3}{5} - \frac{4}{5}x) = 2, \quad x = 2.$$

Substitute $x = 2$ into (1.3) to solve for y:

$$y = \frac{3}{5} - \frac{4}{5} \times 2 = -1.$$

So the solution is $x = 2$, $y = -1$.

The ideas in the two approaches are the same: to reduce the number of unknowns in the equation. Some textbooks call the first approach "elimination by addition/subtraction", and the second one "elimination by substitution". In this course, we will follow the first method because it is more convenient to deal with larger systems.

Example 1.2.2. Solve the following 3×3 system

$$\begin{aligned} x + 2y + 2z &= 4 && ① \\ x + 3y + 3z &= 5 && ② \\ 2x + 6y + 5z &= 6 && ③ \end{aligned}$$

$-1\times$ ①+②:

$$\begin{aligned} x + 2y + 2z &= 4 && ① \\ y + z &= 1 && ② \\ 2x + 6y + 5z &= 6 && ③ \end{aligned}$$

$-2\times$ ①+③:

$$\begin{aligned} x + 2y + 2z &= 4 && ① \\ y + z &= 1 && ② \\ 2y + z &= -2 && ③ \end{aligned}$$

$-2\times$ ②$+$ ③:

$$x + 2y + 2z = 4 \qquad ①$$
$$y + z = 1 \qquad ②$$
$$-z = -4 \qquad ③$$

Equation ③ implies that $z = 4$.

Substitute $z = 4$ into equation ② to get $y = -3$.

Finally, substituting $z = 4,\ y = -3$ into equation ①, we find that $x = 2$.

1.2.2 Possibilities of solution sets

Let us examine geometrically a system of the form

$$a_{11}x_1 + a_{12}x_2 = b_1$$
$$a_{21}x_1 + a_{22}x_2 = b_2$$

Each equation can be represented graphically as a line in the x_1x_2-plane. The ordered pair (x_1, x_2) will be a solution of the system if and only if it lies on both lines. For example, consider the three systems

$$\text{(i)} \begin{cases} 2x - 3y = -5, \\ x + y = 5. \end{cases} \quad \text{(ii)} \begin{cases} 2x - 3y = -5, \\ 2x - 3y = 0. \end{cases} \quad \text{(iii)} \begin{cases} 2x - 3y = -5, \\ -2x + 3y = 5. \end{cases}$$

In system (i), the two lines intersect at the point $(2, 3)$. Thus, $\{(2, 3)\}$ is the solution set of (i).

In system (ii), the two lines are parallel and do not intersect, so (ii) is inconsistent and its solution set is empty.

In system (iii), two lines coincide and their intersection is just the whole line. Let $y = t$ where t is any real number, then we can solve $x = \frac{3t-5}{2}$. Therefore the solution set can be represented by $\{(\frac{3t-5}{2}, t) \mid t \in \mathbb{R}\}$.

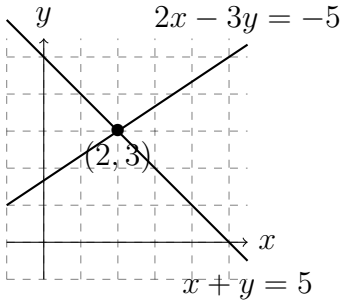


Figure 1: Intersection between lines

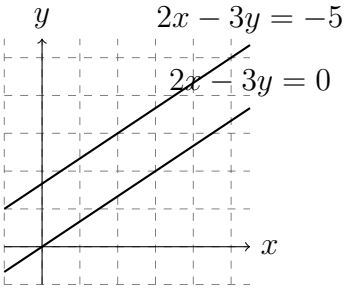


Figure 2: Parallel lines

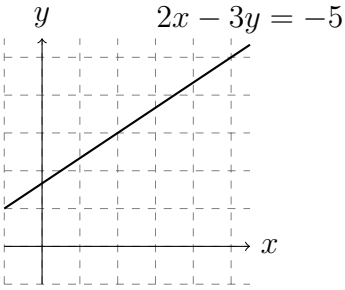


Figure 3: Same line

Theorem 1.2.1. *A system of linear equations can have (i) a unique solution, or (ii) no solution, or (iii) infinitely many solutions.*

Remark 1.2.3. It is impossible for a system of linear equation to have and only have 2 solutions. The detailed proof of the theorem will be shown later.

1.2.3 Equivalent systems and equation operations

Definition 1.2.1 (**Equivalent Systems**). Two systems of linear equations are **equivalent** if their solution sets are equal.

Definition 1.2.2 (**Equation Operations**). The following three types of operations applied on system of linear equations are called **equation operations**:

I. Swap/interchange two equations: $\textcircled{i} \leftrightarrow \textcircled{j}$.

II. Multiply an equation $\textcircled{i}$ by a nonzero constant: $\alpha \times \textcircled{i}$ where $\alpha \neq 0$.

III. Add a multiple of an equation $\textcircled{i}$ to another equation $\textcircled{j}$: $\alpha \times \textcircled{i} + \textcircled{j}$ where $\alpha \in \mathbb{R}$. Note that the equation $\textcircled{i}$ will not be changed, and the equation $\textcircled{j}$ will be replaced by $\alpha \times \textcircled{i} + \textcircled{j}$.

Remark 1.2.4. Each equation operation will transform the linear system into a new one, and we can always find the **"inverse"** operation of the same type to transform the new system back to the original one:

Type	Operation	Inverse operation
I	$\textcircled{i} \leftrightarrow \textcircled{j}$	$\textcircled{i} \leftrightarrow \textcircled{j}$
II	$\alpha \times \textcircled{i},\ \alpha \neq 0$	$\frac{1}{\alpha} \times \textcircled{i},\ \alpha \neq 0$
III	$\alpha \times \textcircled{i} + \textcircled{j}$	$-\alpha \times \textcircled{i} + \textcircled{j}$

Theorem 1.2.2 (**Equation Operations Preserve Solution Sets**). *If we apply the three types of equation operations to a system of linear equations, then the original system and the transformed system are equivalent.*

Proof. (Sketch) Let S be a system of linear equations and S' be another system of linear equations obtained by applying equation operations on S.

Let $(x_1, \ldots, x_n)$ be a solution to S. Because S' is obtained from S by applying equation operations, it is obvious that $(x_1, \ldots, x_n)$ also satisfies S'.

Conversely, suppose that $(x_1, \ldots, x_n)$ is a solution to S'. Note that S be can obtained by applying the inverse equation operations on S, so $(x_1, \ldots, x_n)$ also satisfies S. Hence S and S' have the same solution set.

For details, See Beezer, Theorem EOPSS (Ver 3.5, print version p. 10). $\square$

1.2.4 More examples

Given a system of equations, we can use equation operations to obtain an equivalent system that is easier to solve.

Example 1.2.3.

$$\begin{aligned} x_1 - 5x_2 + 3x_3 &= 1 && ① \\ 2x_1 - 4x_2 + x_3 &= 0 && ② \\ x_1 + x_2 - 2x_3 &= -1 && ③ \end{aligned}$$

$-2 \times ① + ②$:

$$\begin{aligned} x_1 - 5x_2 + 3x_3 &= 1 && ① \\ 6x_2 - 5x_3 &= -2 && ② \\ x_1 + x_2 - 2x_3 &= -1 && ③ \end{aligned}$$

$-1 \times ① + ③$:

$$\begin{aligned} x_1 - 5x_2 + 3x_3 &= 1 && ① \\ 6x_2 - 5x_3 &= -2 && ② \\ 6x_2 - 5x_3 &= -2 && ③ \end{aligned}$$

$-1 \times ② + ③$:

$$\begin{aligned} x_1 - 5x_2 + 3x_3 &= 1 && ① \\ 6x_2 - 5x_3 &= -2 && ② \\ 0 &= 0 && ③ \end{aligned}$$

The last equation ③ is always true and can be ignored. Let $x_3 = a$, where a is any real number, and express x_1, x_2 in terms of x_3 by "back substitution".

Substitute $x_3 = a$ into equation ② to get

$$x_2 = -\frac{1}{3} + \frac{5}{6}a.$$

Substitute x_3 and x_2 into equation ① to get

$$x_1 = -\frac{2}{3} + \frac{7}{6}a.$$

Hence the solution set is

$$\{(-\frac{2}{3} + \frac{7}{6}a, -\frac{1}{3} + \frac{5}{6}a, a) \mid a \in \mathbb{R}\}.$$

Example 1.2.4.

$$\begin{aligned} x_1 - 5x_2 + 3x_3 &= 1 && ① \\ 2x_1 - 4x_2 + x_3 &= 0 && ② \\ x_1 + x_2 - 2x_3 &= -2 && ③ \end{aligned}$$

This system differs from the above one only in one constant. By applying exactly the same equation operations we get

$$\begin{aligned} x_1 - 5x_2 + 3x_3 &= 1 && ① \\ 6x_2 - 5x_3 &= -2 && ② \\ 0 &= -1 && ③ \end{aligned}$$

The last equation $0 = -1$ is a contradiction, so this system has no solution.

Example 1.2.5.

$$\begin{aligned} x_2 + x_3 + 2x_4 + 2x_5 &= 2 && ① \\ x_1 + 2x_2 + 3x_3 + 2x_4 + 3x_5 &= 4 && ② \\ -2x_1 - x_2 - 3x_3 + 3x_4 + x_5 &= 3 && ③ \end{aligned}$$

① $\leftrightarrow$ ② (because the first equation has no x_1 term):

$$\begin{aligned} x_1 + 2x_2 + 3x_3 + 2x_4 + 3x_5 &= 4 && ① \\ x_2 + x_3 + 2x_4 + 2x_5 &= 2 && ② \\ -2x_1 - x_2 - 3x_3 + 3x_4 + x_5 &= 3 && ③ \end{aligned}$$

$2 \times ① + ③$:

$$\begin{aligned} x_1 + 2x_2 + 3x_3 + 2x_4 + 3x_5 &= \ \ 4 & ① \\ x_2 + \ \ x_3 + 2x_4 + 2x_5 &= \ \ 2 & ② \\ 3x_2 + 3x_3 + 7x_4 + 7x_5 &= 11 & ③ \end{aligned}$$

$-3 \times ② + ③$:

$$\begin{aligned} x_1 + 2x_2 + 3x_3 + 2x_4 + 3x_5 &= 4 & ① \\ x_2 + \ \ x_3 + 2x_4 + 2x_5 &= 2 & ② \\ x_4 + \ \ x_5 &= 5 & ③ \end{aligned}$$

Now we can then solve the system by back substitution. Let $x_3 = a$ and $x_5 = b$ where a and b are arbitrary real numbers, and substitute them into equation ③ to get

$$x_4 = 5 - x_5 = 5 - b.$$

Substitute x_3, x_4, x_5 into equation ② to get

$$x_2 = 2 - x_3 - 2x_4 - 2x_5 = -8 - a.$$

Substitute x_2, x_3, x_4, x_5 into equation ① to get

$$x_1 = 4 - 2x_2 - 3x_3 - 2x_4 - 3x_5 = 10 - a - b.$$

Hence the solution set to the system is

$$\{(10 - a - b, -8 - a, a, 5 - b, b) \mid a, b \in \mathbb{R}\}.$$

Remark 1.2.5.

1. We use equation operations to eliminate variables and transformed the system into a new one which looks like a "stair" (also called "upper triangular form", or "echelon form"), and then solve each variable by "back substitution". Here, "back substitution" means the system is solved in reverse order, from bottom to top, by substitution each newly computed value into the equation above.

2. In Example 1.2.5, the transformed upper triangular form is

$$\begin{aligned} x_1 + 2x_2 + 3x_3 + 2x_4 + 3x_5 &= 4 && ① \\ x_2 + x_3 + 2x_4 + 2x_5 &= 2 && ② \\ x_4 + x_5 &= 5 && ③ \end{aligned}$$

The leftmost variables in each equation (which are x_1, x_2, x_4) are called "dependent variables" or "lead variables", and the remaining variables (which are x_3, x_5) are called "independent variables" or "free variables".

To write all the solutions, we express the dependent variables in terms of free variables. More details will be discussed in next chapter.

3. To solve systems of linear equations, the name of the variables $x_1, x_2, \cdots, x_n$ are not important. The solutions sets will not be influenced if the variables are renamed as $y_1, y_2, \cdots, y_n$.

What matter are the coefficients of the system!

This observation motivates us to introduce a simpler notation only involving the coefficients to represent a linear system. For example,

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 4 \\ -5x_1 + 6x_2 - 7x_3 &= -8 \\ 9x_1 + 10x_2 - 11x_3 &= -12 \end{aligned}$$

can be denoted by

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ -5 & 6 & -7 & -8 \\ 9 & 10 & -11 & -12 \end{array}\right]$$

This notation is called "matrix".

2 Matrix and Gaussian Elimination

As mentioned in Remark 1.2.5, we will define a new tool "matrix" to represent a linear system, and discuss the procedure of Gaussian elimination for solving a system.

2.1 Matrix representation for systems of equations

Definition 2.1.1 (Matrix). A **matrix** is a rectangular array of numbers. A matrix with m rows and n columns can be denoted by

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

Notations:

- Some textbooks use parentheses ( ) instead of brackets [ ] for matrix. We adopt both.
- $a_{ij} = [A]_{ij}$ is the general (i, j)-th entry of A in row i and column j.
- $m \times n$ is the size of A. If $m = n$, A is called a square matrix.
- $\begin{bmatrix} a_{i1} & a_{i2} & a_{i3} & \cdots & a_{in} \end{bmatrix}$ is the i-th row of A, and it is a $1 \times n$ matrix.
- $\begin{bmatrix} a_{1j} \\ a_{2j} \\ a_{3j} \\ \vdots \\ a_{mj} \end{bmatrix}$ is the j-th column of A, and it is a $m \times 1$ matrix.

Example 2.1.1. $B = \begin{bmatrix} -1 & 2 & 5 & 3 \\ 1 & 0 & -6 & 1 \\ -4 & 2 & 2 & -2 \end{bmatrix}$ is a 3×4 matrix. $[B]_{2,3} = -6$ and $[B]_{3,4} = -2$.

Definition 2.1.2 (Augmented Matrix Representation). Consider a system of linear equations,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \cdots + a_{3n}x_n &= b_3 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

The $m \times n$ matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

is called the **coefficient matrix** of the system.

The $m \times (n+1)$ matrix

$$\left[\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} & b_2 \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} & b_3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} & b_m \end{array}\right]$$

is called the **augmented matrix representation** of the system.

Remark 2.1.1.

1. Sometimes we put horizontal lines or vertical lines to divide a matrix into different areas (e.g., the vertical line in the augmented matrix). It is the same as the matrix without lines.

2. When we do equation operations to the system, only the constants a_{ij}'s and b_i's are changed. The symbols $+, =$ and variables x_i's are not affected, so we can do those operations directly and equivalently on each row of the augmented matrix. Let's give these operations on matrix another name: "elementary row operations".

Definition 2.1.3 (Elementary Row Operations). The following three types of operations applied on matrix are called **elementary row operations** or simply **row operations**:

I. Swap/interchange two rows: $R_i \leftrightarrow R_j$, where R_i means the i-th row.

II. Multiply a row R_i by a nonzero constant: αR_i, where $\alpha \neq 0$.

III. Add a multiple of a row R_i to another row R_j: $\alpha R_i + R_j$, where $\alpha \in \mathbb{R}$. Note that R_i will not be changed, and R_j will be replaced by $\alpha R_i + R_j$.

Definition 2.1.4 (Row-Equivalent Matrices). Two matrices A and B are called **row-equivalent** if one can be obtained from the other by a sequence of row operations.

Remark 2.1.2. Note that the row operations are reversible, so there is no distinction between "A is row-equivalent to B" and "B is row-equivalent to A".

Theorem 2.1.1 (Row-Equivalent Matrices represent Equivalent Systems). *Let two linear systems be represented by two augmented matrices A and B. Suppose A and B are row-equivalent, then the two linear systems are also equivalent (which means they have same solution sets).*

Proof. It is obvious. See Beezer, Theorem REMES (Ver 3.5 print version p.20).

□

Strategy for solving system using matrix:

1. Represent the system by an augmented matrix.
2. Perform row operations (which will preserve solution sets) to get a "simpler" augmented matrix.
3. convert the "simpler" augmented matrix back to a "simpler" system of equations, and then solve it.

Example 2.1.2.

$$\begin{aligned} x_1 + 2x_2 + 2x_3 &= 4 \\ x_1 + 3x_2 + 3x_3 &= 5 \\ 2x_1 + 6x_2 + 5x_3 &= 6 \end{aligned}$$

Form the augmented matrix,

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 4 \\ 1 & 3 & 3 & 5 \\ 2 & 6 & 5 & 6 \end{array}\right]$$

Apply row operations,

$$\xrightarrow{-1R_1+R_2} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 \\ 2 & 6 & 5 & 6 \end{array}\right] \xrightarrow{-2R_1+R_3} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 2 & 1 & -2 \end{array}\right]$$

$$\xrightarrow{-2R_2+R_3} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & -4 \end{array}\right] \xrightarrow{-1R_3} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

The new "simpler" augmented matrix represents the the following system

$$\begin{aligned} x_1 + 2x_2 + 2x_3 &= 4 \\ x_2 + x_3 &= 1 \\ x_3 &= 4 \end{aligned}$$

The third equation implies $x_3 = 4$.

Substitution x_3 into the second equation to get $x_2 = -3$.

Substitute x_2 and x_3 into the first equation to get $x_1 = 2$.

2.2 RREF and Gaussian elimination

In Example 2.1.2, we applied row operations to transform a "complicated" augmented matrix into a "simpler" one:

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 4 \\ 1 & 3 & 3 & 5 \\ 2 & 6 & 5 & 6 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

This "simpler" matrix is called in row echelon form.

Definition 2.2.1 (REF & RREF). A matrix is said to be in **row echelon form (REF)** if

(i) all "zero rows" (which consist of only zero entries) are at the bottom;

(ii) the "leading coefficient" of a nonzero row (which is the first nonzero entry from the left) is always strictly to the right of the leading coefficient of the row above it.

A matrix is said to be in **reduced row echelon form (RREF)** if conditions (i)-(ii) are satisfied and furthermore,

(iii) the leading coefficient of each nonzero row is 1 (which is called "leading 1");

(iv) each leading 1 is the only nonzero number in its column (the column containing a leading 1 is called a "pivot column").

REF looks like $\begin{bmatrix} \boxed{2} & * & \cdots & 4 & * & \cdots & 8 & * & \cdots \\ 0 & 0 & \cdots & \boxed{3} & * & \cdots & 5 & * & \cdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & \boxed{7} & * & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 & \cdots \end{bmatrix}$

$$\text{RREF looks like} \begin{bmatrix} \boxed{1} & * & \cdots & 0 & * & \cdots & 0 & * & \cdots \\ 0 & 0 & \cdots & \boxed{1} & * & \cdots & 0 & * & \cdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & \boxed{1} & * & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 & \cdots \end{bmatrix}$$

The leading coefficients are boxed. * means an arbitrary number.

Remark 2.2.1. Some textbooks use conditions (i), (ii) and (iii) to define REF, but in this lecture notes we only use (i) and (ii) because to find the solutions from REF of the linear system, it is not necessary to make the leading coefficients be 1.

Example 2.2.1. The matrices below are in **RREF**.

1.
$$\begin{bmatrix} \boxed{1} & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & \boxed{1} & 3 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Column 1, 3, 6 are pivot columns.

2.
$$\begin{bmatrix} \boxed{1} & 0 & 5 & 3 & 0 & 0 & 5 \\ 0 & \boxed{1} & 3 & 6 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 & \boxed{1} & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & \boxed{1} & 3 \end{bmatrix}$$

Column 1, 2, 5, 6 are pivot columns.

3.
$$\begin{bmatrix} \boxed{1} & 0 & 0 \\ 0 & \boxed{1} & 0 \\ 0 & 0 & \boxed{1} \end{bmatrix}$$

Column 1, 2, 3 are pivot columns.

4.

$$\begin{bmatrix} 0 & \boxed{1} & 1 & 0 & 0 & 1 & 0 & 9 & 6 \\ 0 & 0 & 0 & 0 & \boxed{1} & 1 & 0 & 8 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 & \boxed{1} & 3 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Column 2, 5, 7 are pivot columns.

Example 2.2.2. The following matrices are **not RREF**, explain why.

1.

$$\begin{bmatrix} \boxed{1} & 0 & 1 & 0 & 1 & 0 \\ 0 & \boxed{1} & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Condition (i) fails: zero row 3 is not at the bottom, but above a nonzero row 4. It is even not REF.

2.

$$\begin{bmatrix} \boxed{1} & 0 & 0 & 1 & 2 \\ 0 & 0 & \boxed{1} & 0 & 1 \\ 0 & \boxed{1} & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Condition (ii) fails: the leading 1 in row 3 is NOT on the right of the leading 1 in row 2. It is not REF.

3.

$$\begin{bmatrix} \boxed{1} & 0 & 2 & 0 \\ 0 & \boxed{1} & 3 & 0 \\ 0 & 0 & 0 & \boxed{3} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Condition (iii) fails. It is REF according to the definition in this lecture notes.

4.
$$\begin{bmatrix} 0 & \boxed{1} & 0 & 0 & 1 & 2 \\ 0 & 0 & \boxed{1} & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & \boxed{1} & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Condition (iv) fails. It is REF.

Theorem 2.2.1 (Existence and Uniqueness of RREF). *By applying elementary row opeartions, each matrix can be transformed to*

1. *a REF matrix. (REF existent)*
2. *a unique RREF matrix. (RREF existent and unique)*

Proof. It suffices to show the existence and uniqueness of RREF. Existence part is summarized in the following "algorithm", known as "Gaussian elimination" (and also " Gauss-Jordan elimination"). Uniqueness part will be omitted. See Beezer. p.23-27.

Gaussian elimination & Gauss-Jordan elimination:

Step 1. If the first column of the matrix is a zero column, cross it off mentally. Continue this until the first column of the remaining matrix is nonzero.

Step 2. Use Type I operation ($R_i \leftrightarrow R_j$) to put a nonzero entry to the top of the first column.

Step 3. Use that nonzero entry and Type III operation ($\alpha R_i + R_j$) to eliminate other entries in the first column.

Step 4. Cross off the first row and the first column mentally. Repeat Step 1-3 for the submatrix that remains.

Step 1-4 transforms the matrix to REF, and this process is call "Gaussian elimination".

Step 5. Use Type II operation (αR_i) to make the leading coefficients of each nonzero row be leading 1.

Step 6. Beginning with the last nonzero row, use the leading 1 and Type III operation ($\alpha R_i + R_j$) to eliminate other entries above it in its column. Repeat this step for the next-to-last row and so on, until each leading 1 becomes the only nonzero entry in its column.

Step 1-6 transforms the matrix to RREF, and this process is call "Gauss-Jordan elimination".

□

Example 2.2.3. Find the RREF of

$$A = \begin{bmatrix} 0 & 0 & 2 & 2 & 6 & 2 & 3 \\ 2 & 4 & 1 & 3 & 7 & 3 & -1 \\ 1 & 2 & 2 & 3 & 8 & 2 & 1 \\ 1 & 2 & -1 & 0 & -1 & 2 & -1 \end{bmatrix}$$

- Consider column 1 and move a nonzero entry to row 1 by swapping rows. $R_1 \leftrightarrow R_2$, $R_1 \leftrightarrow R_3$ and $R_1 \leftrightarrow R_4$ all work, and we choose $R_1 \leftrightarrow R_3$ (because the following calculations will be easier with entry 1 or -1.)

$$A \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} \boxed{1} & 2 & 2 & 3 & 8 & 2 & 1 \\ 2 & 4 & 1 & 3 & 7 & 3 & -1 \\ 0 & 0 & 2 & 2 & 6 & 2 & 3 \\ 1 & 2 & -1 & 0 & -1 & 2 & -1 \end{bmatrix}$$

- Then use the leading coefficient (boxed number) to eliminate other entries in column 1.

$$\xrightarrow[-R_1+R_4]{-2R_1+R_2} \begin{bmatrix} \boxed{1} & 2 & 2 & 3 & 8 & 2 & 1 \\ 0 & 0 & -3 & -3 & -9 & -1 & -3 \\ 0 & 0 & 2 & 2 & 6 & 2 & 3 \\ 0 & 0 & -3 & -3 & -9 & 0 & -2 \end{bmatrix}$$

- Ignore the row 1 and column 1 and consider the remaining 3×6 submatrix. Since the first column is zero, so we move to next column and then we only need to focus on the ‘right-lower part:

$$\left[\begin{array}{cc|ccccc} \boxed{1} & 2 & 2 & 3 & 8 & 2 & 1 \\ \hline 0 & 0 & \boxed{-3} & -3 & -9 & -1 & -3 \\ 0 & 0 & 2 & 2 & 6 & 2 & 3 \\ 0 & 0 & -3 & -3 & -9 & 0 & -2 \end{array}\right]$$

- Turn the boxed number -3 into a leading 1 by $-\frac{1}{3}R_2$, and use the leading 1 to remove other entries in its column.

$$\xrightarrow{-\frac{1}{3}R_2} \begin{bmatrix} \boxed{1} & 2 & 2 & 3 & 8 & 2 & 1 \\ 0 & 0 & \boxed{1} & 1 & 3 & \frac{1}{3} & 1 \\ 0 & 0 & 2 & 2 & 6 & 2 & 3 \\ 0 & 0 & -3 & -3 & -9 & 0 & -2 \end{bmatrix}$$

$$\xrightarrow[3R_2+R_4]{-2R_2+R_3} \begin{bmatrix} \boxed{1} & 2 & 2 & 3 & 8 & 2 & 1 \\ 0 & 0 & \boxed{1} & 1 & 3 & \frac{1}{3} & 1 \\ 0 & 0 & 0 & 0 & 0 & \frac{4}{3} & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

- Ignore the first 2 rows and the first 3 columns, and continue the above procedure:

$$\xrightarrow{R_3 \leftrightarrow R_4} \begin{bmatrix} \boxed{1} & 2 & 2 & 3 & 8 & 2 & 1 \\ 0 & 0 & \boxed{1} & 1 & 3 & \frac{1}{3} & 1 \\ 0 & 0 & 0 & 0 & 0 & \boxed{1} & 1 \\ 0 & 0 & 0 & 0 & 0 & \frac{4}{3} & 1 \end{bmatrix}$$

$$\xrightarrow{-\frac{4}{3}R_3+R_4} \begin{bmatrix} \boxed{1} & 2 & 2 & 3 & 8 & 2 & 1 \\ 0 & 0 & \boxed{1} & 1 & 3 & \frac{1}{3} & 1 \\ 0 & 0 & 0 & 0 & 0 & \boxed{1} & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{3} \end{bmatrix}$$

$$\xrightarrow{-3R_4} \begin{bmatrix} \boxed{1} & 2 & 2 & 3 & 8 & 2 & 1 \\ 0 & 0 & \boxed{1} & 1 & 3 & \frac{1}{3} & 1 \\ 0 & 0 & 0 & 0 & 0 & \boxed{1} & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

- Starting from the last nonzero row, use each leading 1 to eliminate the nonzero entries above it.

$$\xrightarrow[\substack{-R_4+R_2 \\ -R_4+R_1}]{-R_4+R_3} \begin{bmatrix} \boxed{1} & 2 & 2 & 3 & 8 & 2 & 0 \\ 0 & 0 & \boxed{1} & 1 & 3 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

$$\xrightarrow[-2R_3+R_1]{-\frac{1}{3}R_3+R_2} \begin{bmatrix} \boxed{1} & 2 & 2 & 3 & 8 & 0 & 0 \\ 0 & 0 & \boxed{1} & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

$$\xrightarrow{-2R_2+R_1} \begin{bmatrix} \boxed{1} & 2 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & \boxed{1} & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix} (RREF)$$

Exercise: Find the RREF of

$$\begin{bmatrix} 1 & 2 & 1 & 4 \\ 3 & 8 & 7 & 20 \\ 2 & 7 & 9 & 23 \end{bmatrix}, \quad \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 5 & 7 \end{bmatrix}$$

The following example shows the difference between solving equations by Gaussian elimination and Gauss-Jordan elimination.

Example 2.2.4. Solve

$$-7x_1 - 6x_2 - 12x_3 = -33$$
$$5x_1 + 5x_2 + 7x_3 = 24$$
$$x_1 + 4x_3 = 5$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} -7 & -6 & -12 & -33 \\ 5 & 5 & 7 & 24 \\ 1 & 0 & 4 & 5 \end{array}\right].$$

Use Gaussian elimination,

$$\xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 4 & 5 \\ 5 & 5 & 7 & 24 \\ -7 & -6 & -12 & -33 \end{array}\right] \xrightarrow[7R_1+R_3]{-5R_1+R_2} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 4 & 5 \\ 0 & 5 & -13 & -1 \\ 0 & -6 & 16 & 2 \end{array}\right]$$

$$\xrightarrow{\frac{1}{5}R_2} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 4 & 5 \\ 0 & 1 & \frac{-13}{5} & \frac{-1}{5} \\ 0 & -6 & 16 & 2 \end{array}\right] \xrightarrow{6R_2+R_3} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 4 & 5 \\ 0 & \boxed{1} & \frac{-13}{5} & \frac{-1}{5} \\ 0 & 0 & \frac{2}{5} & \frac{4}{5} \end{array}\right]$$

Then use back substitution to find the solutions:

$$x_3 = 2,\ x_2 = 5,\ x_1 = -3.$$

If we continue to convert the above REF to RREF,

$$\xrightarrow{\frac{5}{2}R_3} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 4 & 5 \\ 0 & \boxed{1} & \frac{-13}{5} & \frac{-1}{5} \\ 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow{\frac{13}{5}R_3+R_2} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 4 & 5 \\ 0 & \boxed{1} & 0 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\xrightarrow{-4R_3+R_1} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & -3 \\ 0 & \boxed{1} & 0 & 5 \\ 0 & 0 & \boxed{1} & 2 \end{array}\right]$$

Then by observation, the solution is

$$x_1 = -3,\ x_2 = 5,\ x_3 = 2.$$

Remark 2.2.2. Gauss-Jordan elimination converts the augmented matrix into R-REF, which automatically gives the solution; while Gaussian elimination puts the matrix in REF, and back substitution is needed to solve the solutions.

Example 2.2.5.

$$\begin{aligned} x_1 - x_2 + 2x_3 &= 1 \\ 2x_1 + x_2 + x_3 &= 8 \\ x_1 + x_2 &= 5 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 2 & 1 & 1 & 8 \\ 1 & 1 & 0 & 5 \end{array}\right] \xrightarrow[-R_1+R_3]{-2R_1+R_2} \left[\begin{array}{ccc|c} \boxed{1} & -1 & 2 & 1 \\ 0 & 3 & -3 & 6 \\ 0 & 2 & -2 & 4 \end{array}\right]$$

$$\xrightarrow{\frac{1}{3}R_2} \left[\begin{array}{ccc|c} \boxed{1} & -1 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 2 & -2 & 4 \end{array}\right] \xrightarrow[-2R_2+R_3]{R_2+R_1} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 1 & 3 \\ 0 & \boxed{1} & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

There are infinitely many solutions which can be represented by

$$x_1 = 3 - a,\ x_2 = 2 + a,\ x_3 = a,$$

where a is an arbitrary number.

Example 2.2.6.

$$\begin{aligned} 2x_1 + x_2 + 7x_3 - 7x_4 &= 2 \\ -3x_1 + 4x_2 - 5x_3 - 6x_4 &= 3 \\ x_1 + x_2 + 4x_3 - 5x_4 &= 2 \end{aligned}$$

$$\left[\begin{array}{cccc|c} 2 & 1 & 7 & -7 & 2 \\ -3 & 4 & -5 & -6 & 3 \\ 1 & 1 & 4 & -5 & 2 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{cccc|c} 1 & 1 & 4 & -5 & 2 \\ -3 & 4 & -5 & -6 & 3 \\ 2 & 1 & 7 & -7 & 2 \end{array}\right]$$

$$\xrightarrow[-2R_1+R_3]{3R_1+R_2} \left[\begin{array}{cccc|c} \boxed{1} & 1 & 4 & -5 & 2 \\ 0 & 7 & 7 & -21 & 9 \\ 0 & -1 & -1 & 3 & -2 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{cccc|c} \boxed{1} & 1 & 4 & -5 & 2 \\ 0 & -1 & -1 & 3 & -2 \\ 0 & 7 & 7 & -21 & 9 \end{array}\right]$$

$$\xrightarrow{7R_2+R_3} \left[\begin{array}{cccc|c} \boxed{1} & 0 & 3 & -2 & 0 \\ 0 & \boxed{\text{-1}} & -1 & 3 & -2 \\ 0 & 0 & 0 & 0 & -5 \end{array}\right]$$

The third row is a contradiction, so there is no solution.

2.3 Type of solution sets

As we have seen in the previous sections, there are three possibilities for the solution sets of linear systems: unique solution, infinitely many solutions, and no solution. We will prove this conclusion using RREF.

Definition 2.3.1 (Independent and Dependent Variables). Suppose the augmented matrix A of a consistent linear system is row equivalent to a matrix B in RREF. Suppose j is the index of a pivot column of B. Then the variable x_j is called a **"dependent variable"** or **"lead variable"**. A variable that is not dependent is called an **"independent variable"** or **"free variable"**.

Example 2.3.1.

$$\begin{aligned} x_1 + 4x_2 - x_4 + 7x_6 &= 3 \\ 2x_1 + 8x_2 - x_3 + 3x_4 + 9x_5 - 13x_6 &= 9 \\ 2x_3 - 3x_4 - 4x_5 + 12x_6 &= 1 \\ -x_1 - 4x_2 + 2x_3 + 4x_4 + 8x_5 - 31x_6 &= 4 \end{aligned}$$

The augmented matrix is

$$\left[\begin{array}{cccccc|c} 1 & 4 & 0 & -1 & 0 & 7 & 3 \\ 2 & 8 & -1 & 3 & 9 & -13 & 9 \\ 0 & 0 & 2 & -3 & -4 & 12 & 1 \\ -1 & -4 & 2 & 4 & 8 & -31 & 4 \end{array}\right]$$

The matrix is row equivalent to the following RREF (**exercise**: check this!)

$$\left[\begin{array}{cccccc|c} \boxed{1} & 4 & 0 & 0 & 2 & 1 & 4 \\ 0 & 0 & \boxed{1} & 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & \boxed{1} & 2 & -6 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The indices of pivot columns is $\{1,\ 3,\ 4\}$, so the dependent variables are x_1, x_3, x_4. The remaining variables, x_2, x_5, x_6, are free variables.

We express dependent variables in terms of free variables and the solutions set can be represented by

$$\{(4 - 4x_2 - 2x_5 - x_6,\ x_2,\ 2 - x_5 + 3x_6,\ 1 - 2x_5 + 6x_6,\ x_5,\ x_6) \,|\, x_2, x_5, x_6 \in \mathbb{R}\}.$$

Example 2.3.2. Suppose the augmented matrix of a linear system is row equivalent to the following RREF

$$\left[\begin{array}{ccccc|c} \boxed{1} & -1 & 0 & 0 & 3 & 6 \\ 0 & 0 & \boxed{1} & 0 & -2 & 1 \\ 0 & 0 & 0 & \boxed{1} & 4 & 9 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Columns 1, 3 and 4 are pivot columns, so x_1, x_3, x_4 are dependent variables, and x_2, x_5 are free variables. Solution sets is

$$\{(6 + x_2 - 3x_5,\ x_2,\ 1 + 2x_5,\ 9 - 4x_5,\ x_5) \,|\, x_2,\ x_5 \in \mathbb{R}\}.$$

Theorem 2.3.1 (Consistency of Linear System). *Let A be the augmented matrix of a $m \times n$ linear system. Suppose B is the matrix in RREF row equivalent to A, then*

- *the system is* **inconsistent** *if and only if column $n+1$ (i.e., the last column) of B is a pivot column;*
- *the system is* **consistent** *if and only if column $n+1$ of B is not a pivot column.*
- *Another way of expressing the theorem is to say that the system is* **consistent** *if and only if the last* **non-zero row** *of B is not $(0\ 0\ \cdots 0 \mid 1)$.*

Proof. Obvious from the structure of RREF. See Beezer, Ver 3.5 print version p.38. □

Theorem 2.3.2 (Consistent Systems, r and n). *Suppose a $m \times n$ linear system is consistent. Suppose that the augmented matrix is row equivalent to B in RREF with r pivot columns. Then $r \leq n$. Furthermore,*

- *If $r = n$, then the system has a unique solution.*
- *If $r < n$, then the system has infinitely many solutions.*

Proof. Notice that B has $n+1$ columns, so B has at most $n+1$ pivot columns, i.e., $r \leq n+1$.

If $r = n+1$, then every column of B is a pivot column, and in particular, the last column is a pivot column. So the previous theorem tells us that the system is inconsistent, contradicted to our hypothesis. So $r \leq n$.

If $r = n$, then there is $n - r = 0$ free variable, so the solution is unique and can be found from the $RREF$.

If $r < n$, then there are $n - r > 0$ free variables. We can choose the free variables to be arbitrary numbers and create infinitely many solutions.

□

Theorem 2.3.3 (Possible Solution Sets for Linear Systems). *A system of linear equations can have (i) a unique solution, or (ii) infinitely many solutions, or (iii) no solution.*

Proof. It follows from the above theorem. □

Corollary 2.3.1 (Special case $n > m$). Suppose a consistent system of linear equations has m equations in n variables. If $n > m$, then the system has infinitely many solutions.

Proof. Suppose B is the RREF of the augmented matrix and B has r nonzero rows. Because B has m rows in total, so $r \leq m$. Since $n > m$ and the system is consistent, there must be $n - r > 0$ free variables and the system admits infinitely many solutions.

□

Summary of solving linear systems:

(1) Represent the system by an augmented matrix.

(2) Use Gaussian elimination (or Gauss-Jordan elimination) to find the REF (or RREF).

(3) If there is a row $[0\ 0\ \cdots 0 \mid d]$ with $d \neq 0$, the system has no solution.

(4) If there is no row $[0\ 0\ \cdots 0 \mid d]$ with $d \neq 0$, there are two possibilities:

   (a) it has unique solution if no free variable;

   (b) it has infinitely many solutions if there are free variables.

3 Matrix Algebra

In this section, we will define arithmetic operations on matrices, including matrix addition, matrix subtraction, scalar multiplication, transpose and matrix multiplication.

Throughout this section, unless otherwise stated,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix},$$

$[A]_{ij} = a_{ij}$, $[B]_{ij} = b_{ij}$ denote the (i, j)-th entry of A and B. Sometimes we also write $A_{m\times n}$ to mean a matrix A in size $m \times n$.

3.1 Matrix addition, subtraction and scalar multiplication

Definition 3.1.1 (Matrix Equality). The $m \times n$ matrices A and B are **equal**, written $A = B$ provided $[A]_{ij} = [B]_{ij}$ for all $1 \le i \le m$, $1 \le j \le n$, i.e.

$$a_{ij} = b_{ij} \text{ for all } i, j.$$

Definition 3.1.2 (Matrix Addition). Given $m \times n$ matrices A and B, define the **sum** of A and B as an $m \times n$ matrix, written $A + B$, according to

$$[A + B]_{ij} = [A]_{ij} + [B]_{ij} \qquad 1 \le i \le m,\ 1 \le j \le n,$$

i.e.,

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{bmatrix}$$

Definition 3.1.3 (**Matrix Scalar Multiplication**). Given $m \times n$ matrix A and a scalar $\alpha \in \mathbb{R}$, the **scalar multiple** of A by α is the $m \times n$ matrix, written αA and defined according to

$$[\alpha A]_{ij} = \alpha\,[A]_{ij} \qquad\qquad 1 \le i \le m,\ 1 \le j \le n,$$

i.e.,

$$\alpha A = \begin{bmatrix} \alpha a_{11} & \alpha a_{12} & \cdots & \alpha a_{1n} \\ \alpha a_{21} & \alpha a_{22} & \cdots & \alpha a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ \alpha a_{m1} & \alpha a_{m2} & \cdots & \alpha a_{mn} \end{bmatrix}.$$

Example 3.1.1.

$$\begin{bmatrix} 2 & -3 & 4 \\ 1 & 0 & -7 \end{bmatrix} + \begin{bmatrix} 6 & 2 & -4 \\ 3 & 5 & 2 \end{bmatrix} = \begin{bmatrix} 2+6 & -3+2 & 4+(-4) \\ 1+3 & 0+5 & -7+2 \end{bmatrix} = \begin{bmatrix} 8 & -1 & 0 \\ 4 & 5 & -5 \end{bmatrix},$$

$$7\begin{bmatrix} 2 & -3 & 4 \\ 1 & 0 & -7 \end{bmatrix} = \begin{bmatrix} 2\times 7 & -3\times 7 & 4\times 7 \\ 1\times 7 & 0\times 7 & -7\times 7 \end{bmatrix} = \begin{bmatrix} 14 & -21 & 28 \\ 7 & 0 & -749 \end{bmatrix},$$

$$\begin{bmatrix} 2 & -3 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 6 & 2 & -4 \\ 3 & 5 & 2 \end{bmatrix} \quad \text{is not well-defined.}$$

Definition 3.1.4 (**Zero Matrix**). The $m \times n$ **zero matrix** is written as $\mathcal{O} = \mathcal{O}_{m\times n}$ and defined by $[\mathcal{O}]_{ij} = 0$, for all $1 \le i \le m$, $1 \le j \le n$, i.e.

$$\mathcal{O} = \mathcal{O}_{m\times n} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}.$$

Definition 3.1.5 (Additive Inverse). The additive inverse of a matrix A, denoted by $-A$, is defined by $-A = (-1)A$, i.e.

$$-A = \begin{bmatrix} -a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & -a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ -a_{m1} & -a_{m2} & \cdots & -a_{mn} \end{bmatrix}.$$

Theorem 3.1.1. *The following properties hold for any scalars $\alpha, \beta \in \mathbb{R}$ and any matrices A, B, C for which the indicated operations are defined.*

1. *Commutativity*

   $A + B = B + A.$

2. *Additive Associativity*

   $A + (B + C) = (A + B) + C.$

3. *Zero Matrix*

   $A + \mathcal{O} = A.$

4. *Additive Inverses*

   $A + (-A) = \mathcal{O}.$

5. *Scalar Multiplication Associativity*

   $\alpha(\beta A) = (\alpha\beta)A.$

6. *Distributivity across Matrix Addition*

   $\alpha(A + B) = \alpha A + \alpha B.$

7. *Distributivity across Scalar Addition*

   $(\alpha + \beta)A = \alpha A + \beta A.$

8. *One*

   $1A = A.$

Proof. As an example, we prove property 7, $(\alpha + \beta)A = \alpha A + \beta A$.

For *any* i and j, $1 \leq i \leq m$, $1 \leq j \leq n$,

$$\begin{aligned}
[(\alpha+\beta)A]_{ij} &= (\alpha+\beta)\,[A]_{ij} \\
&= \alpha\,[A]_{ij} + \beta\,[A]_{ij} \\
&= [\alpha A]_{ij} + [\beta A]_{ij} \\
&= [\alpha A + \beta A]_{ij}\,.
\end{aligned}$$

Hence by the definition of equality of matrices, $(\alpha+\beta)A = \alpha A + \beta A$. $\square$

Definition 3.1.6 (Matrix Subtraction). Given $m \times n$ matrices A and B, define the **difference** of A and B as an $m \times n$ matrix, written $A - B$, according to

$$A - B = A + (-B),$$

i.e.,

$$A - B = \begin{bmatrix} a_{11} - b_{11} & a_{12} - b_{12} & \cdots & a_{1n} - b_{1n} \\ a_{21} - b_{21} & a_{22} - b_{22} & \cdots & a_{2n} - b_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} - b_{m1} & a_{m2} - b_{m2} & \cdots & a_{mn} - b_{mn} \end{bmatrix}$$

3.2 Matrix multiplication and linear system

We will define the most important operation: the multiplication of two matrices, and use it to represent a linear system.

Definition 3.2.1. [**Column Vectors and Row vectors**]

- An $(n \times 1)$-matrix is called a column vector of size n.
- A $(1 \times m)$-matrix is called a row vector of size m.

Remark 3.2.1. The set of all $n \times 1$ matrices of real numbers is called Euclidean n-space and is usually denoted by $\mathbb{R}^n$. Since we will be working almost exclusively with column vectors in this course, when we use the word "vector", we mean "column vector" unless otherwise stated.

Definition 3.2.2 (**Matrix Multiplication**).

(a) Let $A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}$ be a $1 \times n$ matrix (i.e., a row vector), and $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$ be a $n \times 1$ matrix (i.e., a column vector). We define the product AB to be a 1×1 matrix

$$AB = \begin{bmatrix} a_1b_1 + a_2b_2 + \cdots + a_nb_n \end{bmatrix}$$

Remark: We can also write the answer as a number $AB = a_1b_1 + a_2b_2 + \cdots + a_nb_n$, and it is called the "dot product" or "inner product" of the vectors A and B.

(b) Let $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$ be an $m \times n$ matrix,

and $B = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$ be an $n \times p$ matrix.

Then the **matrix product** of A and B is the $m \times p$ matrix

$$AB = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mp} \end{bmatrix}$$

with the (i, j)-th entry

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} = a_{i1} b_{1j} + a_{i2} b_{2j} + \cdots + a_{in} b_{nj}, \quad \text{for all } i, j.$$

Remark: the (i, j)-th entry of AB is just the product of row i of A and column j of B.

Example 3.2.1. Let

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 & 6 \\ 0 & -4 & 1 & 2 & 3 \\ -5 & 1 & 2 & -3 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 6 & 2 & 1 \\ -1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 3 \\ 6 & 4 & -1 & 2 \\ 1 & -2 & 3 & 0 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 28 & 17 & 20 & 10 \\ 20 & -13 & -3 & -1 \\ -18 & -44 & 12 & -3 \end{bmatrix}.$$

Let's show the calculation of some entries. For example,

$$[AB]_{11} = \sum_{k=1}^{n} a_{1k} b_{k1} = 1 \times 1 + 2 \times (-1) + (-1) \times 1 + 4 \times 6 + 6 \times 1 = 28.$$

$$[AB]_{23} = \sum_{k=1}^{n} a_{2k} b_{k3} = 0 \times 2 + (-4) \times 3 + 1 \times 2 + 2 \times (-1) + 3 \times 3 = -3.$$

Theorem 3.2.1. *The following properties hold for any scalars $\alpha, \beta \in \mathbb{R}$ and any matrices A, B, C for which the indicated operations are defined.*

(i) Multiplication of Matrix and Zero Matrix

$A\mathcal{O} = \mathcal{O},\ \mathcal{O}A = \mathcal{O}.$

(ii) Distributivity of Matrix Multiplication across Matrix Addition

$A(B + C) = AB + AC,\ (A + B)C = AC + BC.$

(iii) Matrix Multiplication and Scalar Multiplication

$(\alpha\beta)A = \alpha(\beta A),\ \alpha(AB) = (\alpha A)B = A(\alpha B).$

(iv) Associativity of Matrix Multiplication

$A(BC) = (AB)C.$

Proof. We only prove (ii) and (iv).

(ii). For any matrices $A_{m\times n},\ B_{n\times p},\ C_{n\times p}$,

$$\begin{aligned}
[A(B+C)]_{ij} &= \sum_{k=1}^{n} [A]_{ik} [B+C]_{kj} \\
&= \sum_{k=1}^{n} [A]_{ik} ([B]_{kj} + [C]_{kj}) \\
&= \sum_{k=1}^{n} [A]_{ik} [B]_{kj} + [A]_{ik} [C]_{kj} \\
&= \sum_{k=1}^{n} [A]_{ik} [B]_{kj} + \sum_{k=1}^{n} [A]_{ik} [C]_{kj} \\
&= [AB]_{ij} + [AC]_{ij} \\
&= [AB + AC]_{ij}
\end{aligned}$$

Since all the entries of $A(B+C)$ and $AB+AC$ are equal, so $A(B+C) = AB+AC$.

(iv). For any matrices $A_{m\times n},\ B_{n\times p},\ C_{p\times q}$,

$$\begin{aligned}
[A(BC)]_{ij} &= \sum_{k=1}^{n} [A]_{ik}\,[BC]_{kj} \\
&= \sum_{k=1}^{n} [A]_{ik} \left(\sum_{\ell=1}^{p} [B]_{k\ell}\,[C]_{\ell j} \right) \\
&= \sum_{k=1}^{n} \sum_{\ell=1}^{p} [A]_{ik}\,[B]_{k\ell}\,[C]_{\ell j}
\end{aligned}$$

We can switch the order of the summation since these are finite sums,

$$= \sum_{\ell=1}^{p} \sum_{k=1}^{n} [A]_{ik}\,[B]_{k\ell}\,[C]_{\ell j}$$

As $[C]_{\ell j}$ does not depend on the index k, we can use distributivity to move it outside of the inner sum,

$$\begin{aligned}
&= \sum_{\ell=1}^{p} [C]_{\ell j} \left(\sum_{k=1}^{n} [A]_{ik}\,[B]_{k\ell} \right) \\
&= \sum_{\ell=1}^{p} [C]_{\ell j}\,[AB]_{i\ell} \\
&= \sum_{\ell=1}^{p} [AB]_{i\ell}\,[C]_{\ell j} \\
&= [(AB)C]_{ij}
\end{aligned}$$

So the matrices $(AB)C$ and $A(BC)$ are equal, entry-by-entry, and by the definition of matrix equality we can say they are equal matrices.

□

Remark 3.2.2.

(a) $\boxed{AB \neq BA, \text{ in general.}}$ Matrix multiplication is not commutative!

- If A is a 2×3 matrix, B is a 3×4 matrix, then AB is a 2×4 matrix, but BA is not defined.

- If A is a 2×3 matrix, B is a 3×2 matrix, then AB is a 2×2 matrix, but BA is a 3×3 matrix.
- Even if AB and BA are well-defined and have the same size, they may be unequal. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 0 & 2 \\ 0 & 4 \end{bmatrix}, \quad BA = \begin{bmatrix} 0 & 0 \\ 3 & 4 \end{bmatrix}.$$

(b) $\boxed{AB = \mathcal{O} \not\Rightarrow A = \mathcal{O} \text{ or } B = \mathcal{O}, \text{ in general.}}$

Next we will represent systems of linear equations using matrix multiplication. Consider the following linear system,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \cdots + a_{3n}x_n &= b_3 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= b_m \end{aligned} \tag{3.4}$$

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b2 \\ b_3 \\ \vdots \\ b_m \end{bmatrix}$$

Then it is easy to check that the linear system (3.4) is equivalent to the matrix equation

$$A\mathbf{x} = \mathbf{b}.$$

Definition 3.2.3 (Matrix Representation of Linear Systems). The linear system (3.4) can be represented by the matrix equation

$$A\mathbf{x} = \mathbf{b}.$$

Example 3.2.2.

(a) In Example 2.2.4, the linear system

$$\begin{aligned} -7x_1 - 6x_2 - 12x_3 &= -33 \\ 5x_1 + 5x_2 + 7x_3 &= 24 \\ x_1 + 4x_3 &= 5 \end{aligned}$$

can be represented by

$$\begin{bmatrix} -7 & -6 & -12 \\ 5 & 5 & 7 \\ 1 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -33 \\ 24 \\ 5 \end{bmatrix}.$$

The solution can be represented in vector form:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 5 \\ 2 \end{bmatrix}.$$

(b) In Example 2.3.1, the linear system

$$\begin{aligned} x_1 + 4x_2 - x_4 + 7x_6 &= 3 \\ 2x_1 + 8x_2 - x_3 + 3x_4 + 9x_5 - 13x_6 &= 9 \\ 2x_3 - 3x_4 - 4x_5 + 12x_6 &= 1 \\ -x_1 - 4x_2 + 2x_3 + 4x_4 + 8x_5 - 31x_6 &= 4 \end{aligned}$$

can be represented by

$$\begin{bmatrix} 1 & 4 & 0 & -1 & 0 & 7 \\ 2 & 8 & -1 & 3 & 9 & -13 \\ 0 & 0 & 2 & -3 & -4 & 12 \\ -1 & -4 & 2 & 4 & 8 & -31 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 3 \\ 9 \\ 1 \\ 4 \end{bmatrix}$$

The solution can be represented in vector form:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 4 - 4x_2 - 2x_5 - x_6 \\ x_2 \\ 2 - x_5 + 3x_6 \\ 1 - 2x_5 + 6x_6 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -4 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -2 \\ 0 \\ -1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_6 \begin{bmatrix} -1 \\ 0 \\ 3 \\ 6 \\ 0 \\ 1 \end{bmatrix}$$

where $x_2,\ x_5$ and x_6 are arbitrary numbers.

Example 3.2.3. Suppose that A and B are $m \times n$ matrices such that $A\mathbf{x} = B\mathbf{x}$ for every $\mathbf{x} \in \mathbb{R}^n$. Then $A = B$.

Proof. We are assuming $A\mathbf{x} = B\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$, so we can employ this equality for *any* choice of the vector $\mathbf{x}$. However, we will limit our use of this equality to the standard unit vectors.

Denote the (i, j)-th entries of A and B by a_{ij} and b_{ij}. For $1 \leq j \leq n$, we define **the standard unit vector $\mathbf{e}_j$** as the column vector with the j-th entry to be 1 and other entries to be zero.

For each $1 \leq j \leq n$, we can check

$$A\mathbf{e}_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}, \quad B\mathbf{e}_j = \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{mj} \end{bmatrix}$$

are the j-th columns of A and B respectively, which must be equal according to the given assumption. So we have $A = B$.

$\square$

3.3 Transposes and symmetric matrices

We describe one more common operation which can be performed on matrices. Informally, to transpose a matrix is to build a new matrix by swapping its rows and columns.

Definition 3.3.1 (Transpose). Given an $m \times n$ matrix A, its **transpose** is the $n \times m$ matrix A^t given by

$$\left[A^t\right]_{ij} = [A]_{ji}, \quad 1 \leq i \leq n,\, 1 \leq j \leq m,$$

i.e.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nm} \end{bmatrix}^t = \begin{bmatrix} a_{11} & a_{21} & a_{31} & \dots & a_{m1} \\ a_{12} & a_{22} & a_{32} & \dots & a_{m2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \dots & a_{mn} \end{bmatrix}.$$

Example 3.3.1. Transpose of a 3×4 matrix

$$D = \begin{bmatrix} 3 & 7 & 2 & -3 \\ -1 & 4 & 2 & 8 \\ 0 & 3 & -2 & 5 \end{bmatrix}, \text{ then } D^t = \begin{bmatrix} 3 & -1 & 0 \\ 7 & 4 & 3 \\ 2 & 2 & -2 \\ -3 & 8 & 5 \end{bmatrix}.$$

Definition 3.3.2 (Symmetric Matrix). A matrix A is **symmetric** if $A^t = A$, i.e.

$$[A]_{ij} = [A]_{ji}, \text{ for all } i, j.$$

Remark 3.3.1 (Symmetric matrices must be square matrices)**.** Let A be a $m \times n$ matrix, then A^t is a $n \times m$ matrix. If A is symmetric, i.e., $A^t = A$, then they must have the same size, so $m = n$.

Example 3.3.2. The following matrix is symmetric:

$$\begin{bmatrix} 2 & 3 & -9 & 5 & 7 \\ 3 & 1 & 6 & -2 & -3 \\ -9 & 6 & 0 & -1 & 9 \\ 5 & -2 & -1 & 4 & -8 \\ 7 & -3 & 9 & -8 & -3 \end{bmatrix}$$

Remark 3.3.2. For a $n \times n$ square matrix A, the collection of all entries

$$[A]_{11},\ [A]_{22},\ [A]_{33},\ \cdots,\ [A]_{nn}$$

is called the diagonal (or main diagonal) of A. Thus, that A is symmetric means the entries of A are symmetric about the diagonal.

Theorem 3.3.1. *The following properties hold for any scalars $\alpha \in \mathbb{R}$ and any matrices A, B for which the indicated operations are defined.*

(i) Transpose and Matrix Addition

$(A + B)^t = A^t + B^t.$

(ii) Transpose and Matrix Scalar Multiplication

$(\alpha A)^t = \alpha A^t.$

(iii) Transpose of Transpose

$(A^t)^t = A.$

(iv) Transpose and Matrix Multiplication

$(AB)^t = B^t A^t.$

Proof. (i) For all i, j,

$$\begin{aligned}\left[(A+B)^t\right]_{ij} &= [A+B]_{ji} \\ &= [A]_{ji} + [B]_{ji} \\ &= \left[A^t\right]_{ij} + \left[B^t\right]_{ij} \\ &= \left[A^t + B^t\right]_{ij}.\end{aligned}$$

So $(A+B)^t = A^t + B^t$.

(iv). Suppose A is an $m \times n$ matrix, and B is an $n \times p$ matrix, then $(AB)^t$ and $B^t A^t$ are both $p \times m$ matrices.

For all $1 \leq i \leq m$, $1 \leq j \leq p$,

$$\begin{aligned}\left[(AB)^t\right]_{ji} &= [AB]_{ij} \\ &= \sum_{k=1}^{n} [A]_{ik} [B]_{kj} \\ &= \sum_{k=1}^{n} [B]_{kj} [A]_{ik} \\ &= \sum_{k=1}^{n} \left[B^t\right]_{jk} \left[A^t\right]_{ki} \\ &= \left[B^t A^t\right]_{ji}.\end{aligned}$$

So $(AB)^t = B^t A^t$.

□

Definition 3.3.3 (**Skew-symmetric/Anti-symmetric Matrix**). A matrix A is **anti-symmetric** if $A^t = -A$, i.e.,

$$[A]_{ij} = -[A]_{ji}, \text{ for all } i, j.$$

Remark 3.3.3. If A is anti-symmetric, then A must be a square matrix, and its diagonal entries are all zero, i.e.,

$$[A]_{ii} = 0, \quad \text{for all } i.$$

Example 3.3.3. The following matrix is anti-symmetric:

$$\begin{bmatrix} 0 & 3 & -9 & 5 & 7 \\ -3 & 0 & 6 & -2 & -3 \\ 9 & -6 & 0 & -1 & 9 \\ -5 & 2 & 1 & 0 & -8 \\ -7 & 3 & -9 & 8 & 0 \end{bmatrix}$$

3.4 Special matrices

In this section, we will discuss some special matrices, including diagonal matrix, upper/lower triangular matrix, identity matrix, and elementary matrix.

Definition 3.4.1 (Diagonal Matrix). A $n \times n$ square matrix A is called **diagonal** if all its off-diagonal entries are zero , i.e.,

$$[A]_{ij} = 0 \text{ for all } i \neq j.$$

The general form is

$$A = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix},$$

and we can denote $A = \text{diag}(a_{11},\ a_{22},\ \cdots,\ a_{nn})$.

Theorem 3.4.1. *The product of two $n \times n$ diagonal matrices is still diagonal,*

$$\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} b_{11} & 0 & \cdots & 0 \\ 0 & b_{22} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & b_{nn} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & 0 & \cdots & 0 \\ 0 & a_{22}b_{22} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{nn}b_{nn} \end{bmatrix}.$$

Definition 3.4.2 (Upper/Lower Triangular Matrix). A $n \times n$ square matrix A is called **upper triangular** if all its entries below the diagonal are zero, i.e.,

$$[A]_{ij} = 0 \text{ for all } i > j.$$

The general form is

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}.$$

The definition of lower triangular matrix is similar.

Theorem 3.4.2. *The product of two $n \times n$ upper/lower triangular matrices is still upper/lower triangular,*

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ 0 & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & b_{nn} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & * & \cdots & * \\ 0 & a_{22}b_{22} & \cdots & * \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{nn}b_{nn} \end{bmatrix}.$$

Remark 3.4.1.

(1) If A is upper triangular, then its transpose A^t is lower triangular.

(2) In some textbooks, the concept of upper/lower triangular matrix is defined generally (without change) for a $m \times n$ matrix.

Definition 3.4.3 (Identity Matrix). An **identity matrix** of order n, denoted by I or I_n, is a diagonal $n \times n$ matrix with all diagonal entries equal to 1.

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}, \text{ or simply } \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}.$$

Theorem 3.4.3 (**Matrix Multiplication and Identity Matrix**). *Suppose that A is an $m \times n$ matrix. Then*

(i) $AI_n = A$

(ii) $I_m A = A$

Proof. We only prove (i). For $1 \leq i \leq m$, $1 \leq j \leq n$,

$$[AI_n]_{ij} = \sum_{k=1}^{n} [A]_{ik} [I_n]_{kj} .$$

Note that $[I_n]_{kj} = 0$ for all $k \neq j$, so in the above summation, there is only one nonzero term when $k = j$,

$$[AI_n]_{ij} = [A]_{ij} [I_n]_{jj} = [A]_{ij} .$$

So $AI_n = A$. □

Just as the number 1 acts as an identity for the multiplication of real numbers, multiplying a matrix by I does not change the matrix. That is why we name I as "identity matrix".

Definition 3.4.4 (**Elementary Matrix**). An **elementary matrix** is a square matrix obtained by applying a single elementary row operation to the identity matrix. The elementary matrix is said to be of type I, II, or III, according to the type of the elementary row operation.

Example 3.4.1 (3×3 **elementary matrix in type I**). .

$$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is an elementary matrix of type I, since it was obtained by applying the elementary row operation $R_1 \leftrightarrow R_2$ to I_3.

Let A be a 3×3 matrix, then

$$E_1 A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$AE_1 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{12} & a_{11} & a_{13} \\ a_{22} & a_{21} & a_{23} \\ a_{32} & a_{31} & a_{33} \end{bmatrix}$$

Hence, multiplying A on the left by E_1 is equivalent to applying the same elementary row operation $R_1 \leftrightarrow R_2$ to A.

E_1 can be also obtained from I_3 by en elementary column operation $C_1 \leftrightarrow C_2$, and multiplying A on the right by E_1 is equivalent to applying the same elementary column operation to A.

Example 3.4.2 (3×3 **elementary matrix in type II).** .

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

is an elementary matrix of type II, since it was obtained by applying the elementary row operation $4R_3$ to I_3.

Let A be a 3×3 matrix, then

$$E_2 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 4a_{31} & 4a_{32} & 4a_{33} \end{bmatrix}$$

$$AE_2 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & 4a_{13} \\ a_{21} & a_{22} & 4a_{23} \\ a_{31} & a_{32} & 4a_{33} \end{bmatrix}$$

Hence, multiplying A on the left by E_2 is equivalent to applying the same elementary row operation $4R_3$ to A.

E_2 can be also obtained from I_3 by en elementary column operation $4C_3$, and multiplying A on the right by E_2 is equivalent to applying the same elementary column operation to A.

Example 3.4.3 (3×3 **elementary matrix in type III).** .

$$E_3 = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is an elementary matrix of type III, since it was obtained by applying the elementary row operation $4R_3 + R_1$ to I_3.

Let A be a 3×3 matrix, then

$$E_3A = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} + 4a_{31} & a_{12} + 4a_{32} & a_{13} + 4a_{33} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$AE_3 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & 4a_{11} + a_{13} \\ a_{21} & a_{22} & 4a_{21} + a_{23} \\ a_{31} & a_{32} & 4a_{31} + a_{33} \end{bmatrix}$$

Hence, multiplying A on the left by E_3 is equivalent to applying the same elementary row operation $4R_3 + R_1$ to A.

E_3 can be also obtained from I_3 by en elementary column operation $4C_1 + C_3$, and multiplying A on the right by E_3 is equivalent to applying the same elementary column operation to A.

Theorem 3.4.4 (Row/Column Operation and Matrix Multiplication). *Suppose that E is an $n \times n$ elementary matrix obtained from I_n by either a row operation or a column operation.*

(i) If A is an $n \times p$ matrix, then EA has the effect of performing that same row operation on A.

(ii) If B is an $m \times n$ matrix, then BE has the effect of performing that same column operation on B.

Remark 3.4.2. The above theorem establishes the relation between elementary row/column operation and matrix multiplication, using the bridge of elementary matrix.

Corollary 3.4.1. If a matrix A is row equivalent to B, then there exists a finite sequence of elementary matrices $E_1,\ E_2,\ \cdots,\ E_k$ such that

$$E_k E_{k-1} \cdots E_1 A = B.$$

Proof. By definition of row equivalence for matrices, A can be transformed to B by performing a finite sequence of elementary row operations. Denote the corresponding elementary matrices by $E_1,\ E_2,\ \cdots,\ E_k$, then the conclusion follows from the previous theorem. □

3.5 Block matrices

A matrix A can be partitioned into smaller submatrices by drawing horizontal lines between rows and vertical lines between columns. The smaller submatrices are called "blocks". For example, the matrix

$$A = \begin{bmatrix} 1 & 2 & 5 & 6 & 7 \\ 3 & 4 & 8 & 9 & 10 \\ 11 & 12 & 21 & 22 & 23 \\ 13 & 14 & 24 & 25 & 26 \\ 15 & 16 & 27 & 28 & 29 \end{bmatrix}$$

can be divided into four submatrices $A_{11},\ A_{12},\ A_{21}$, and A_{22},

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \left[\begin{array}{cc|ccc} 1 & 2 & 5 & 6 & 7 \\ 3 & 4 & 8 & 9 & 10 \\ \hline 11 & 12 & 21 & 22 & 23 \\ 13 & 14 & 24 & 25 & 26 \\ 15 & 16 & 27 & 28 & 29 \end{array}\right]$$

In general, a matrix A with s row partitions and t column partitions is

$$\begin{array}{c} \\ m_1 \\ m_2 \\ \vdots \\ m_s \end{array} \begin{array}{c} \begin{array}{cccc} p_1 & p_2 & \cdots & p_t \end{array} \\ \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1t} \\ A_{21} & A_{22} & \cdots & A_{2t} \\ \vdots & \vdots & & \vdots \\ A_{s1} & A_{s2} & \cdots & A_{st} \end{bmatrix} \end{array}$$

where the (i, j)-th block A_{ij} contains m_i rows and p_j columns.

I. Addition of block matrices: To calculate the sum of two block matrices with the same partitions, just add each block.

Example 3.5.1.

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \left[\begin{array}{cc|ccc} 1 & 2 & 5 & 6 & 7 \\ 3 & 4 & 8 & 9 & 10 \\ \hline 11 & 12 & 13 & 14 & 15 \end{array}\right]$$

$$B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \left[\begin{array}{cc|ccc} 1 & 0 & 0 & 2 & 3 \\ 3 & 1 & 2 & 1 & -1 \\ \hline 0 & -2 & 3 & 2 & 1 \end{array}\right]$$

Then

$$A + B = \begin{bmatrix} A_{11} + B_{11} & A_{12} + B_{12} \\ A_{21} + B_{21} & A_{22} + B_{22} \end{bmatrix} = \left[\begin{array}{cc|ccc} 2 & 2 & 5 & 8 & 10 \\ 6 & 5 & 10 & 10 & 9 \\ \hline 11 & 10 & 16 & 16 & 16 \end{array}\right]$$

II. Scalar multiplication of block matrices: To calculate the product of a scalar and a block matrix, just multiply each block by the scalar.

Example 3.5.2.

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \left[\begin{array}{cc|ccc} 1 & 2 & 5 & 6 & 7 \\ 3 & 4 & 8 & 9 & 10 \\ \hline 11 & 12 & 13 & 14 & 15 \end{array}\right]$$

Then

$$2A = \begin{bmatrix} 2A_{11} & 2A_{12} \\ 2A_{21} & 2A_{22} \end{bmatrix} = \left[\begin{array}{cc|ccc} 2 & 4 & 10 & 12 & 14 \\ 6 & 8 & 16 & 18 & 20 \\ \hline 22 & 24 & 26 & 28 & 30 \end{array}\right]$$

III. Block matrices multiplication:

Example 3.5.3.

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \left[\begin{array}{cc|cc} 1 & 2 & 5 & 6 \\ 3 & 4 & 8 & 9 \\ \hline 11 & 12 & 13 & 14 \end{array}\right]$$

$$B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \left[\begin{array}{ccc|cc} 1 & 0 & 0 & 2 & 3 \\ 3 & 1 & 2 & 1 & -1 \\ \hline 0 & -2 & 3 & 2 & 1 \\ 1 & 2 & 0 & 1 & 3 \end{array}\right]$$

To compute AB, we first treat the blocks A_{ij} and B_{ij} as numbers, and use the ordinary matrix multiplication

$$AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix} = \left[\begin{array}{ccc|cc} 13 & 4 & 19 & 20 & 24 \\ 24 & 6 & 32 & 35 & 40 \\ \hline 61 & 14 & 63 & 74 & 76 \end{array}\right]$$

which is the same as the answer obtained by directly multiplying A with B.

In general, if the blocks have the proper sizes, the block multiplication can be carried out in the same manner as ordinary matrix multiplication. That is, if A is partitioned as

$$\begin{array}{c} \\ m_1 \\ m_2 \\ \vdots \\ m_s \end{array} \begin{array}{c} \begin{array}{cccc} p_1 & p_2 & \cdots & p_t \end{array} \\ \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1t} \\ A_{21} & A_{22} & \cdots & A_{2t} \\ \vdots & \vdots & & \vdots \\ A_{s1} & A_{s2} & \cdots & A_{st} \end{bmatrix} \end{array},$$

and B is partitioned as

$$\begin{array}{c} \\ p_1 \\ p_2 \\ \vdots \\ p_t \end{array} \begin{array}{c} \begin{array}{cccc} n_1 & n_2 & \cdots & n_r \end{array} \\ \begin{bmatrix} B_{11} & B_{12} & \cdots & B_{1r} \\ B_{21} & B_{22} & \cdots & B_{2r} \\ \vdots & \vdots & & \vdots \\ B_{t1} & B_{t2} & \cdots & B_{tr} \end{bmatrix} \end{array},$$

then the block multiplication AB is

$$
\begin{array}{c}
 \\ m_1 \\ m_2 \\ \vdots \\ m_s
\end{array}
\begin{array}{c}
\begin{array}{cccc} n_1 & n_2 & \cdots & n_r \end{array} \\
\begin{bmatrix}
C_{11} & C_{12} & \cdots & C_{1r} \\
C_{21} & C_{22} & \cdots & C_{2r} \\
\vdots & \vdots & & \vdots \\
C_{s1} & C_{s2} & \cdots & C_{sr}
\end{bmatrix}
\end{array},
$$

where

$$C_{ij} = \sum_{k=1}^{t} A_{ik} B_{kj}.$$

The only requirement is that the blocks are "compatible", in the sense that the sizes of the blocks must be such that all matrix products of blocks that occur make sense. It means that for each k, the number of columns in A_{ik} equals the number of rows in B_{kj}, which is denoted by p_k.

Example 3.5.4. Let A be an $m \times n$ matrix, and B be an $n \times r$ matrix partitioned into columns $B = [\mathbf{b}_1,\ \mathbf{b}_2,\ \cdots,\ \mathbf{b}_r]$, then

$$AB = A\left[\mathbf{b}_1,\ \mathbf{b}_2,\ \cdots,\ \mathbf{b}_r\right] = \left[A\mathbf{b}_1,\ A\mathbf{b}_2,\ \cdots,\ A\mathbf{b}_r\right]$$

Example 3.5.5. Let A be an $m \times n$ matrix partitioned into rows $A = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_m \end{bmatrix}$, and B be an $n \times r$ matrix, then

$$AB = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_m \end{bmatrix} B = \begin{bmatrix} \mathbf{a}_1 B \\ \mathbf{a}_2 B \\ \vdots \\ \mathbf{a}_m B \end{bmatrix}.$$

Example 3.5.6. Consider the $m \times n$ linear system $A\mathbf{x} = \mathbf{b}$. Partition A into columns

$A = \begin{bmatrix} \mathbf{a}_1, & \mathbf{a}_2, & \cdots, & \mathbf{a}_n \end{bmatrix}$, then $A\mathbf{x} = \mathbf{b}$ can be rewritten as

$$\mathbf{b} = \begin{bmatrix} \mathbf{a}_1, & \mathbf{a}_2, & \cdots, & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

The right hand side is called a "linear combination" of the vectors $\mathbf{a}_1, \cdots, \mathbf{a}_n$.

Example 3.5.7. Let

$$A = \begin{bmatrix} 1 & -3 \\ -2 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 0 & -2 \\ 2 & 0 & 4 \end{bmatrix}$$

Solve the matrix equation $AX = B$.

Solution. Partition X and B into columns, $X = [\mathbf{X}_1, \ \mathbf{X}_2, \ \mathbf{X}_3]$, $B = [\mathbf{B}_1, \ \mathbf{B}_2, \ \mathbf{B}_3]$. Then

$$\begin{aligned} AX = B &\Leftrightarrow \begin{bmatrix} A\mathbf{X}_1, & A\mathbf{X}_2, & A\mathbf{X}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{B}_1, & \mathbf{B}_2, & \mathbf{B}_3 \end{bmatrix} \\ &\Leftrightarrow A\mathbf{X}_1 = \mathbf{B}_1, \ A\mathbf{X}_2 = \mathbf{B}_2, \ A\mathbf{X}_3 = \mathbf{B}_3. \end{aligned}$$

We need to solve the above 3 linear systems. Since their coefficients matrices are the same A, we can put A and B in an augmented matrix:

$$\begin{aligned} \left[\begin{array}{c|c} A & B \end{array}\right] &= \left[\begin{array}{cc|ccc} 1 & -3 & -1 & 0 & -2 \\ -2 & 6 & 2 & 0 & 4 \end{array}\right] \\ &\xrightarrow{2R_1+R_2} \left[\begin{array}{cc|ccc} 1 & -3 & -1 & 0 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \end{aligned}$$

Hence the solutions to $A\mathbf{X}_1 = \mathbf{B}_1, \ A\mathbf{X}_2 = \mathbf{B}_2, \ A\mathbf{X}_3 = \mathbf{B}_3$ are

$$\mathbf{X}_1 = \begin{bmatrix} 3c_1 - 1 \\ c_1 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 3c_2 \\ c_2 \end{bmatrix}, \quad \mathbf{X}_3 = \begin{bmatrix} 3c_3 - 2 \\ c_3 \end{bmatrix},$$

where c_1, c_2 and c_3 are arbitrary numbers.

Hence

$$X = \begin{bmatrix} 3c_1 - 1 & 3c_2 & 3c_3 - 2 \\ c_1 & c_2 & c_3 \end{bmatrix}.$$

□

4 Homogeneous System and Nonsingular Matrix

In this section, we will discuss nonsingular matrices, and the structure of solution sets to homogeneous systems and nonhomogeneous systems.

4.1 Homogeneous System and Null Space

A general $m \times n$ linear system, denoted by $\mathcal{LS}(A, \mathbf{b})$ or $A\mathbf{x} = \mathbf{b}$, is

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \cdots + a_{3n}x_n &= b_3 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

where A is the coefficient matrix, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$.

Definition 4.1.1 (Homogeneous System). A linear system $A\mathbf{x} = \mathbf{0}$ is called **homogeneous**, i.e.,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0 \end{aligned}$$

For a general system $A\mathbf{x} = \mathbf{b}$, we say the corresponding homogeneous system is $A\mathbf{x} = \mathbf{0}$.

Example 4.1.1. Given

$$\begin{aligned} x_1 - 2x_2 + 3x_3 - 4x_4 &= 1 \\ x_2 - x_4 &= 2 \\ x_1 + 3x_2 - 5x_3 + 5x_4 &= 3, \end{aligned}$$

the corresponding homogeneous system is

$$\begin{aligned} x_1 - 2x_2 + 3x_3 - 4x_4 &= 0 \\ x_2 - x_4 &= 0 \\ x_1 + 3x_2 - 5x_3 + 5x_4 &= 0. \end{aligned}$$

Theorem 4.1.1 (**Homogeneous Systems are Consistent**). *Suppose that a system of linear equations is homogeneous. Then the system is consistent.*
In fact, $\mathbf{x} = \mathbf{0}$ *(i.e.* $x_1 = x_2 = \cdots = x_n = 0$*) is always a solution. This solution is called a* **trivial solution**.

Example 4.1.2.

(a)

$$\begin{aligned} -7x_1 - 6x_2 - 12x_3 &= 0 \\ 5x_1 + 5x_2 + 7x_3 &= 0 \\ x_1 + 4x_3 &= 0 \end{aligned}$$

The reduced row echelon form of the augmented matrix is

$$\left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & 0 \\ 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & \boxed{1} & 0 \end{array}\right]$$

It has no free variable. Hence it has only the trivial solution.

(b)

$$\begin{aligned} x_1 - x_2 + 2x_3 &= 0 \\ 2x_1 + x_2 + x_3 &= 0 \\ x_1 + x_2 &= 0 \end{aligned}$$

The reduced row echelon form of the augmented matrix is

$$\left[\begin{array}{ccc|c} \boxed{1} & 0 & 1 & 0 \\ 0 & \boxed{1} & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

It has 1 free variable x_3. The solution set is

$$\begin{aligned} S =& \{ \begin{bmatrix} -x_3 \\ x_3 \\ x_3 \end{bmatrix} \mid x_3 \in \mathbb{R} \} \\ =& \{ x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \mid x_3 \in \mathbb{R} \} \end{aligned}$$

Notice that the augmented matrix for the a homogeneous system is $[A|\mathbf{0}]$, and any row operations will not change the last zero column. Therefore, for homogeneous systems, we can replace the augmented matrix by the coefficient matrix. Just remember there is actually a zero column as the last column.

Example 4.1.3.

$$\begin{aligned} 2x_1 + x_2 + 7x_3 - 7x_4 &= 0 \\ -3x_1 + 4x_2 - 5x_3 - 6x_4 &= 0 \\ x_1 + x_2 + 4x_3 - 5x_4 &= 0 \end{aligned}$$

We can apply row operations directly to the coefficient matrix

$$A = \begin{bmatrix} 2 & 1 & 7 & -7 \\ -3 & 4 & -5 & -6 \\ 1 & 1 & 4 & -5 \end{bmatrix}$$

The RREF is

$$\begin{bmatrix} \boxed{1} & 0 & 3 & -2 \\ 0 & \boxed{1} & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The corresponding augmented matrix is

$$\left[\begin{array}{cccc|c} \boxed{1} & 0 & 3 & -2 & 0 \\ 0 & \boxed{1} & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The system has 2 free variables x_3 and x_4. The solution set is

$$\begin{aligned} S =& \{ \begin{bmatrix} -3x_3 + 2x_4 \\ -x_3 + 3x_4 \\ x_3 \\ x_4 \end{bmatrix} \mid x_3,\, x_4 \in \mathbb{R} \} \\ =& \{ x_3 \begin{bmatrix} -3 \\ -1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ 3 \\ 0 \\ 1 \end{bmatrix} \mid x_3,\, x_4 \in \mathbb{R} \} \end{aligned}$$

Remark 4.1.1. In the above example, it is not necessary to express the solution as a sum of one part involving x_3 and another part involving x_4. Here we do this decomposition just in order to view the solution structure more clearly. As we will define later, this expression is called a linear combination.

Theorem 4.1.2 (Special case $n > m$). *Suppose a homogeneous system $A\mathbf{x} = \mathbf{0}$ has m equations and n variables with $n > m$. Then it has infinitely many solutions.*

Proof. The system is homogeneous, so it is consistent. Since $n > m$, by Corollary 2.3.1, there must exist free variables and the system has infinitely many solutions. $\square$

Definition 4.1.2 (Null Space of a Matrix). Let A be an $m \times n$ matrix. The **null space** of A, denoted by $\mathcal{N}(A)$, is the set of all solutions $\mathbf{x}$ to the homogeneous system $A\mathbf{x} = \mathbf{0}$. That is,

$$\mathcal{N}(A) = \{\mathbf{x} \in \mathbb{R}^n | A\mathbf{x} = \mathbf{0}\}.$$

Remark 4.1.2. If A is an $m \times n$ matrix, all the solutions to $A\mathbf{x} = 0$ form a subset of $\mathbb{R}^n$. As we will show in the future, this subset actually forms a subspace of $\mathbb{R}^n$. That is why we call it a null "space".

Example 4.1.4. Let

$$A = \begin{bmatrix} 1 & 4 & 0 & -1 & 0 & 7 & -9 \\ 2 & 8 & -1 & 3 & 9 & -13 & 7 \\ 0 & 0 & 2 & -3 & -4 & 12 & -8 \\ -1 & -4 & 2 & 4 & 8 & -31 & 37 \end{bmatrix}, \ \mathbf{x} = \begin{bmatrix} 3 \\ 0 \\ -5 \\ -6 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \ \mathbf{y} = \begin{bmatrix} -4 \\ 1 \\ -3 \\ -2 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \ \mathbf{z} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 2 \end{bmatrix}$$

We can check $A\mathbf{x} = \mathbf{0}$, $A\mathbf{y} = \mathbf{0}$ and $A\mathbf{z} = \begin{bmatrix} -17 \\ 16 \\ -16 \\ 73 \end{bmatrix} \neq \mathbf{0}$,

Therefor, $\mathbf{x}, \ \mathbf{y} \in \mathcal{N}(A)$ and $\mathbf{z} \notin \mathcal{N}(A)$.

Example 4.1.5. Find the null space of

$$A = \begin{bmatrix} 2 & -1 & 7 & -3 & -8 \\ 1 & 0 & 2 & 4 & 9 \\ 2 & 2 & -2 & -1 & 8 \end{bmatrix}.$$

Solution. It is equivalent to find the solution set to the homogeneous system $A\mathbf{x} = \mathbf{0}$. We row-reduce the augmented matrix to obtain

$$\left[\begin{array}{ccccc|c} \boxed{1} & 0 & 2 & 0 & 1 & 0 \\ 0 & \boxed{1} & -3 & 0 & 4 & 0 \\ 0 & 0 & 0 & \boxed{1} & 2 & 0 \end{array}\right]$$

There are 2 free variables x_3 and x_5, and the solution set is

$$\begin{aligned} \mathcal{N}(A) =& \{ \begin{bmatrix} -2x_3 - x_5 \\ 3x_3 - 4x_5 \\ x_3 \\ -2x_5 \\ x_5 \end{bmatrix} \mid x_3,\, x_5 \in \mathbb{R}^n \} \\ =& \{ x_3 \begin{bmatrix} -2 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -1 \\ -4 \\ 0 \\ -2 \\ 1 \end{bmatrix} \mid x_3,\, x_5 \in \mathbb{R}^n \}. \end{aligned}$$

□

Example 4.1.6. Find the null space of

$$C = \begin{bmatrix} -4 & 6 & 1 \\ -1 & 4 & 1 \\ 5 & 6 & 7 \\ 4 & 7 & 1 \end{bmatrix}$$

Solution. To solve $C\mathbf{x} = \mathbf{0}$, we row-reduce the augmented matrix to obtain

$$\left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & 0 \\ 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

There is no free variable and the only solution is the trivial solution $\mathbf{x} = \mathbf{0}$. So

$$\mathcal{N}(C) = \{\mathbf{0}\} = \{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \}.$$

$\square$

4.2 Solution Structure of Nonhomogeneous System

The next theorem is about the relation between the solution sets to the nonhomogeneous system $A\mathbf{x} = \mathbf{b}$ and its corresponding homogeneous system $A\mathbf{x} = \mathbf{0}$.

Theorem 4.2.1 (Particular Solution Plus Homogeneous Solutions). *Let* $\mathbf{w}$ *be one solution to the linear system* $A\mathbf{x} = \mathbf{b}$ *(which is called a "particular solution"). Then* $\mathbf{y}$ *is a solution to* $A\mathbf{x} = \mathbf{b}$ *if and only if* $\mathbf{y} - \mathbf{w}$ *is a solution to* $A\mathbf{x} = \mathbf{0}$*, i.e.,* $\mathbf{y} = \mathbf{w} + \mathbf{z}$ *for some* $\mathbf{z} \in \mathcal{N}(A)$*.*

Proof.

($\Rightarrow$) Suppose $\mathbf{y}$ is a solution to $A\mathbf{x} = \mathbf{b}$, then

$$A(\mathbf{y} - \mathbf{w}) = A\mathbf{y} - A\mathbf{w} = \mathbf{b} - \mathbf{b} = \mathbf{0}.$$

Hence $\mathbf{y} - \mathbf{w}$ is a solution to $A\mathbf{x} = \mathbf{0}$.

($\Leftarrow$) Suppose $\mathbf{y} - \mathbf{w}$ is a solution to $A\mathbf{x} = \mathbf{0}$, then

$$A\mathbf{y} = A(\mathbf{y} - \mathbf{w}) + A\mathbf{w} = \mathbf{0} + \mathbf{b} = \mathbf{b}.$$

Hence $\mathbf{y}$ is a solution to $A\mathbf{x} = \mathbf{b}$. □

Remark 4.2.1. The theorem tells that, pick a particular solution $\mathbf{w}$ to $A\mathbf{x} = \mathbf{b}$, then every solution to $A\mathbf{x} = \mathbf{b}$ can be expressed as $\mathbf{w} + \mathbf{z}$ where $\mathbf{z}$ is a solution to $A\mathbf{x} = \mathbf{0}$. In other words:

"totality of all solutions to $A\mathbf{x} = \mathbf{b}$"
$=$"one particular solution to $A\mathbf{x} = \mathbf{b}$" + "totality of all solutions to $A\mathbf{x} = \mathbf{0}$"

We can also write the solution set of $A\mathbf{x} = \mathbf{b}$ as

$$\mathbf{w} + \mathcal{N}(A) = \{\mathbf{w} + \mathbf{z} \mid \mathbf{z} \in \mathcal{N}(A)\}$$

where z is any particular solution to $A\mathbf{x} = \mathbf{b}$.

Example 4.2.1.

$$2x_1 + x_2 + 7x_3 - 7x_4 = 8$$

$$-3x_1 + 4x_2 - 5x_3 - 6x_4 = -12$$

$$x_1 + x_2 + 4x_3 - 5x_4 = 4$$

The augmented matrix can be row-reduced to

$$\left[\begin{array}{cccc|c} \boxed{1} & 0 & 3 & -2 & 4 \\ 0 & \boxed{1} & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

So the solution set is

$$S = \{ \begin{bmatrix} 4 - 3x_3 + 2x_4 \\ -x_3 + 3x_4 \\ x_3 \\ x_4 \end{bmatrix} \mid x_3,\, x_4 \in \mathbb{R} \}$$

$$= \{ \begin{bmatrix} 4 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ -1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ 3 \\ 0 \\ 1 \end{bmatrix} \mid x_3,\, x_4 \in \mathbb{R} \}$$

One particular solution can be chosen as $\mathbf{w} = \begin{bmatrix} 4 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ by putting $x_3 = x_4 = 0$.

The solution set to the corresponding homogeneous system is

$$\mathcal{N}(A) = \{ x_3 \begin{bmatrix} -3 \\ -1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ 3 \\ 0 \\ 1 \end{bmatrix} \mid x_3,\, x_4 \in \mathbb{R} \}$$

It can be seen that $S = \mathbf{w} + \mathcal{N}(A)$.

4.3 Nonsingular Matrix

We will introduce the definition of nonsingular matrices, and investigate their properties. Throughout this subsection, the matrices are all $n \times n$ square matrices.

Definition 4.3.1 (Nonsingular Matrices). Let A be an $n \times n$ square matrix. If the homogeneous system $A\mathbf{x} = \mathbf{0}$ has *only* trivial solution $\mathbf{x} = \mathbf{0}$, then A is called a **nonsingular** matrix. Otherwise we say A is a **singular** matrix.

Example 4.3.1.

(a) Let

$$A = \begin{bmatrix} -7 & -6 & -12 \\ 5 & 5 & 7 \\ 1 & 0 & 4 \end{bmatrix}.$$

By Example 4.1.2 (a), the homogeneous system $A\mathbf{x} = \mathbf{0}$ has *only* trivial solution $\mathbf{x} = \mathbf{0}$, so A is nonsingular.

(b) Let

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

By Example 4.1.2 (b), the homogeneous system $A\mathbf{x} = \mathbf{0}$ has nontrivial solutions $x \neq \mathbf{0}$, so A is singular.

Theorem 4.3.1 (Nonsingular Matrices have Trivial Null Spaces). *Let A be an $n \times n$ square matrix. Then A is nonsingular if and only if the null space of A contains only zero vector, i.e., $\mathcal{N}(A) = \{\mathbf{0}\}$.*

Proof. It just follows from the definitions of nonsingular matrix and null space.

A is nonsingular if and only if the solution set to $A\mathbf{x} = \mathbf{0}$ contains *only* trivial solution $\mathbf{x} = \mathbf{0}$, and this solution set is just denoted by $\mathcal{N}(A)$. $\square$

Theorem 4.3.2 (Nonsingular Matrices Row Reduce to Identity Matrices). *Let A be an $n \times n$ square matrix. Then A is nonsingular if and only if A is row equivalent to the identity matrix I_n.*

Proof.

($\Leftarrow$) Suppose A is row equivalent to I_n. Then for the homogeneous system $A\mathbf{x} = \mathbf{0}$, the augmented matrix $[A|\mathbf{0}]$ is row-reduced to $[I_n|\mathbf{0}]$. There is no free variable, and the system has just trivial solution, so A is nonsingular.

($\Rightarrow$) Suppose A is nonsingular. Then the homogeneous system $A\mathbf{x} = \mathbf{0}$ has a unique trivial solution, and has no free variable. We apply elementary row operations to transform $[A|\mathbf{0}]$ into the reduced row echelon form $[B|\mathbf{0}]$. Since there is no free variable, B must have n pivot columns among its total of n columns. This is enough to force B to be I_n (why?). $\square$

Example 4.3.2.

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

is row equivalent to the reduced row echelon form

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Since B is not I_3, A is singular.

Example 4.3.3.

$$A = \begin{bmatrix} -7 & -6 & -12 \\ 5 & 5 & 7 \\ 1 & 0 & 4 \end{bmatrix}$$

is row equivalent to the reduced row echelon form

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So A is nonsingular.

Theorem 4.3.3 (Nonsingular Matrices and Nonhomogeneous Systems). *Let A be an $n \times n$ square matrix. Then A is nonsingular if and only if the system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every choice of constant vector $\mathbf{b}$.*

Proof.

($\Leftarrow$) Suppose $A\mathbf{x} = \mathbf{b}$ has a unique solution for every choice of constant vector $\mathbf{b}$. We choose a special vector $\mathbf{b} = \mathbf{0}$, then by the assumption, the system $A\mathbf{x} = \mathbf{0}$ has a unique solution, which must be $\mathbf{x} = \mathbf{0}$. So A is nonsingular.

($\Rightarrow$) Suppose A is nonsingular, then it follows from Theorem 4.3.2 that A is row equivalent to I_n. For every system $A\mathbf{x} = \mathbf{b}$, the augmented matrix $[A|\mathbf{b}]$ can be row-reduced to $[I_n|\mathbf{c}]$. There are n lead variables, no free variable, and the unique solution is just given by $\mathbf{x} = \mathbf{c}$. $\square$

Theorem 4.3.4 (Nonsingular Matrix Equivalences). *Let A be an $n \times n$ square matrix. The following statements are equivalent.*

(i) A is nonsingular.

(ii) A is row equivalent to I_n.

(iii) The null space of A contains only the zero vector, $\mathcal{N}(A) = \{\mathbf{0}\}$.

(iv) The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every choice of $\mathbf{b}$.

Remark 4.3.1. By Corollary 3.4.1, the above condition (ii) is also equivalent to that there exists a finite sequence of elementary matrices $E_1,\ E_2,\ \cdots,\ E_k$ such that

$$E_k E_{k-1} \cdots E_1 A = I_n.$$

Denote $B = E_k E_{k-1} \cdots E_1$, then we get

$$BA = I_n.$$

This identity is related to matrix inverse. We will define invertible matrix in next section, and show that nonsingular is equivalent to invertible.

5 Invertible Matrix

This section is about matrix inverse and the equivalence between nonsingular matrix and invertible matrix.

5.1 Inverse of a Matrix

Motivation: To solve a single linear equation

$$ax = b,$$

if $a \neq 0$, we can multiply the equation by a^{-1} to obtain the solution $x = a^{-1}b = \dfrac{b}{a}$. Note that a^{-1} is called the (multiplicative) inverse of a satisfying that

$$a^{-1}a = aa^{-1} = 1.$$

Now consider a general $n \times n$ linear system

$$A\mathbf{x} = \mathbf{b}.$$

Is it possible to find a matrix B such that

$$AB = BA = I_n?$$

If yes, then multiply the original system by B on the left,

$$\begin{aligned} BA\mathbf{x} &= B\mathbf{b}, \\ I_n\mathbf{x} &= B\mathbf{b}, \\ \mathbf{x} &= B\mathbf{b}. \end{aligned}$$

The unique solution is given by $\mathbf{x} = B\mathbf{b}$. The matrix B is called the (multiplicative) inverse of A.

Example 5.1.1. Consider

$$A\mathbf{x} = \mathbf{b},$$

with

$$A = \begin{bmatrix} -7 & -6 & -12 \\ 5 & 5 & 7 \\ 1 & 0 & 4 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -33 \\ 24 \\ 5 \end{bmatrix}.$$

We define a 3×3 matrix B,

$$B = \begin{bmatrix} -10 & -12 & -9 \\ \frac{13}{2} & 8 & \frac{11}{2} \\ \frac{5}{2} & 3 & \frac{5}{2} \end{bmatrix}.$$

and check that

$$AB = BA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then multiply the original system by B on the left to obtain the unique solution

$$\mathbf{x} = B\mathbf{b} = \begin{bmatrix} -3 \\ 5 \\ 2 \end{bmatrix}.$$

Definition 5.1.1 (Matrix Inverse). An $n \times n$ matrix A is called **invertible** if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n.$$

The matrix B is called the **inverse** of A, and we write $B = A^{-1}$.

Remark 5.1.1. Notice that if B is the inverse of A, then A is also the inverse of B, so we can say A and B are inverses of each other.

Example 5.1.2. (Matrix with an inverse)

Show that $AB = BA = I$ for the following matrices (check it by yourself!)

(a)

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}, \; B = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}.$$

(b)

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}, \; B = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 2 \\ 1 & -1 & -1 \end{bmatrix}.$$

(c)

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 & 1 \\ -2 & -3 & 0 & -5 & -1 \\ 1 & 1 & 0 & 2 & 1 \\ -2 & -3 & -1 & -3 & -2 \\ -1 & -3 & -1 & -3 & 1 \end{bmatrix}, \; B = \begin{bmatrix} -3 & 3 & 6 & -1 & -2 \\ 0 & -2 & -5 & -1 & 1 \\ 1 & 2 & 4 & 1 & -1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & -1 & -2 & 0 & 1 \end{bmatrix}.$$

Example 5.1.3. (Matrix without an inverse)

Show that the following matrices are not invertible.

(a)

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

(b)

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

(c)

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}.$$

Proof. (a) Note that $A = \mathcal{O}$, so for any 2×2 matrix B, $AB = \mathcal{O}$ cannot equal I_2, so A is not invertible.

(b) For any 2×2 matrix $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, we have

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}.$$

AB cannot equal I_2, so A is not invertible.

(c) For any 2×2 matrix $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, we have

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+2c & b+2d \\ 3(a+2c) & 3(b+2d) \end{bmatrix}.$$

In order that $AB = I_2$, we need $a + 2c = 1$ and $3(a + 2c) = 0$, which cannot both happen. So AB cannot equal I_2, and A is not invertible.

□

Theorem 5.1.1 (Matrix Inverse is Unique). *Suppose A is invertible. Then the inverse of A is unique.*

Proof. Suppose B and C are both inverses of A, which means $AB = BA = I_n$ and $AC = CA = I_n$. Then

$$CAB = C(AB) = CI_n = C$$
$$CAB = (CA)B = I_nB = B$$

So $B = C$. □

Theorem 5.1.2 (Inverse of Inverse). *Suppose A is invertible. Then A^{-1} is invertible and*

$$(A^{-1})^{-1} = A.$$

Proof. By definition of inverse, we have

$$AA^{-1} = A^{-1}A = I_n.$$

So the matrix A^{-1} is also invertible, and its inverse is A. We write $(A^{-1})^{-1} = A$. $\square$

Theorem 5.1.3 (Inverse of Transpose). *Suppose A is invertible. Then A^t is invertible and*

$$(A^t)^{-1} = (A^{-1})^t.$$

Proof. Note that

$$A^t(A^{-1})^t = (A^{-1}A)^t = I_n^t = I_n,$$

and

$$(A^{-1})^t A^t = (AA^{-1})^t = I_n^t = I_n.$$

So the matrix A^t is invertible, and its inverse is $(A^{-1})^t$. We write $(A^t)^{-1} = (A^{-1})^t$. $\square$

Theorem 5.1.4 (Inverse of Scalar Multiple). *Suppose A is invertible and α is a nonzero scalar. Then αA is invertible and*

$$(\alpha A)^{-1} = \frac{1}{\alpha}A^{-1}.$$

Proof. Note that

$$(\alpha A)\left(\frac{1}{\alpha}A^{-1}\right) = \left(\alpha\frac{1}{\alpha}\right)\left(AA^{-1}\right) = I_n,$$

and

$$\left(\frac{1}{\alpha}A^{-1}\right)(\alpha A) = \left(\frac{1}{\alpha}\alpha\right)\left(A^{-1}A\right) = I_n.$$

So the matrix αA is invertible, and its inverse is $\frac{1}{\alpha}A^{-1}$. We write $(\alpha A)^{-1} = \frac{1}{\alpha}A^{-1}$. $\square$

Theorem 5.1.5 (Inverse of AB). *Suppose A and B are $n \times n$ invertible matrices. Then AB is invertible and*

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Proof. Note that

$$\begin{aligned}(AB)(B^{-1}A^{-1}) &= A(BB^{-1})A^{-1} \\ &= AI_nA^{-1} \\ &= AA^{-1} \\ &= I_n,\end{aligned}$$

and

$$\begin{aligned}(B^{-1}A^{-1})(AB) &= B^{-1}(A^{-1}A)B \\ &= B^{-1}I_nB \\ &= B^{-1}B \\ &= I_n.\end{aligned}$$

So the matrix AB is invertible, and its inverse is $B^{-1}A^{-1}$. We write $(AB)^{-1} = B^{-1}A^{-1}$. □

Remark 5.1.2.

(a) When we dress in the morning, we put on our socks first, followed by our shoes. In the evening we must reverse the order, first remove our shoes, then our socks. Try to connect the conclusion of the above theorem with this everyday example.

(b) $(A+B)^{-1} \neq A^{-1} + B^{-1}$ in general. Any example?

Corollary 5.1.1.

(a) Suppose $A_1, A_2, \cdots, A_k$ are $n \times n$ invertible matrices. Then $A_1A_2\cdots A_k$ is invertible and

$$(A_1A_2\cdots A_k)^{-1} = A_k^{-1}\cdots A_2^{-1}A_1^{-1}.$$

(b) Suppose A is invertible and k is a positive integer. Then A^k is invertible and

$$(A^k)^{-1} = (A^{-1})^k.$$

5.2 Equivalence between Nonsingular and Invertible

In this subsection, we will show that nonsingular and invertible are equivalent.

Theorem 5.2.1 (Elementary Matrices are Invertible). *If E is an elementary matrix, then E is invertible and E^{-1} is an elementary matrix of the same type.*

Proof.

(a) Type I. If E is the elementary matrix formed from I_n by applying $R_i \leftrightarrow R_j$,

$$E = \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 0 & & 1 & & \\ & & & \ddots & & & \\ & & 1 & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix},$$

then E can be transformed back into I_n by the same operation. Therefore, $EE = I_n$ and hence $E^{-1} = E$.

(b) Type II. If E is the elementary matrix formed from I_n by applying αR_i with $\alpha \neq 0$,

$$E = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & \alpha & & \\ & & & \ddots & \\ & & & & 1 \end{bmatrix},$$

then E can be transformed back into I_n by applying $\dfrac{1}{\alpha}R_i$.

$$F = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & \frac{1}{\alpha} & & \\ & & & \ddots & \\ & & & & 1 \end{bmatrix},$$

We can check $EF = FE = I_n$, so $E^{-1} = F$.

(c) Type III. If E is the elementary matrix formed from I_n by applying $\alpha R_i + R_j$,

$$E = \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & \vdots & \ddots & & \\ & & \alpha & \cdots & 1 & \\ & & & & & \ddots \\ & & & & & & 1 \end{bmatrix},$$

then E can be transformed back into I_n by applying $-\alpha R_i + R_j$.

$$F = \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & \vdots & \ddots & & \\ & & -\alpha & \cdots & 1 & \\ & & & & & \ddots \\ & & & & & & 1 \end{bmatrix},$$

We can check $EF = FE = I_n$, so $E^{-1} = F$.

$\square$

Theorem 5.2.2 (Nonsingular and Invertible). *Let A be an $n \times n$ square matrix. Then A is nonsingular if and only if A is invertible.*

Proof.

($\Rightarrow$) Suppose A is nonsingular. Then by the equivalence for nonsingular matrix (Theorem 4.3.4), A is row equivalent to I_n, that is, there exists a sequence of elementary matrices $E_1,\ E_2,\ \cdots,\ E_k$ such that

$$E_k E_{k-1} \cdots E_1 A = I_n.$$

Since each elementary matrix is invertible, multiply the above equation by $E_1^{-1} \cdots E_{k-1}^{-1} E_k^{-1}$ on the left, then

$$A = E_1^{-1} \cdots E_{k-1}^{-1} E_k^{-1}.$$

So A is a product of elementary matrices, and it is invertible with

$$A^{-1} = (E_1^{-1} \cdots E_{k-1}^{-1} E_k^{-1})^{-1} = E_k E_{k-1} \cdots E_1.$$

($\Leftarrow$) Suppose A is invertible. If $\mathbf{x}$ is a solution to the homogeneous system

$$A\mathbf{x} = \mathbf{0},$$

multiply it by A^{-1} on the left to get

$$\mathbf{x} = A^{-1}\mathbf{0} = \mathbf{0}.$$

Hence the homogeneous system $A\mathbf{x} = \mathbf{0}$ has only trivial solution, which implies A is nonsingular.

□

For $\alpha, \beta \in \mathbb{R}$, then $\alpha\beta \neq 0$ if and only if $\alpha \neq 0$ and $\beta \neq 0$. We have a similar result for nonsingular matrix.

Theorem 5.2.3 (Nonsingular Product has Nonsingular Terms). *Suppose A and B are $n \times n$ matrices. The product AB is nonsingular if and only if A and B are both nonsingular.*

Proof.

($\Leftarrow$) Suppose that both A and B are nonsingular.

Method 1: use invertibility. By the equivalence of nonsingular and invertible, A and B are both invertible. So AB is also invertible, and hence nonsingular.

Method 2: use definition. Suppose $\mathbf{x}$ is a solution to $\mathcal{LS}(AB, \mathbf{0})$, i.e.,

$$(AB)\mathbf{x} = \mathbf{0},$$

then $B\mathbf{x}$ is a solution to $\mathcal{LS}(A, \mathbf{0})$. Since A is nonsingular, we get $B\mathbf{x} = \mathbf{0}$. Again, since B is nonsingular, we get $\mathbf{x} = \mathbf{0}$. So the only solution to $\mathcal{LS}(AB, \mathbf{0})$ is trivial solution, and we conclude that AB is nonsingular.

($\Rightarrow$) Suppose that AB is nonsingular.

Part 1: to prove B is nonsingular. Let's use contradiction argument. Suppose B is singular, which means there is a nontrivial solution $\mathbf{z}$ to $\mathcal{LS}(B, \mathbf{0})$. Then

$$(AB)\mathbf{z} = A(B\mathbf{z}) = A\mathbf{0} = \mathbf{0}$$

Therefore $\mathbf{z}$ is also a nontrivial solution to $\mathcal{LS}(AB, \mathbf{0})$. Thus AB is singular. Contradiction arises. So B must be nonsingular.

Part 2: to prove A is nonsingular. By contradiction argument, suppose A is singular, then there is a nontrivial solution $\mathbf{y}$ to $A\mathbf{y} = \mathbf{0}$. Now consider the linear system

$$B\mathbf{w} = \mathbf{y}.$$

Since B is nonsingular, the system has a unique solution $\mathbf{w}$. Note that $\mathbf{w}$ is not $\mathbf{0}$, otherwise, $\mathbf{y} = B\mathbf{w} = \mathbf{0}$, contrary to $\mathbf{y}$ being nontrivial. So $\mathbf{w} \neq \mathbf{0}$ satisfies

$$(AB)\mathbf{w} = A(B\mathbf{w}) = A\mathbf{y} = \mathbf{0}.$$

Then AB is singular, contradicting the assumption. So A must be nonsingular.

□

Remark 5.2.1. In the future we will find another description of nonsingular/invertible matrix, using determinant. Then the above theorem will become very obvious if explained by determinant.

Theorem 5.2.4 (One-Sided Inverse is Sufficient). *Suppose A and B are $n \times n$ matrices such that $AB = I_n$. Then $BA = I_n$, and A, B are inverses of each other.*

Proof. Since $AB = I_n$ and I_n is nonsingular, so A and B are both nonsingular and invertible. Multiplying $AB = I_n$ by A^{-1} on the left, we get $B = A^{-1}$. As a result, $BA = I_n$.

□

Remark 5.2.2.

If $AB = I_n$, B is called a **right-inverse** of A.

If $BA = I_n$, B is called a **left-inverse** of A.

The above theorem tells us that right-inverse, left-inverse, and inverse, are the same things.

Theorem 5.2.5 (Nonsingular Matrix Equivalences). *Let A be an $n \times n$ square matrix. The following 7 statements are equivalent.*

1. *A is nonsingular.*
2. *A is row equivalent to I_n.*
3. *The null space of A contains only the zero vector, $\mathcal{N}(A) = \{\mathbf{0}\}$.*
4. *The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every choice of $\mathbf{b}$.*
5. *A is invertible.*
6. *There exists B such that $AB = I_n$.*
7. *There exists B such that $BA = I_n$.*

In the case that A is invertible, the unique solution to $A\mathbf{x} = \mathbf{b}$ is given by

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

5.3 Computing the Inverse of a Matrix

Given an $n \times n$ invertible matrix A, how would we find its inverse, that is, the matrix B such that

$$AB = I_n?$$

Theorem 5.3.1 (Compute Inverse). *Let A be an $n \times n$ matrix. Suppose we can perform elementary row operations to the augmented matrix $[A|I_n]$ to obtain $[I_n|B]$, then A is invertible and $A^{-1} = B$.*

Proof. Let $E_1, E_2, \cdots, E_k$ be the elementary matrices corresponding to the operations such that

$$E_k \cdots E_2 E_1 [A|I_n] = [I_n|B].$$

Then

$$I_n = E_k \cdots E_2 E_1 A,$$

$$B = E_k \cdots E_2 E_1 I_n = E_k \cdots E_2 E_1.$$

Hence A is invertible, and

$$A^{-1} = E_k \cdots E_2 E_1 = B.$$

□

Recall that A is invertible if and only if A is row equivalent to I_n, so we have the following "algorithm" to determine the invertibility of A, and to find its inverse if it is invertible.

$$[\,A \,|\, I_n\,] \xrightarrow[\text{row operations}]{\text{elementary}} [\,\mathrm{RREF}(A) \,|\, B\,]$$

(i) If $\mathrm{RREF}(A) = I_n$, then A is invertible and $A^{-1} = B$.

(ii) If $\mathrm{RREF}(A) \neq I_n$, then A is not invertible.

Example 5.3.1.

(a) $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$.

$$[A|I_2] = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right] \xrightarrow{-R_1+R_2} \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{array}\right]$$

$$\xrightarrow{-R_2+R_1} \left[\begin{array}{cc|cc} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 1 \end{array}\right]$$

So A is invertible and

$$A^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}.$$

(b) $B = \begin{bmatrix} -1 & -1 \\ 2 & 2 \end{bmatrix}$.

$$[B|I_2] = \left[\begin{array}{cc|cc} -1 & -1 & 1 & 0 \\ 2 & 2 & 0 & 1 \end{array}\right] \xrightarrow{2R_1+R_2} \left[\begin{array}{cc|cc} -1 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{array}\right]$$

B is not row equivalent to I_2, so B is not invertible.

(c) $C = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$.

$$[C|I_3] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{-R_1+R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & -2 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & -1 & -2 & -1 & 1 & 0 \end{array}\right] \xrightarrow{R_2+R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & -1 & -1 & 1 & 1 \end{array}\right]$$

$$\xrightarrow{-R_2+R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & -1 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & -1 & -1 & 1 & 1 \end{array}\right] \xrightarrow[-R_3]{R_3+R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & -1 & -1 \end{array}\right]$$

So C is invertible and

$$C^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 2 \\ 1 & -1 & -1 \end{bmatrix}.$$

6 Vector Space

In Chapter 3, we defined column vectors and row vectors (Definition 3.2.1), and two operations: addition (Definition 3.1.2), scalar multiplication (Definition 3.1.3). These two operations are used in many diverse contexts in mathematics and usually obey the same set of algebraic rules (Theorem 3.1.1). Thus, a general theory of mathematical systems involving addition and scalar multiplication will be applicable to many areas in mathematics. Mathematical systems with this structure are called vector spaces or linear spaces. In this chapter, the definition of a vector space is given and some of the general theory of vector spaces is developed.

6.1 Definition and examples of vector spaces

In this section, we present the formal definition of a vector space. Before doing this, however, it is instructive to look at a number of examples. We begin with the Euclidean vector spaces $\mathbb{R}^n$.

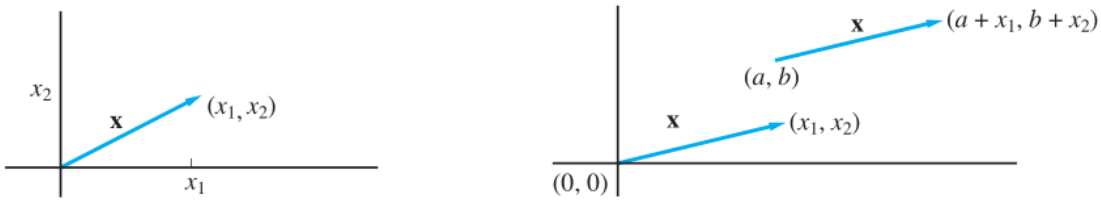
I. Euclidean vector space $\mathbb{R}^n$

The Euclidean vector space $\mathbb{R}^n$ is the set of all column vectors (or row vectors, depending on the context) with n entries of real numbers. A vector $\mathbf{x} \in \mathbb{R}^n$ can be denoted by:

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ or } \begin{bmatrix} x_1, x_2, \cdots, x_n \end{bmatrix}.$$

For simplicity, let us consider first $\mathbb{R}^2$. Nonzero vectors in $\mathbb{R}^2$ can be represented geometrically by directed line segments. This geometric representation will help us to visualize how addition and scalar multiplication work in $\mathbb{R}^2$. Given a nonzero vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, we can associate it with the directed line segment in the plane from the point $(0, 0)$ to the point (x_1, x_2). If we equate line segments that have the same length

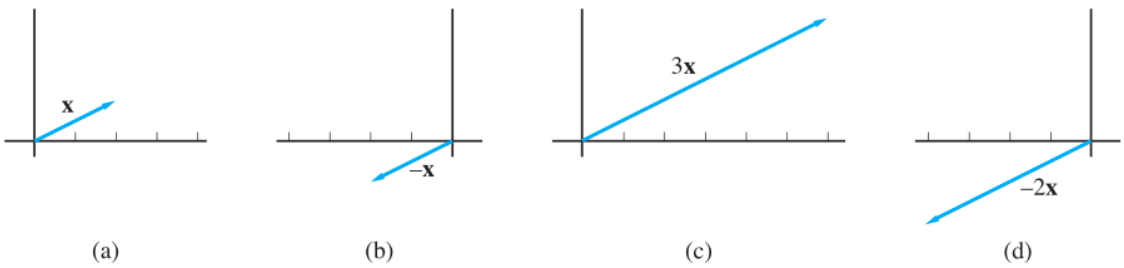
and direction, then $\mathbf{x}$ can be represented by any line segment from the point (a, b) to the point $(a + x_1, b + x_2)$.



For each vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, the length of $\mathbf{x}$ is the length of the line segment from $(0, 0)$ to (x_1, x_2), that is, $\sqrt{x_1^2 + x_2^2}$. If α is a scalar (i.e., real number), $\alpha\mathbf{x}$ is defined by

$$\alpha \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \end{bmatrix}.$$

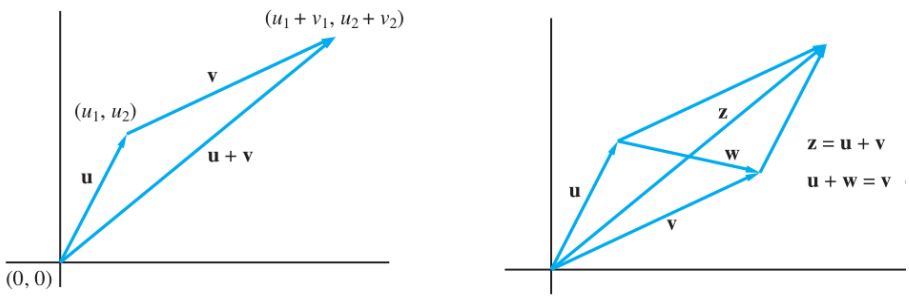
The length of $\alpha\mathbf{x}$ is $|\alpha|$ times that of $\mathbf{x}$, and the direction of $\alpha\mathbf{x}$ is the same as $\mathbf{x}$ if $\alpha > 0$, and opposite if $\alpha < 0$. See the following figure for $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.



For two vectors $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$, the sum is defined by

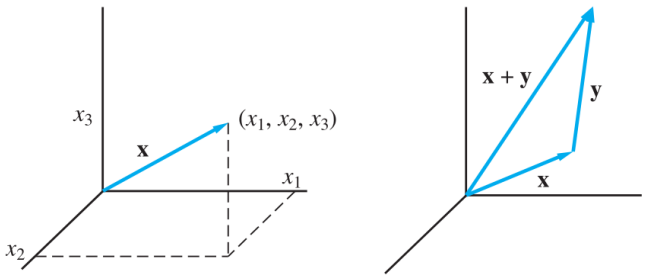
$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}.$$

(i) Head-to-tail method: place the tail (initial point) of $\mathbf{v}$ at the head (terminal point) of $\mathbf{u}$, then $\mathbf{u} + \mathbf{v}$ is represented by vector from the tail of $\mathbf{u}$ to the head of $\mathbf{v}$.



(ii) Parallelogram law: place the tails of $\mathbf{u}$ and $\mathbf{v}$ together to form a parallelogram, then the diagonals of the parallelogram will represent $\mathbf{u}+\mathbf{v}$ and $\mathbf{v}-\mathbf{u}$.

In a similar manner, vectors in $\mathbb{R}^3$ can be represented by directed line segments in 3−space.



In general, for any vectors $\mathbf{u}$, $\mathbf{v} \in \mathbb{R}^n$, and any scalar $\alpha \in \mathbb{R}$, the scalar multiplication and addition are defined by

$$\alpha\mathbf{u} = \begin{bmatrix} \alpha u_1 \\ \alpha u_2 \\ \vdots \\ \alpha u_n \end{bmatrix}, \quad \mathbf{u}+\mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}.$$

II. Matrix space M_{mn} or $\mathbb{R}^{m\times n}$

The set of all $m \times n$ matrices with real entries is denoted by $\mathbb{R}^{m\times n}$ or M_{mn}. The scalar multiplication and addition on $\mathbb{R}^{m\times n}$ obey certain algebraic rules (Theorem 3.1.1), which will be summarized as axioms to define an abstract vector space.

III. Vector space

For math major only. Non-math major can skip this part.

Definition 6.1.1 (Vector Space). A set V together with operations of addition and scalar multiplication is said to form a **vector space** (or **linear space**) if the following axioms are satisfied.

1. *Closure under addition*

   If $\mathbf{u}$, $\mathbf{v} \in V$, then $\mathbf{u} + \mathbf{v} \in V$.

2. *Closure under scalar multiplication*

   If $\alpha \in \mathbb{R}$ and $\mathbf{u} \in V$, then $\alpha\mathbf{u} \in V$.

3. *Commutativity*

   If $\mathbf{u}$, $\mathbf{v} \in V$, then $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.

4. *Additive Associativity*

   If $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w} \in V$, then $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$.

5. *Zero Vector*

   There is a vector, $\mathbf{0}$, called the **zero vector**, such that $\mathbf{u}+\mathbf{0} = \mathbf{u}$ for all $\mathbf{u} \in V$.

6. *Additive Inverse*

   If $\mathbf{u} \in V$, then there exists a vector $-\mathbf{u} \in V$ so that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.

7. *Scalar Multiplication Associativity*

   If α, $\beta \in \mathbb{R}$ and $\mathbf{u} \in V$, then $\alpha(\beta\mathbf{u}) = (\alpha\beta)\mathbf{u}$.

8. *Distributivity across Vector Addition*

   If $\alpha \in \mathbb{R}$ and $\mathbf{u}$, $\mathbf{v} \in V$, then $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$.

9. *Distributivity across Scalar Addition*

   If α, $\beta \in \mathbb{R}$ and $\mathbf{u} \in V$, then $(\alpha + \beta)\mathbf{u} = \alpha\mathbf{u} + \beta\mathbf{u}$.

10. *Scalar one*

    If $\mathbf{u} \in V$, then $1\mathbf{u} = \mathbf{u}$.

The elements in V are called **vectors**, and are usually denoted by boldface letters $\mathbf{u}$ or $\vec{u}$. The **scalars** refer to real numbers in this course (although they can be used for complex numbers in a general case).

Example 6.1.1. Euclidean space $\mathbb{R}^n$ and matrix space M_{mn}(or $\mathbb{R}^{m\times n}$) with the usual addition and scalar multiplication are vector spaces.

Example 6.1.2. The vector space of functions

Let F be the set of all functions from $\mathbb{R}$ to $\mathbb{R}$, i.e.,

$$F = \{f \mid f \text{ is a function from } \mathbb{R} \text{ to } \mathbb{R}\}.$$

We say two functions f and g are equal, denoted by $f = g$, if and only if $f(x) = g(x)$ for all $x \in \mathbb{R}$.

- Vector Addition: $f + g$ is the function defined by

$$(f + g)(x) = f(x) + g(x), \quad \text{for all } x \in \mathbb{R}.$$

- Scalar Multiplication: αf is the function defined by

$$(\alpha f)(x) = \alpha f(x), \quad \text{for all } x \in \mathbb{R}.$$

The zero vector in F is the constant zero function

$$f(x) = 0, \quad \text{for all } x \in \mathbb{R}.$$

It is easy to show that F with the above addition and scalar multiplication satisfies all the axioms in Definition 6.1.1, so F is a vector space. For example, the commutativity $f + g = g + f$ is true because

$$(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x), \text{ for all } x \in \mathbb{R}.$$

The additive inverse $-f$ is just the following function

$$(-f)(x) = -f(x), \text{ for all } x \in \mathbb{R}.$$

Example 6.1.3. The vector space of polynomials, P_n

Let P_n be the set of all polynomials about the variable x of degree less than or equal to n, i.e.,

$$P_n = \{a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \mid a_i \in \mathbb{R} \text{ for all } 0 \le i \le n\}.$$

We say two polynomials $p = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ and $q = b_0 + b_1x + b_2x^2 + \cdots + b_nx^n$ are equal, denoted by $p = q$, if and only if $a_i = b_i$ for all $0 \leq i \leq n$.

- Vector Addition: $p + q$ is the polynomial defined by

$$\begin{aligned}&(a_0 + a_1x + a_2x^2 + \cdots + a_nx^n) + (b_0 + b_1x + b_2x^2 + \cdots + b_nx^n)\\ =&(a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \cdots + (a_n + b_n)x^n\end{aligned}$$

- Scalar Multiplication: αp is the polynomial defined by

$$\alpha(a_0 + a_1x + a_2x^2 + \cdots + a_nx^n) = (\alpha a_0) + (\alpha a_1)x + (\alpha a_2)x^2 + \cdots + (\alpha a_n)x^n$$

The zero vector in P_n is the constant zero

$$\mathbf{0} = 0 + 0x + 0x^2 + \cdots + 0x^n.$$

P_n with the above addition and scalar multiplication is a vector space.

Remark 6.1.1. Notice that to construct a vector space, I am free to choose my set and define my operations any way I please, as long as they satisfy all the axioms for vector space. Just for fun, check the following example is a vector space, and it is different from the standard space $\mathbb{R}^2$.

Example 6.1.4. One crazy vector space

Let $C = \{(x_1,\, x_2) \,|\, x_1,\, x_2 \in \mathbb{R}\}$.

- Vector Addition: $(x_1,\, x_2) + (y_1,\, y_2) = (x_1 + y_1 + 1,\, x_2 + y_2 + 1)$.
- Scalar Multiplication: $\alpha(x_1,\, x_2) = (\alpha x_1 + \alpha - 1,\, \alpha x_2 + \alpha - 1)$.

1. *Closure under addition* is obvious.
2. *Closure under scalar multiplication* is obvious.

3. *Commutativity*

$$\begin{aligned}\mathbf{u}+\mathbf{v} &= (x_1,\, x_2)+(y_1,\, y_2) = (x_1+y_1+1,\, x_2+y_2+1)\\ &= (y_1+x_1+1,\, y_2+x_2+1) = (y_1,\, y_2)+(x_1,\, x_2)\\ &= \mathbf{v}+\mathbf{u}\end{aligned}$$

4. *Additive Associativity*

$$\begin{aligned}\mathbf{u}+(\mathbf{v}+\mathbf{w}) &= (x_1,\, x_2)+((y_1,\, y_2)+(z_1,\, z_2))\\ &= (x_1,\, x_2)+(y_1+z_1+1,\, y_2+z_2+1)\\ &= (x_1+(y_1+z_1+1)+1,\, x_2+(y_2+z_2+1)+1)\\ &= (x_1+y_1+z_1+2,\, x_2+y_2+z_2+2)\\ &= ((x_1+y_1+1)+z_1+1,\, (x_2+y_2+1)+z_2+1)\\ &= (x_1+y_1+1,\, x_2+y_2+1)+(z_1,\, z_2)\\ &= ((x_1,\, x_2)+(y_1,\, y_2))+(z_1,\, z_2)\\ &= (\mathbf{u}+\mathbf{v})+\mathbf{w}\end{aligned}$$

5. *Zero Vector*

   The zero vector is $\mathbf{0} = (-1,\, -1)$ (**not** $(0, 0)$!!!)

$$\mathbf{u}+\mathbf{0} = (x_1,\, x_2)+(-1,\, -1) = (x_1+(-1)+1,\, x_2+(-1)+1) = (x_1,\, x_2) = \mathbf{u}$$

6. *Additive Inverse*

   For each vector $\mathbf{u} = (x_1,\, x_2)$, the additive inverse is $-\mathbf{u} = (-x_1-2,\, -x_2-2)$ because

$$\begin{aligned}\mathbf{u}+(-\mathbf{u}) &= (x_1,\, x_2)+(-x_1-2,\, -x_2-2)\\ &= (x_1+(-x_1-2)+1,\, -x_2+(x_2-2)+1) = (-1,\, -1) = \mathbf{0}\end{aligned}$$

7. *Scalar Multiplication Associativity*

$$\begin{aligned}
\alpha(\beta\mathbf{u}) &= \alpha(\beta(x_1,\, x_2)) \\
&= \alpha(\beta x_1 + \beta - 1,\, \beta x_2 + \beta - 1) \\
&= (\alpha(\beta x_1 + \beta - 1) + \alpha - 1,\, \alpha(\beta x_2 + \beta - 1) + \alpha - 1) \\
&= ((\alpha\beta x_1 + \alpha\beta - \alpha) + \alpha - 1,\, (\alpha\beta x_2 + \alpha\beta - \alpha) + \alpha - 1) \\
&= (\alpha\beta x_1 + \alpha\beta - 1,\, \alpha\beta x_2 + \alpha\beta - 1) \\
&= (\alpha\beta)(x_1,\, x_2) \\
&= (\alpha\beta)\mathbf{u}
\end{aligned}$$

8. *Distributivity across Vector Addition*

$$\begin{aligned}
&\alpha(\mathbf{u} + \mathbf{v}) \\
&\quad = \alpha\left((x_1,\, x_2) + (y_1,\, y_2)\right) \\
&\quad = \alpha(x_1 + y_1 + 1,\, x_2 + y_2 + 1) \\
&\quad = (\alpha(x_1 + y_1 + 1) + \alpha - 1,\, \alpha(x_2 + y_2 + 1) + \alpha - 1) \\
&\quad = (\alpha x_1 + \alpha y_1 + \alpha + \alpha - 1,\, \alpha x_2 + \alpha y_2 + \alpha + \alpha - 1) \\
&\quad = (\alpha x_1 + \alpha - 1 + \alpha y_1 + \alpha - 1 + 1,\, \alpha x_2 + \alpha - 1 + \alpha y_2 + \alpha - 1 + 1) \\
&\quad = ((\alpha x_1 + \alpha - 1) + (\alpha y_1 + \alpha - 1) + 1,\, (\alpha x_2 + \alpha - 1) + (\alpha y_2 + \alpha - 1) + 1) \\
&\quad = (\alpha x_1 + \alpha - 1,\, \alpha x_2 + \alpha - 1) + (\alpha y_1 + \alpha - 1,\, \alpha y_2 + \alpha - 1) \\
&\quad = \alpha(x_1,\, x_2) + \alpha(y_1,\, y_2) \\
&\quad = \alpha\mathbf{u} + \alpha\mathbf{v}
\end{aligned}$$

9. *Distributivity across Scalar Addition*

$$
\begin{aligned}
&(\alpha+\beta)\mathbf{u} \\
&\quad = (\alpha+\beta)(x_1,\, x_2) \\
&\quad = ((\alpha+\beta)x_1 + (\alpha+\beta) - 1,\, (\alpha+\beta)x_2 + (\alpha+\beta) - 1) \\
&\quad = (\alpha x_1 + \beta x_1 + \alpha + \beta - 1,\, \alpha x_2 + \beta x_2 + \alpha + \beta - 1) \\
&\quad = (\alpha x_1 + \alpha - 1 + \beta x_1 + \beta - 1 + 1,\, \alpha x_2 + \alpha - 1 + \beta x_2 + \beta - 1 + 1) \\
&\quad = ((\alpha x_1 + \alpha - 1) + (\beta x_1 + \beta - 1) + 1,\, (\alpha x_2 + \alpha - 1) + (\beta x_2 + \beta - 1) + 1) \\
&\quad = (\alpha x_1 + \alpha - 1,\, \alpha x_2 + \alpha - 1) + (\beta x_1 + \beta - 1,\, \beta x_2 + \beta - 1) \\
&\quad = \alpha(x_1,\, x_2) + \beta(x_1,\, x_2) \\
&\quad = \alpha\mathbf{u} + \beta\mathbf{u}
\end{aligned}
$$

10. *Scalar one*

$$1\mathbf{u} = 1(x_1,\, x_2) = (x_1 + 1 - 1,\, x_2 + 1 - 1) = (x_1,\, x_2) = \mathbf{u}$$

That is it, C is a vector space, as crazy as that may seem.

We close this section with several theorems that states the fundamental properties of vector spaces.

Theorem 6.1.1 (Cancellation Law for Vector Addition). *If* $\mathbf{v}$, $\mathbf{u}$ *and* $\mathbf{w}$ *are vectors in a vector space* V *such that*

$$\mathbf{v} + \mathbf{w} = \mathbf{u} + \mathbf{w},$$

then $\mathbf{v} = \mathbf{u}$.

Proof. By Property 6, there exists a vector $-\mathbf{w}$ such that $\mathbf{w} + (-\mathbf{w}) = \mathbf{0}$. Thus

$$(\mathbf{v} + \mathbf{w}) + (-\mathbf{w}) = (\mathbf{u} + \mathbf{w}) + (-\mathbf{w})$$

$$\mathbf{v} + (\mathbf{w} + (-\mathbf{w})) = \mathbf{u} + (\mathbf{w} + (-\mathbf{w})) \qquad \text{Property 4}$$

$$\mathbf{v} + \mathbf{0} = \mathbf{u} + \mathbf{0} \qquad \text{Property 6}$$

$$\mathbf{v} = \mathbf{u} \qquad \text{Property 5.}$$

□

Theorem 6.1.2 (Uniqueness of Zero Vector). *Let V be a vector space. The vector $\mathbf{0}$ described in Property 5 is unique.*

Proof. Suppose both $\mathbf{0}_1$ and $\mathbf{0}_2$ satisfy the property 5. Let $\mathbf{w}$ be an element in V.

$$\mathbf{0}_1 + \mathbf{w} = \mathbf{w} = \mathbf{0}_2 + \mathbf{w} \qquad \text{Property 5}$$

$$\mathbf{0}_1 = \mathbf{0}_2 \qquad \text{by the previous theorem}$$

□

Theorem 6.1.3 (Uniqueness of Additive Inverse). *Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors in a vector space V. If both $\mathbf{v}$ and $\mathbf{u}$ satisfies*

$$\mathbf{v} + \mathbf{w} = \mathbf{0},$$

$$\mathbf{u} + \mathbf{w} = \mathbf{0},$$

i.e., both $\mathbf{u}$ and $\mathbf{v}$ are additive inverse of $\mathbf{w}$ in Property 6, then

$$\mathbf{v} = \mathbf{u}.$$

This shows that the additive inverse is unique.

Proof.

$$\mathbf{v} + \mathbf{w} = \mathbf{0} = \mathbf{u} + \mathbf{w}.$$

By Theorem 6.1.1,

$$\mathbf{v} = \mathbf{u}.$$

□

Theorem 6.1.4. *Let V be a vector space, $\alpha \in \mathbb{R}$ and $\mathbf{v} \in V$. Then we have*

1. $0\mathbf{v} = \mathbf{0}$ *(The first 0 is the scalar 0, the second $\mathbf{0}$ is the zero vector).*

2. $\alpha\mathbf{0} = \mathbf{0}$.

3. $(-\alpha)\mathbf{v} = -(\alpha\mathbf{v}) = \alpha(-\mathbf{v})$.

Proof.

1.

$$\begin{aligned} 0\mathbf{v} + \mathbf{v} =& 0\mathbf{v} + 1\mathbf{v} && \text{Property 10} \\ =& (0+1)\mathbf{v} && \text{Property 9} \\ =& \mathbf{v} \\ =& \mathbf{0} + \mathbf{v} && \text{Property 5} \end{aligned}$$

Use Theorem 6.1.1 and cancel $\mathbf{v}$, then we get $0\mathbf{v} = \mathbf{0}$.

2.

$$\begin{aligned} \alpha\mathbf{0} &= \alpha(0\mathbf{0}) && \text{Above part 1} \\ &= (\alpha 0)\mathbf{0} && \text{Property 7} \\ &= 0\mathbf{0} \\ &= \mathbf{0} \end{aligned}$$

3. First,

$$\begin{aligned} \alpha\mathbf{v} + (-\alpha)\mathbf{v} &= (\alpha + (-\alpha))\mathbf{v} && \text{Property 9.} \\ &= 0\mathbf{v} \\ &= \mathbf{0} \end{aligned}$$

By definition and uniqueness of additive inverse, we have $(-\alpha)\mathbf{v} = -(\alpha\mathbf{v})$.

Next,

$$\begin{aligned}
\alpha\mathbf{v} + \alpha(-\mathbf{v}) &= \alpha(\mathbf{v} + (-\mathbf{v})) \qquad \text{Property 8.} \\
&= \alpha\mathbf{0} \qquad \text{Property 6} \\
&= \mathbf{0} \qquad \text{Above part 2}
\end{aligned}$$

By definition and uniqueness of additive inverse, we have

$$\alpha(-\mathbf{v}) = -(\alpha\mathbf{v}).$$

$\square$

6.2 Subspaces

Question: given a vector space V, is it possible to form another vector space by taking a nonempty subset S of V and using the operations of V?

First, in order to be a vector space, the subset S must be closed under the operations. That is, the sum of two elements of S must always be an element of S, and the product of a scalar and an element of S must always be an element of S.

Second, the other axioms $3-10$ will be automatically satisfied on S, because the operations are "inherited" from V, and their properties will remain valid on S.

Definition 6.2.1 (Subspace). Let V be vector space. A subset W of V is said to be a **subspace** of V if the following conditions are satisfied:

1. W is nonempty;
2. $\mathbf{u}+\mathbf{v} \in W$ for all $\mathbf{u}, \mathbf{v} \in W$;
3. $\alpha\mathbf{u} \in W$ for all $\alpha \in \mathbb{R}$, $\mathbf{u} \in W$.

Example 6.2.1. Let $V = \mathbb{R}^2$, $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \,\middle|\, y = 2x \right\}$. Show that W is a subspace of V.

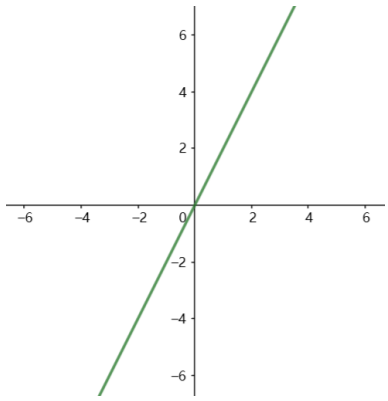
Proof.

1. $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \in W$ so W is nonempty.
2. For all $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in W$, we have $u_2 = 2u_1$ and $v_2 = 2v_1$, so $u_2 + v_2 = 2(u_1 + v_1)$, which implies $\mathbf{u}+\mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix} \in W$.
3. For all $\alpha \in \mathbb{R}, \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in W$, we have $u_2 = 2u_1$, so $\alpha u_2 = 2(\alpha u_1)$, which

implies $\alpha\mathbf{u} = \begin{bmatrix} \alpha u_1 \\ \alpha u_2 \end{bmatrix} \in W$.

Hence, W is a subspace of V. $\square$

Remark 6.2.1. In the above example, W is a line in $\mathbb{R}^2$ passing through the origin. Similarly we can show any line in $\mathbb{R}^2$ passing through the origin is a subspace of $\mathbb{R}^2$.



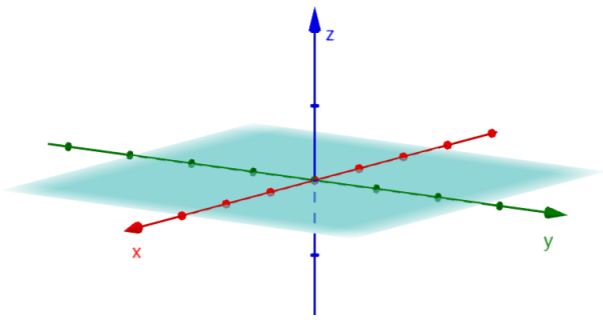
Example 6.2.2. Let $V = \mathbb{R}^3$, $W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \,\middle|\, z = 0 \right\}$. Show that W is a subspace of V.

Proof.

1. $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \in W$ so W is nonempty.
2. For all $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ 0 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ 0 \end{bmatrix} \in W$, we have $\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ 0 \end{bmatrix} \in W$.
3. For all $\alpha \in \mathbb{R}, \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ 0 \end{bmatrix} \in W$, we have $\alpha\mathbf{u} = \begin{bmatrix} \alpha u_1 \\ \alpha u_2 \\ 0 \end{bmatrix} \in W$.

Hence, W is a subspace of V. $\square$

Remark 6.2.2. In the above example, W is the special $xy-$plane in $\mathbb{R}^3$. It will be shown later any plane in $\mathbb{R}^3$ passing through the origin is a subspace of $\mathbb{R}^3$.



Proposition 6.2.1. Let W be a subspace of V. Then $\mathbf{0} \in W$.

Proof. By Definition 6.2.1 condition 1, W is nonempty. Pick some $\mathbf{w} \in W$. Then by Definition 6.2.1 condition 3, $0\mathbf{w} = \mathbf{0} \in W$.

□

Remark 6.2.3. Any subspace must contain $\mathbf{0}$. If a subset does not contain $\mathbf{0}$, it cannot be a subspace.

Example 6.2.3. Let V be any vector space.

$W = \{\mathbf{0}\}$ is the minimal subspace of V.

$W = V$ is the maximal subspace of V.

Example 6.2.4. $V = \mathbb{R}^3$, $W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \,\middle|\, x + 2y + 3z = 0 \right\}$.

Obviously $\mathbf{0} \in W$, so W is nonempty.

For all $\alpha \in \mathbb{R}$, $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in W$. Then

$$u_1 + 2u_2 + 3u_3 = 0,$$

$$v_1 + 2v_2 + 3v_3 = 0.$$

So

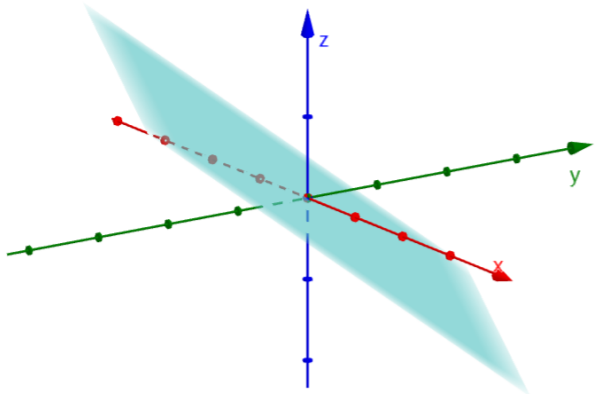
$$(u_1 + v_1) + 2(u_2 + v_2) + 3(u_3 + v_3) = 0,$$

$$\alpha u_1 + 2\alpha u_2 + 3\alpha u_3 = 0$$

Hence

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{bmatrix} \in W, \quad \alpha\mathbf{u} = \begin{bmatrix} \alpha u_1 \\ \alpha u_2 \\ \alpha u_3 \end{bmatrix} \in W.$$

Thus W is a subspace of V.



Theorem 6.2.1. *Let A be an $m \times n$ matrix, then $W = \mathcal{N}(A)$ is a subspace of $\mathbb{R}^n$.*

Proof. Because $\mathbf{0} \in \mathcal{N}(A)$, so $\mathcal{N}(A)$ is nonempty.

For all $\alpha \in \mathbb{R}$, $\mathbf{u}, \mathbf{v} \in \mathcal{N}(A)$. Then

$$A\mathbf{u} = \mathbf{0}, \ A\mathbf{v} = \mathbf{0}.$$

Then

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0},$$

$$A(\alpha\mathbf{u}) = \alpha A\mathbf{u} = \alpha\mathbf{0} = \mathbf{0}.$$

So

$$\mathbf{u}+\mathbf{v} \in \mathcal{N}(A)\,,\ \alpha\mathbf{u} \in \mathcal{N}(A)\,.$$

Thus $W = \mathcal{N}(A)$ is a subspace, and that is why we call it a null "space". $\square$

Remark 6.2.4.

Example 6.2.1, $W = \mathcal{N}(A)$ with $A = \begin{bmatrix} -2 & 1 \end{bmatrix}$.

Example 6.2.2, $W = \mathcal{N}(A)$ with $A = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$.

Example 6.2.4, $W = \mathcal{N}(A)$ with $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$.

Example 6.2.5. for math major only.

Let S_n be the set of symmetric matrices of M_{nn}. Then S_n is a subspace of M_{nn}.

Because $\mathcal{O} \in W$, so W is nonempty.

For all $\alpha \in \mathbb{R}$, $A, B \in W$. Then $A^t = A$, $B^t = B$.

$$(A+B)^t = A^t + B^t = A + B,$$

$$(\alpha A)^t = \alpha A^t = \alpha A.$$

Thus $A + B \in S_n,\ \alpha A \in S_n$. Hence S_n is a subspace.

Example 6.2.6. for math major only.

Let

$$E = \{f(x) \in P_n \mid f(1) = 0\}.$$

Then E is a subspace of P_n.

Because $\mathbf{0} \in E$, so E is nonempty.

For all $\alpha \in \mathbb{R}$, $f(x), g(x) \in E$. Then $f(1) = g(1) = 0$.

$$f(1) + g(1) = 0,\ \alpha f(1) = 0.$$

So $f + g \in E,\ \alpha f \in E$. Hence E is a subspace.

Example 6.2.7. for math major only.

Let

$$E = \{f(x) \in P_n \,|\, f(x) = f(-x)\}.$$

Then E is a subspace of P_n.

Because $\mathbf{0} \in E$, so E is nonempty.

For all $\alpha \in \mathbb{R}$, $f(x), g(x) \in E$. Then $f(x) = f(-x)$, $g(x) = g(-x)$.

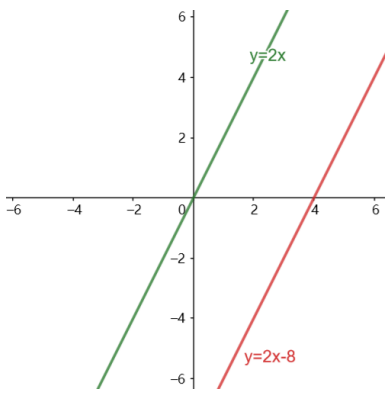
$$(f+g)(-x) = f(-x) + g(-x) = f(x) + g(x) = (f+g)(x),$$

$$(\alpha f)(-x) = \alpha f(-x) = \alpha f(x) = (\alpha f)(x),$$

So $f + g \in E$, $\alpha f \in E$. Hence E is a subspace.

Next we give some non examples. To show that W is **not** a subspace of V, it suffices to show that it violates one condition in Definition 6.2.1. Usually we start by checking if $\mathbf{0} \in W$ (see Proposition 6.2.1).

Example 6.2.8. Let $V = \mathbb{R}^2$, $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \,\middle|\, y = 2x - 8 \right\}$.



Method 1: Obviously $\mathbf{0}$ is not in W. So by proposition 6.2.1, W is not a subspace.

Method 2: We can find an explicit example violating condition 2.

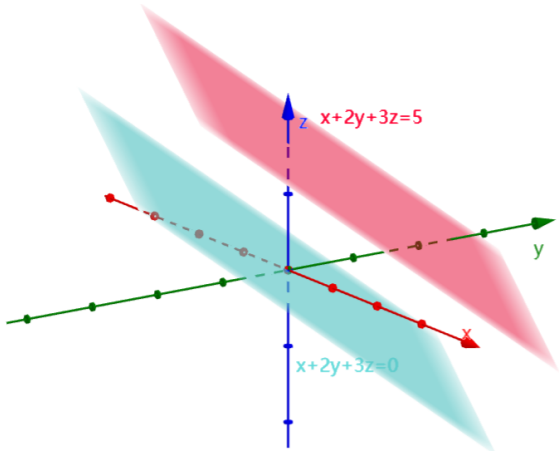
$\mathbf{u} = \begin{bmatrix} 0 \\ -8 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 4 \\ 0 \end{bmatrix} \in W$, but $\mathbf{u} + \mathbf{v} = \begin{bmatrix} 4 \\ -8 \end{bmatrix} \notin W$. Hence W is not a subspace.

Method 3: We can also find an explicit example violating condition 3. $\mathbf{u} = \begin{bmatrix} 0 \\ -8 \end{bmatrix} \in W$, but $2\mathbf{u} = \begin{bmatrix} 0 \\ -16 \end{bmatrix} \notin W$. Hence W is not a subspace.

Example 6.2.9. $V = \mathbb{R}^3$, $W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \,\middle|\, x + 2y + 3z = 5 \right\}$.

Obviously $\mathbf{0} \notin W$, so W is not a subspace.

Can you find some examples violating condition 2 or 3?



Remark 6.2.5.

(i) A line in $\mathbb{R}^2$ represented by $ax + by = c$ is a subspace if and only if $c = 0$.

(ii) A plane in $\mathbb{R}^3$ represented by $ax + by + cz = d$ is a subspace if and only if $d = 0$.

Example 6.2.10. $V = \mathbb{R}^n$, $W = \{\mathbf{v} \in V \mid \sum_{i=1}^{n} [\mathbf{v}]_i = 1\}$. $[\mathbf{v}]_i$ means the ith entry of $\mathbf{v}$.

Method 1: Obviously $\mathbf{0}$ is not in W. So by Proposition 6.2.1, W is not a subspace.

Method 2: We can find an explicit example violating condition 2. Let

$$\mathbf{v} = \mathbf{w} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Then both $\mathbf{v}$ and $\mathbf{w}$ are in W.

$$\mathbf{v} + \mathbf{w} = \begin{bmatrix} 2 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \notin W.$$

Therefore W is not a subspace.

Method 3: Can you find an explicit example violating condition 3?

Example 6.2.11. $V = \mathbb{R}^n$, $W = \{\mathbf{v} \in V \mid [\mathbf{v}]_1 \geq 0\}$.

Let

$$\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in W.$$

Then $[(-1)\mathbf{v}]_1 = -1 < 0$. So $(-1)\mathbf{v} \notin W$. Thus W violates condition 3 and hence not a subspace.

Example 6.2.12. $V = \mathbb{R}^2$, $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in V \mid xy \geq 0 \right\}$.

Let $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \in W$. $\mathbf{v} + \mathbf{w} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Because $1 \times (-1) = -1 < 0$. So $\mathbf{v} + \mathbf{w} \notin W$. Thus W violates condition 2 and hence not a subspace.

In fact, we can show that W satisfies condition 3 but fails condition 2.

Example 6.2.13. for math major only.

Let $V = P_n$. Let G be the set of polynomial with degree exactly equal to n.

Let $f(x) = x^n$, $g(x) = -x^n + 1$. Both f and g have degree exactly equal to n. But

$$f(x) + g(x) = 1$$

is a polynomial with degree 0. So $f + g$ is not in G. Thus W violates condition 2 and hence not a subspace.

Example 6.2.14. Determine if W is a subspace of V.

(a) $V = \mathbb{R}^2$, $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \,|\, y = x^2 \right\}$.

No. Check $\begin{bmatrix} 1 \\ 1 \end{bmatrix} \in W$, but $3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \notin W$.

(b) $V = \mathbb{R}^2$, $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \,|\, x^2 + y^2 = 1 \right\}$.

No. Check $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \notin W$.

(c) $V = \mathbb{R}^3$, $W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \,|\, z = xy \right\}$.

No. Check $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \in W$, but $3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \notin W$.

(d) $V = \mathbb{R}^3$, $W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \,|\, x - 2y + z = 0 \text{ and } 3x - y - z = 0 \right\}$.

Yes. Check $W = \mathcal{N}(A)$ with $A = \begin{bmatrix} 1 & -2 & 1 \\ 3 & -1 & -1 \end{bmatrix}$.

(e) for math major only. $V = M_{22}$, $W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{22} \,|\, a + b = c + d \right\}$.

Yes. Use definition to prove it.

(f) for math major only. $V = M_{22}$, $W =$ set of all 2×2 singular matrices.

No. Check $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \in W$, but $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \notin W$.

(g) for math major only. $V = P_n$, $W =$ set of all polynomials $p(x)$ in P_n such that $p(0) = 0$.

Yes. Use definition to prove it.

6.3 Linear Combinations

In this section, we will discuss about linear combinations of vectors. Although most definitions and theorems are stated for general vector space, we will focus on examples in Euclidean space $\mathbb{R}^m$.

Definition 6.3.1 (Linear Combination). Let $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$ be vectors in a vector space V, and α_1, α_2, $\ldots$, α_n be scalars. Then

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n$$

is called a **linear combination** of $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$.

Example 6.3.1. Given vectors $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$, then

(a) $23\mathbf{u}_1 - \sqrt{2}\mathbf{u}_2 + \cdots + \pi\mathbf{u}_n$ is a linear combination of $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$.

(b) $\mathbf{0}$ is a linear combination of $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$ because

$$\mathbf{0} = 0\mathbf{u}_1 + 0\mathbf{u}_2 + \cdots 0\mathbf{u}_n.$$

(c) $\mathbf{u_1}$ is a linear combination of $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$ because

$$\mathbf{u}_1 = 1\mathbf{u}_1 + 0\mathbf{u}_2 + \cdots 0\mathbf{u}_n.$$

Similarly, each of $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$ is a linear combination of $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$.

(d) If $\mathbf{v}$ and $\mathbf{w}$ are both linear combinations of $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$ with

$$\mathbf{v} = \alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n,$$

$$\mathbf{w} = \beta_1\mathbf{u}_1 + \beta_2\mathbf{u}_2 + \cdots + \beta_n\mathbf{u}_n.$$

Then any linear combination of $\mathbf{v}$ and $\mathbf{w}$, denoted by $c\mathbf{v} + d\mathbf{w}$, is also a linear combination of $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$ with

$$c\mathbf{v} + d\mathbf{w} = (c\alpha_1 + d\beta_1)\mathbf{u}_1 + (c\alpha_2 + d\beta_2)\mathbf{u}_2 + \cdots + (c\alpha_n + d\beta_n)\mathbf{u}_n.$$

Example 6.3.2. Let

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 3 \\ -5 \\ 4 \end{bmatrix}.$$

(a) Determine if $\mathbf{v} = \begin{bmatrix} 2 \\ -6 \\ 3 \end{bmatrix}$ can be expressed as a linear combination of $\mathbf{u}_1, \mathbf{u}_2$.

(b) Determine if $\mathbf{w} = \begin{bmatrix} 2 \\ -6 \\ -16 \end{bmatrix}$ can be expressed as a linear combination of $\mathbf{u}_1, \mathbf{u}_2$.

Solution.

(a) Suppose $\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2$, then

$$c_1 \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ -5 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ 3 \end{bmatrix}.$$

It can be rewritten as

$$\begin{bmatrix} 1 & 3 \\ -2 & -5 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ 3 \end{bmatrix}.$$

Then we solve c_1, c_2 from the above linear system. The augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 3 & 2 \\ -2 & -5 & -6 \\ -1 & 4 & 3 \end{array}\right] \xrightarrow{\text{row-reduced}} \left[\begin{array}{cc|c} 1 & 3 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 19 \end{array}\right].$$

There is no solution, so $\mathbf{v}$ cannot be expressed as a linear combination of $\mathbf{u}_1, \mathbf{u}_2$.

(b) Suppose $\mathbf{w} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2$, then

$$c_1 \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ -5 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ -16 \end{bmatrix}.$$

It is equivalent to

$$\begin{bmatrix} 1 & 3 \\ -2 & -5 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ -16 \end{bmatrix}.$$

The augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 3 & 2 \\ -2 & -5 & -6 \\ -1 & 4 & -16 \end{array}\right] \xrightarrow{\text{row-reduced}} \left[\begin{array}{cc|c} 1 & 3 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right].$$

The unique solution is $c_1 = 8, c_2 = -2$, so $\mathbf{w}$ can be expressed as a linear combination of $\mathbf{u}_1, \mathbf{u}_2$ with

$$\mathbf{w} = 8\mathbf{u}_1 - 2\mathbf{u}_2.$$

□

Theorem 6.3.1 (Linear System and Linear Combination).
Let $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$ *and* $\mathbf{b}$ *be vectors in* $\mathbb{R}^m$. *Denote* $A = \left[\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_n\right]$ *which is an* $m \times n$ *matrix. Then* $\mathbf{b}$ *can be expressed as a linear combination of* $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$ *if and only if the linear system* $\mathcal{LS}(A, \mathbf{b})$ *has a solution.*

Proof. $\mathbf{b}$ can be expressed as a linear combination of $\mathbf{u_1}$, $\mathbf{u_2}$, $\ldots$, $\mathbf{u_n}$ if and only if there are some scalars $x_1, x_2, \cdots, x_n$ such that

$$x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + \cdots + x_n\mathbf{u}_n = \mathbf{b}.$$

The above linear system can be rewritten equivalently to

$$\left[\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_n\right] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{b}$$

Thus it happens if and only if the linear system $A\mathbf{x} = \mathbf{b}$ has some solution $\mathbf{x}$.

□

Definition 6.3.2 (Span). Let $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ be vectors in a vector space V. Then the set of all linear combinations of $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ is called the **span** of $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$, denoted by $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$, i.e.,

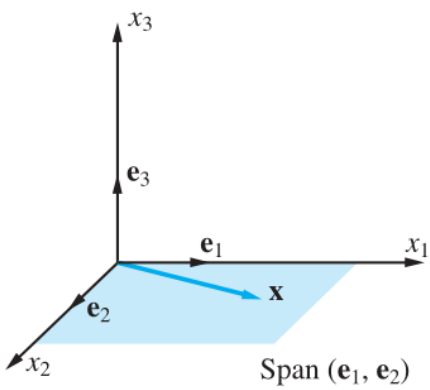
$$\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\} = \{\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n \,|\, \alpha_i \in \mathbb{R},\ 1 \leq i \leq n\}.$$

Example 6.3.3. Consider $\mathbb{R}^3$.

(a) Let $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. The span of $\mathbf{e}_1$ and $\mathbf{e}_2$ is the set of all vectors of the form

$$\alpha\mathbf{e}_1 + \beta\mathbf{e}_2 = \begin{bmatrix} \alpha \\ \beta \\ 0 \end{bmatrix}, \quad \alpha, \beta \in \mathbb{R}.$$

Hence, $\text{Span}\{\mathbf{e}_1, \mathbf{e}_2\}$ is just the x_1x_2-plane.



(b) The span of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ is the set of all vectors of the form

$$\alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 + \alpha_3\mathbf{e}_3 = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}, \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}.$$

Hence, $\text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \mathbb{R}^3$.

Theorem 6.3.2. *Let* $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ *be vectors in a vector space* V*. Then* $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ *is a subspace of* V*.*

Proof. Obviously $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is a nonempty subset of V.

For all $\alpha \in \mathbb{R}$, $\mathbf{v}, \mathbf{w} \in \text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$, there exist scalars $\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_n$ such that

$$\mathbf{v} = \alpha_1\mathbf{u}_1 + \cdots + \alpha_k\mathbf{u}_n,$$

$$\mathbf{w} = \beta_1\mathbf{u}_1 + \cdots + \beta_n\mathbf{u}_n.$$

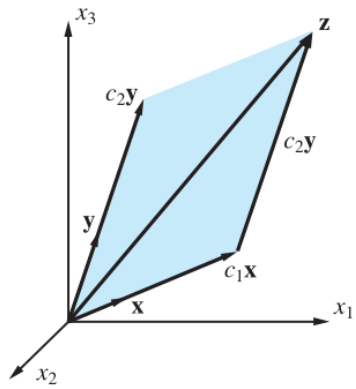
Then

$$\mathbf{v} + \mathbf{w} = (\alpha_1 + \beta_1)\mathbf{u}_1 + \cdots + (\alpha_n + \beta_n)\mathbf{u}_n,$$

$$\alpha\mathbf{v} = (\alpha\alpha_1)\mathbf{u}_1 + \cdots + (\alpha\alpha_n)\mathbf{u}_n.$$

So both $\mathbf{v}+\mathbf{w}$ and $\alpha\mathbf{v}$ are in $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$. Therefore, $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is a subspace of V. □

Remark 6.3.1. Suppose $\mathbf{x}, \mathbf{y}$ are two nonparallel vectors in $\mathbb{R}^3$. If $\mathbf{z} = c_1\mathbf{x} + c_2\mathbf{y}$, then $\mathbf{z}$ is a sum of two vectors parallel to $\mathbf{x}$ and $\mathbf{y}$ and hence must lie on the plane determined by $\mathbf{x}$ and $\mathbf{y}$. So the geometrical representation of $\text{Span}\{\mathbf{x}, \mathbf{y}\}$ is the plane determined by the two vectors.



Example 6.3.4. Let

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -4 \\ -1 \\ 2 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} -3 \\ 1 \\ 3 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -7 \\ 7 \\ 11 \end{bmatrix}.$$

Determine if $\mathbf{v}$ is in $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

Solution. We need to show whether $\mathbf{v}$ can be expressed as a linear combination of $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$. Suppose $\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3$, then

$$c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -4 \\ -1 \\ 2 \end{bmatrix} + c_3 \begin{bmatrix} -3 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -7 \\ 7 \\ 11 \end{bmatrix}.$$

It can be rewritten as

$$\begin{bmatrix} 1 & -4 & -3 \\ 2 & -1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} -7 \\ 7 \\ 11 \end{bmatrix}.$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & -4 & -3 & -7 \\ 2 & -1 & 1 & 7 \\ 1 & 2 & 3 & 11 \end{array}\right] \xrightarrow{\text{row-reduced}} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 5 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

There are infinitely many solutions, so $\mathbf{v} \in \text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

For example, $\mathbf{v} = 5\mathbf{u}_1 + 3\mathbf{u}_2$, or $\mathbf{v} = 3\mathbf{u}_1 + \mathbf{u}_2 + 2\mathbf{u}_3$. □

Remark 6.3.2. Why are there infinitely many ways to express $\mathbf{v}$ as a linear combination of $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ in the above example? Notice that

$$\mathbf{u}_3 = \mathbf{u}_1 + \mathbf{u}_2.$$

So once we find one way $\mathbf{v} = \alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \alpha_3\mathbf{u}_3$, we can construct infinitely many ways $\mathbf{v} = (\alpha_1 + k)\mathbf{u}_1 + (\alpha_2 + k)\mathbf{u}_2 + (\alpha_3 - k)\mathbf{u}_3$, where k is arbitrary.

The linear relation $\mathbf{u}_3 = \mathbf{u}_1 + \mathbf{u}_2$ makes $\mathbf{u}_3$ redundant in $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$, i.e.,

$$\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}.$$

Generally, if a vector space is spanned by a set of vectors, how could we remove those unnecessary vectors to obtain a "minimal" spanning set? This question will be discussed in next section.

6.4 Linear Dependence and Independence

Motivation: Suppose $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ spans a subspace

$$W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}.$$

Can we find a minimal spanning set of W, that is, a subset of $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ containing no unnecessary vectors and spanning the same W?

To see how to find a minimal spanning set, we need to consider how the vectors in the collection depend on each other. This motivates us to introduce the concepts of **linear dependence** and **linear independence**. These simple concepts provide the keys to understanding the structure of vector spaces.

Consider the vectors in Example 6.3.4,

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -4 \\ -1 \\ 2 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} -3 \\ 1 \\ 3 \end{bmatrix}.$$

Denote $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$. Notice that $\mathbf{u}_3$ is a linear combination of $\mathbf{u}_1, \mathbf{u}_2$,

$$\mathbf{u}_3 = \mathbf{u}_1 + \mathbf{u}_2,$$

so any linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ can be reduced to a linear combination of $\mathbf{u}_1, \mathbf{u}_2$:

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \alpha_3\mathbf{u}_3 = (\alpha_1 + \alpha_3)\mathbf{u}_1 + (\alpha_2 + \alpha_3)\mathbf{u}_2.$$

Thus

$$W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}.$$

Actually we can use the relation

$$1\mathbf{u}_1 + 1\mathbf{u}_2 + (-1)\mathbf{u}_3 = 0 \tag{6.5}$$

to express any vector in terms of the other two:

$$\mathbf{u}_1 = -\mathbf{u}_2 + \mathbf{u}_3, \ \mathbf{u}_2 = -\mathbf{u}_1 + \mathbf{u}_3, \ \mathbf{u}_3 = \mathbf{u}_1 + \mathbf{u}_2.$$

It follows that

$$W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \text{Span}\{\mathbf{u}_2, \mathbf{u}_3\} = \text{Span}\{\mathbf{u}_1, \mathbf{u}_3\} = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}.$$

Because of the dependency relation (6.5), the subspace W can be spanned by any two of the given vectors. On the other hand, there is no such dependency relation between $\mathbf{u}_1$ and $\mathbf{u}_2$. Because if there are scalars c_1 and c_2, not both 0, such that

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 = 0, \tag{6.6}$$

then we could solve one vector in terms of the other:

$$\mathbf{u}_1 = -\frac{c_2}{c_1}\mathbf{u}_1 \text{ (if } c_1 \neq 0) \quad \text{or} \quad \mathbf{u}_2 = -\frac{c_1}{c_2}\mathbf{u}_1 \text{ (if } c_2 \neq 0).$$

However, neither of the two vectors is a multiple of the other. Therefore W cannot be spanned by only one vector from $\mathbf{u}_1$ and $\mathbf{u}_2$. The relation (6.6) holds if and only if $c_1 = c_2 = 0$, and we say $\mathbf{u}_1$ and $\mathbf{u}_2$ are linearly independent.

Definition 6.4.1 (Linear Dependence and Independence).
Let $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ be vectors in a vector space V.

(a) $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ is said to be **linearly independent** if

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n = \mathbf{0}$$

holds only for $\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0$;

(b) $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ is said to be **linearly dependent** if there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$, not all zero, such that

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n = \mathbf{0}$$

Remark 6.4.1. We can also denote $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$, and say the set S is linearly independent for case (a), and linearly dependent for case (b) in the above definition.

Remark 6.4.2.

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n = \mathbf{0}$$

is a linear relation between $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$. If all scalars are zero,

$$0\mathbf{u}_1 + 0\mathbf{u}_2 + \cdots + 0\mathbf{u}_n = \mathbf{0}$$

is called a trivial linear relation. Hence,

$$\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n \text{ is linearly independent} \Leftrightarrow \mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n \text{ has only trivial relation.}$$

$$\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n \text{ is linearly dependent} \Leftrightarrow \mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n \text{ has nontrivial relation.}$$

Example 6.4.1.

(a) The vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ are linearly independent because if

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

then

$$\begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This system has only trivial solution $c_1 = c_2 = 0$.

(b) $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ are linearly independent. Check it!

(c) $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \mathbf{u} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$ are linearly dependent since the following nontrivial relation can be easily observed

$$3\mathbf{e}_1 + 4\mathbf{e}_2 + 5\mathbf{e}_3 - \mathbf{u} = \mathbf{0}.$$

(d) $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n, \mathbf{0}$ must be linearly dependent because

$$0\mathbf{u}_1 + 0\mathbf{u}_2 + \cdots + 0\mathbf{u}_n + 1(\mathbf{0}) = \mathbf{0}$$

is a nontrivial relation.

(e) One vector $\mathbf{u}$ is linearly independent

$\Leftrightarrow k\mathbf{u} = \mathbf{0}$ holds only for $k = 0$.

$\Leftrightarrow \mathbf{u} \neq \mathbf{0}$.

Theorem 6.4.1 (Linearly Independence and Homogeneous System). *Let $\mathbf{u_1}, \mathbf{u_2}, \ldots, \mathbf{u_n}$ be vectors in $\mathbb{R}^m$. Denote $A = \left[\mathbf{u}_1, \mathbf{u}_2, \cdots, \mathbf{u}_n\right]$ which is an $m \times n$ matrix. Then $\mathbf{u_1}, \mathbf{u_2}, \ldots, \mathbf{u_n}$ are linearly independent if and only if the homogeneous linear system $\mathcal{LS}(A, \mathbf{0})$ has only trivial solution.*

Proof. $\mathbf{u_1}, \mathbf{u_2}, \ldots, \mathbf{u_n}$ are linearly independent if and only if there is only trivial solution $x_1 = x_2 = \cdots = x_n = 0$ to the vector equation

$$x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + \cdots + x_n\mathbf{u}_n = \mathbf{0},$$

which is equivalent to $A\mathbf{x} = \mathbf{0}$.

□

Example 6.4.2.

(a) Consider $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}$, $\mathbf{u}_3 = \begin{bmatrix} 3 \\ 8 \\ 7 \\ 5 \end{bmatrix}$ in $\mathbb{R}^4$.

Suppose $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 = \mathbf{0}$, then

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 8 \\ 2 & 4 & 7 \\ 1 & 1 & 5 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 3 & 8 & 0 \\ 2 & 4 & 7 & 0 \\ 1 & 1 & 5 & 0 \end{array}\right] \xrightarrow{\text{row-reduced}} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

It has only trivial solution $c_1 = c_2 = c_3 = 0$. So $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$ are linearly independent.

(b) Consider $\mathbf{u}_1 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} 5 \\ -2 \\ 4 \end{bmatrix}$, $\mathbf{u}_3 = \begin{bmatrix} 3 \\ 8 \\ 6 \end{bmatrix}$, $\mathbf{u}_4 = \begin{bmatrix} 2 \\ 7 \\ -4 \end{bmatrix}$ in $\mathbb{R}^3$.

Suppose $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 + c_4\mathbf{u}_4 = \mathbf{0}$, then

$$\begin{bmatrix} 2 & 5 & 3 & 2 \\ 1 & -2 & 8 & 7 \\ 3 & 4 & 6 & -4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The homogeneous system has more unknowns than equations, so it must admit nontrivial solution (Theorem 4.1.2). Therefore, $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$, $\mathbf{u}_4$ are linearly dependent.

The above example (b) is generalized to the following theorem.

Theorem 6.4.2 ($n > m$ **implies Linear Dependence).** *Suppose* $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ *are vectors in* $\mathbb{R}^m$ *and* $n > m$. *Then* $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ *are linearly dependent.*

Proof. Exercise. □

Theorem 6.4.3 (**Nonsingular Matrix Equivalences**). *Let A be an $n \times n$ square matrix. The following statements are equivalent.*

1. *A is nonsingular.*
2. *A is row equivalent to I_n.*
3. *The null space of A contains only the zero vector, $\mathcal{N}(A) = \{\mathbf{0}\}$.*
4. *The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every choice of $\mathbf{b}$.*
5. *A is invertible.*
6. *There exists B such that $AB = I_n$.*
7. *There exists B such that $BA = I_n$.*
8. *The n columns of A are linearly independent.*

Proof. The equivalence between statement 1 and 8 follows from Theorem 6.4.1. □

Theorem 6.4.4. *Let $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ be vectors in a vector space V.*

(a) $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ are linearly dependent if and only if at least one vector among $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ is a linear combination of the others.

(b) $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ are linearly independent if and only if none of the vector among $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ is a linear combination of the others.

Proof. We only show part (a), and part (b) directly follows from (a).

($\Rightarrow$) Suppose $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ are linearly dependent. Then there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$, not all zero, such that

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n = \mathbf{0}.$$

Suppose that $\alpha_i \neq 0$. Then it follows from the above relation that

$$\mathbf{u}_i = -\frac{\alpha_1}{\alpha_i}\mathbf{u}_1 \cdots - \frac{\alpha_{i-1}}{\alpha_i}\mathbf{u}_{i-1} - \frac{\alpha_{i+1}}{\alpha_i}\mathbf{u}_{i+1} - \cdots - \frac{\alpha_n}{\alpha_i}\mathbf{u}_n.$$

So $\mathbf{u}_i$ can be expressed as a linear combination of the other vectors.

($\Leftarrow$) Suppose $\mathbf{u}_i$ is a linear combination of the other vectors with

$$\mathbf{u}_i = \beta_1\mathbf{u}_1 + \cdots + \beta_{i-1}\mathbf{u}_{i-1} + \beta_{i+1}\mathbf{u}_{i+1} + \cdots + \beta_n\mathbf{u}_n.$$

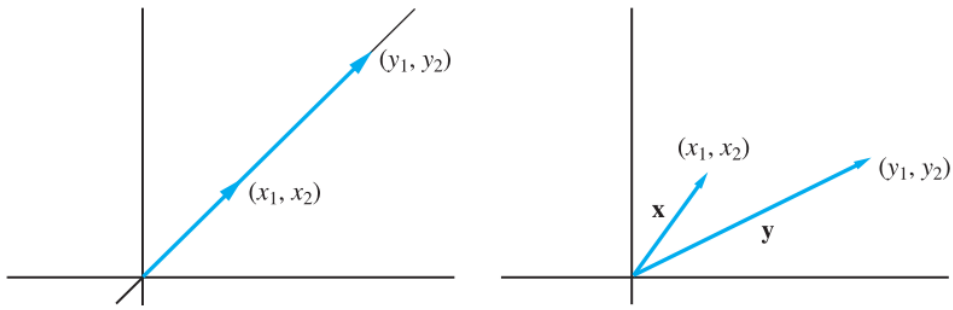
Then we have

$$\beta_1\mathbf{u}_1 + \cdots + \beta_{i-1}\mathbf{u}_{i-1} + (-1)\mathbf{u}_i + \beta_{i+1}\mathbf{u}_{i+1} + \cdots + \beta_n\mathbf{u}_n = \mathbf{0}$$

This provides a nontrivial linear relation for $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ because there is at least one nonzero scalar -1. Therefore, $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ are linearly dependent. $\square$

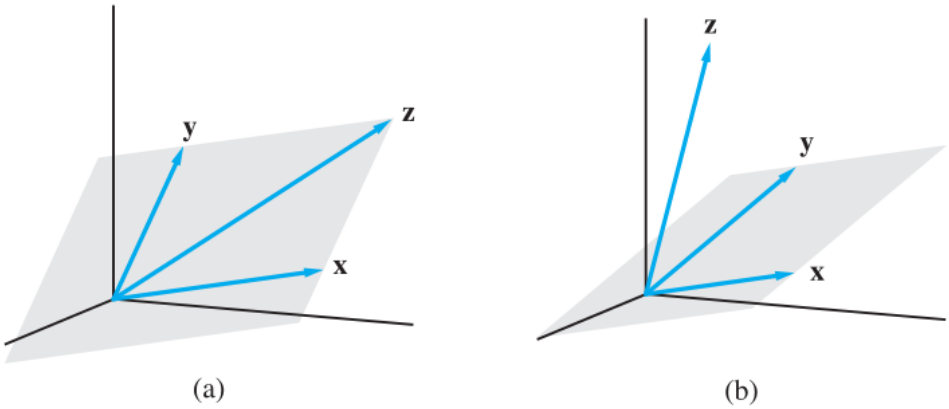
Geometric Interpretation:

(1) In $\mathbb{R}^2$, two vectors $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent if and only if one can be expressed as a scalar multiple of the other, which means $\mathbf{x}$ and $\mathbf{y}$ lie on the same line. In that case, $\mathbf{x}$ and $\mathbf{y}$ are said to be colinear (or spelled as "collinear").



(a) $\mathbf{x}$ and $\mathbf{y}$ linearly dependent (b) $\mathbf{x}$ and $\mathbf{y}$ linearly independent

(2) In $\mathbb{R}^3$, three vectors $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ are linearly dependent if and only if some one can be expressed as a linear combination of the other two, which means $\mathbf{x}$, $\mathbf{y}$, $\mathbf{z}$ lie in the same plane. In that case, $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ are said to be coplanar.



Theorem 6.4.5 (Subset of Linearly Independent Set). *Let* $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ *be a set of vectors in a vector space* V.

(a) If $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ *is linearly independent, then any subset of* $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ *is also linearly independent.*

(b) If some subset of $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ *is linearly dependent, then* $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ *is linearly dependent.*

Proof. We only show part (a), and part (b) directly follows from (a).

(a) Suppose $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is linearly independent. Choose any subset and denote it by $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ without loss of generality. If

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_k\mathbf{u}_k = \mathbf{0},$$

then we have

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_k\mathbf{u}_k + 0\mathbf{u}_{k+1} + \cdots + 0\mathbf{u}_n = \mathbf{0}.$$

Since $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is linearly independent, all the scalars in the above equation must be zero, so $\alpha_1 = \alpha_2 = \cdots = \alpha_k = 0$. Therefore $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k$ has only trivial linear relation, so they are linearly independent.

□

Theorem 6.4.6 (Add One Vector into a Linearly Independent Set).
Let $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ *be linearly independent vectors, and* v *be a vector in* V*.*

(a) $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n, \mathbf{v}$ *are linearly dependent if and only if* $\mathbf{v}$ *can be expressed as a linear combination of* $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$*.*

(b) $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n, \mathbf{v}$ *are linearly independent if and only if* $\mathbf{v}$ *cannot be expressed as a linear combination of* $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$*.*

Proof. We only show part (a), and part (b) directly follows from (a).

($\Rightarrow$) Suppose $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n, \mathbf{v}$ are linearly dependent. Then there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n, \beta$, not all zero, such that

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n + \beta\mathbf{v} = \mathbf{0}.$$

If $\beta = 0$, then $\alpha_1, \alpha_2, \ldots, \alpha_n$ are not all zero and satisfy

$$\alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n = \mathbf{0}.$$

This contradicts the linear independence of $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$. Therefore $\beta \neq 0$, and

$$\mathbf{v} = -\frac{\alpha_1}{\beta}\mathbf{u}_1 - \frac{\alpha_2}{\beta}\mathbf{u}_2 - \cdots - \frac{\alpha_n}{\beta}\mathbf{u}_n$$

is a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$.

($\Leftarrow$) Suppose $\mathbf{v}$ is a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$. By Theorem 6.4.4, $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n, \mathbf{v}$ are linearly dependent. □

Theorem 6.4.7 (Equivalent Statements on Linear Independence). *Let* $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ *be vectors in a vector space* V*. Then the following statements are equivalent.*

(a) $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ *are linearly independent.*

(b) None of the vector among $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ *is a linear combination of the other vectors.*

(c) The subspace $\mathrm{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ *cannot be be spanned by a smaller (proper) subset of* $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$*.*

(d) Every vector in $\mathrm{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ *can be expressed uniquely as a linear combination of* $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$*.*

Proof. The equivalence between (a), (b), (c) has been explained in Theorem 6.4.4. Now we show (a) is equivalent to (d).

(a)⇒(d). Suppose $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ are linearly independent.

Let $\mathbf{v} \in \mathrm{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$, then $\mathbf{v}$ can be written as a linear combination

$$\mathbf{v} = \alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n. \tag{6.7}$$

Suppose that $\mathbf{v}$ can be written as another linear combination

$$\mathbf{v} = \beta_1\mathbf{u}_1 + \beta_2\mathbf{u}_2 + \cdots + \beta_n\mathbf{u}_n,$$

then we have

$$(\alpha_1 - \beta_1)\mathbf{u}_1 + (\alpha_2 - \beta_2)\mathbf{u}_2 + \cdots + (\alpha_n - \beta_n)\mathbf{u}_n = \mathbf{v} - \mathbf{v} = \mathbf{0}.$$

By the linear independence of $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$, all the scalars must be 0, i.e.,

$$\alpha_1 = \beta_1,\ \alpha_2 = \beta_2,\ \ldots,\ \alpha_n = \beta_n.$$

Thus, the representation (6.7) is unique.

(d)⇒(a). Suppose every vector $\mathbf{v} \in \mathrm{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ can be expressed uniquely as

$$\mathbf{v} = \alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \cdots + \alpha_n\mathbf{u}_n. \tag{6.8}$$

By contradiction argument, suppose $\mathbf{u}_1,\ \mathbf{u}_2,\ \ldots,\ \mathbf{u}_n$ are linearly dependent, then there exist scalars $c_1, c_2, \ldots, c_n$, not all zero, such that

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n = \mathbf{0}.$$

Then $\mathbf{v}$ can also be expressed as

$$\mathbf{v} = (\alpha_1 + c_1)\mathbf{u}_1 + (\alpha_2 + c_2)\mathbf{u}_2 + \cdots + (\alpha_n + c_n)\mathbf{u}_n. \tag{6.9}$$

Since c_i are not all zero, the two representations in (6.8) (6.9) must be different. Contradiction arises. So $\mathbf{u}_1,\ \mathbf{u}_2,\ \ldots,\ \mathbf{u}_n$ are linearly independent. □

Remark 6.4.3. For a vector space W, we would like to find its minimal spanning set $\{\mathbf{u}_1,\ \mathbf{u}_2,\ \ldots,\ \mathbf{u}_n\}$, which is a linearly independent set such that every vector in W can be spanned by $\mathbf{u}_1,\ \mathbf{u}_2,\ \ldots,\ \mathbf{u}_n$ in a unique way. $\{\mathbf{u}_1,\ \mathbf{u}_2,\ \ldots,\ \mathbf{u}_n\}$ is called a basis of W. See next section.

Uniquesness of RREF: for math major only.

We close this section with the proof of uniqueness of RREF of a matrix. There are two approaches. Let's first show a constructive proof using the following example.

Example 6.4.3. Linear relations of columns of A uniquely determine the entries of its RREF.

Let

$$A = \begin{bmatrix} 1 & 1 & 3 & 1 & 0 & 0 \\ 2 & 1 & 5 & 1 & 1 & 2 \\ 1 & -1 & 1 & 2 & 1 & -3 \\ 1 & 3 & 5 & 1 & -1 & 1 \end{bmatrix}.$$

We show that the RREF of A must be

$$B = \begin{bmatrix} 1 & 0 & 2 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

Key observation: Let $\mathbf{A}_i$ and $\mathbf{B}_i$ be the i-th column of A and B. By the equivalence of systems $\mathcal{LS}(A, \mathbf{0})$ and $\mathcal{LS}(B, \mathbf{0})$,

$$x_1\mathbf{A}_1 + x_2\mathbf{A}_2 + \cdots + x_6\mathbf{A}_6 = \mathbf{0}$$

if and only if

$$x_1\mathbf{B}_1 + x_2\mathbf{B}_2 + \cdots + x_6\mathbf{B}_6 = \mathbf{0},$$

which means, the linear relation of columns of A and the linear relation of columns of B are exactly the same.

Step 1: $\mathbf{A}_1 \neq \mathbf{0}$ is linearly independent, so $\mathbf{B}_1$ is also linearly independent, and it must be a pivot column:

$$\mathbf{B}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Step 2: We can check $\mathbf{A}_2 \notin \text{Span}\{\mathbf{A}_1\}$ and $\mathbf{A}_1, \mathbf{A}_2$ are linearly independent, so $\mathbf{B}_1, \mathbf{B}_2$ are also linearly independent, and $\mathbf{B}_2$ must be a pivot column:

$$\mathbf{B}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

Step 3: We can check $\mathbf{A}_3 \in \text{Span}\{\mathbf{A}_1, \mathbf{A}_2\}$ with

$$\mathbf{A}_3 = 2\mathbf{A}_1 + \mathbf{A}_2.$$

So

$$\mathbf{B}_3 = 2\mathbf{B}_1 + \mathbf{B}_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

Step 4: We can check $\mathbf{A}_4 \notin \text{Span}\{\mathbf{A}_1, \mathbf{A}_2\}$, and $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_4$ are linearly independent, so $\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_4$ are also linearly independent, and $\mathbf{B}_4$ must be a pivot column:

$$\mathbf{B}_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

Step 5: We can check $\mathbf{A}_5 \notin \text{Span}\{\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_4\}$, and $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_4, \mathbf{A}_5$ are linearly independent, so $\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_4, \mathbf{B}_5$ are also linearly independent, and $\mathbf{B}_5$ must be a pivot column:

$$\mathbf{B}_5 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Step 6: We can check $\mathbf{A}_6 \in \text{Span}\{\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_4, \mathbf{A}_5\}$ with

$$\mathbf{A}_6 = \mathbf{A}_1 + \mathbf{A}_2 - 2\mathbf{A}_4 + \mathbf{A}_5.$$

So

$$\mathbf{B}_6 = \mathbf{B}_1 + \mathbf{B}_2 - 2\mathbf{B}_4 + \mathbf{B}_5 = \begin{bmatrix} 1 \\ 1 \\ -2 \\ 1 \end{bmatrix}.$$

Hence the RREF of A must be

$$B = \begin{bmatrix} 1 & 0 & 2 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

Remark 6.4.4. Similarly for a general matrix A, the RREF of A is uniquely determined by the linear relation of the columns of A. Next we give another proof for the uniqueness.

Theorem 6.4.8. *If two matrices B and C are both RREF of A, then $B = C$.*

Proof. Let B_i, C_i be the i-th column of B, C. We will show $B_i = C_i$ by induction.

First, B_1 and C_1 are either $\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$ or $\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$. Note that the matrices $[B_1]$ and $[C_1]$ are also row equivalent, so they must be equal.

Now suppose the first k columns of B, C are all equal, we are going to show

$$B_{k+1} = C_{k+1}.$$

Consider the matrices $B' = [B_1, \cdots, B_k, B_{k+1}]$, $C' = [C_1, \cdots, C_k, C_{k+1}]$. Note that

(1) B', C' are row equivalent.

(2) B', C' are both in RREF.

Assume $B_{k+1} \neq C_{k+1}$. Let $\mathbf{x}$ be any solution to $B'\mathbf{x} = \mathbf{0}$, then $C'\mathbf{x} = \mathbf{0}$ and $(B' - C')\mathbf{x} = \mathbf{0}$. Since the columns of $B' - C'$ are all zero except the $(k+1)$-th one, x_{k+1} must be zero. Thus any solution to $B'\mathbf{x} = \mathbf{0}$ or $C'\mathbf{x} = \mathbf{0}$ must have $x_{k+1} = 0$. This implies that the $(k+1)$-th columns of B' and C' must contain leading 1's, for otherwise x_{k+1} would be a free variable and can be chosen arbitrarily. Then since the first k columns of B' and C' are identical, the position where this leading 1 appears in column $k+1$ must be the same for both B' and C', and consequently $B_{k+1} = C_{k+1}$, which is a contradiction.

□

6.5 Basis and Dimension

We have shown that a spanning set for a vector space is minimal if its elements are linearly independent. The elements of a minimal spanning set form the basic building blocks for the whole vector space, and consequently, we say that they form a basis for the vector space.

Definition 6.5.1. A set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ is said to be a **basis** for a vector space V if

(i) $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m$ are linearly independent.

(ii) $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\} = V$.

Example 6.5.1 (Standard Basis of $\mathbb{R}^m$).

Let $V = \mathbb{R}^3$, then $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ form a basis for $\mathbb{R}^3$. Check:

(i) $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ are linearly independent.

(ii) $\text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \mathbb{R}^3$.

Similarly, if $V = \mathbb{R}^m$ and $\mathbf{e}_i$ $(1 \leq i \leq m)$ is the vector with all zero entries except the i-th entry being 1, then $\{\mathbf{e}_1, \ldots, \mathbf{e}_m\}$ is a basis for $\mathbb{R}^m$, and it is called the standard basis of $\mathbb{R}^m$.

Example 6.5.2. for math major only. Let $V = M_{22}$,

$$B_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$$

$$B_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, B_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$$

Then $\{B_{11}, B_{12}, B_{21}, B_{22}\}$ is a basis for V.

Check:

(i) $B_{11}, B_{12}, B_{21}, B_{22}$ are linearly independent (exercise).

(ii) For any $A \in V$,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = aB_{11} + bB_{12} + cB_{21} + dB_{22}.$$

So $\text{Span}\{B_{11}, B_{12}, B_{21}, B_{22}\} = V$.

Example 6.5.3. for math major only. Let $V = M_{mn}$, and for $1 \leq i \leq m$, $1 \leq j \leq n$, let B_{ij} be the $m \times n$ matrix with (i, j)-th entry equal to 1 and all other entries equal to 0.

Then $\{B_{ij} | 1 \leq i \leq m,\ 1 \leq j \leq n\}$ is the standard basis for M_{mn}. (exercise).

Example 6.5.4. for math major only. Let $V = P_n$.

$\{1, x, x^2, \ldots, x^n\}$ is the standard basis for P_n.

Check:

(i) $1, x, x^2, \ldots, x^n$ are linearly independent.

(ii) Any polynomial

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

is a linear combinations of $1, x, x^2, \ldots, x^n$.

Remark 6.5.1. Basis of a vector space is not unique. For example, $V = \mathbb{R}^2$, $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\}$ are two bases for $\mathbb{R}^2$. However, we will show that:

"the number of vectors in a basis is unique."

Theorem 6.5.1. *Suppose that* $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\}$ *is a spanning set of a vector space* V*. Then any collection of* n *vectors in* V*, where* $n > m$*, is linearly dependent.*

Proof. Let $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ be n vectors in V, where $n > m$. Since $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ span V, each $\mathbf{u}_i$ can be written as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$:

$$\begin{aligned}
\mathbf{u}_1 &= a_{11}\mathbf{v}_1 + a_{21}\mathbf{v}_2 + \cdots + a_{m1}\mathbf{v}_m \\
\mathbf{u}_2 &= a_{12}\mathbf{v}_1 + a_{22}\mathbf{v}_2 + \cdots + a_{m2}\mathbf{v}_m \\
&\vdots \\
\mathbf{u}_n &= a_{1n}\mathbf{v}_1 + a_{2n}\mathbf{v}_2 + \cdots + a_{mn}\mathbf{v}_m
\end{aligned}$$

Suppose

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n = \mathbf{0},$$

then it is equivalent to

$$\begin{aligned}
\mathbf{0} =& (a_{11}c_1 + a_{12}c_2 + \cdots + a_{1n}c_n)\mathbf{v}_1 \\
&+ (a_{21}c_1 + a_{22}c_2 + \cdots + a_{2n}c_n)\mathbf{v}_2 \\
&+ \cdots \\
&+ (a_{m1}c_1 + a_{m2}c_2 + \cdots + a_{mn}c_n)\mathbf{v}_m.
\end{aligned}$$

Now consider the system of equations

$$\begin{aligned}
a_{11}c_1 + a_{12}c_2 + \cdots + a_{1n}c_n &= 0 \\
a_{21}c_1 + a_{22}c_2 + \cdots + a_{2n}c_n &= 0 \\
&\vdots \\
a_{m1}c_1 + a_{m2}c_2 + \cdots + a_{mn}c_n &= 0
\end{aligned}$$

This is a homogeneous system with more variables than equations $(n > m)$, so it must have a nontrivial solution $\hat{c}_1, \hat{c}_2, \ldots, \hat{c}_n$. Then we have a nontrivial linear relation

$$\hat{c}_1\mathbf{u}_1 + \hat{c}_2\mathbf{u}_2 + \cdots + \hat{c}_n\mathbf{u}_n = \mathbf{0},$$

which implies $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ are linearly dependent. $\square$

Theorem 6.5.2. *If both* $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ *and* $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ *are bases for a vector space* V*, then* $n = m$*.*

Proof. Since $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ span V and $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m$ are linearly independent, it follows from Theorem 6.5.1 that $m \leq n$.

Similarly, since $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m$ span V and $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent, we have $n \leq m$. Therefore, $n = m$.

□

The above theorem shows that the number of vectors in any basis is always the same, no matter which basis we choose. This leads to the following definition.

Definition 6.5.2 (Dimension). Let V be a vector space.

(a) The dimension of V, denoted by $\dim(V)$, is the number of vectors in a basis of V.

(b) V is called finite dimensional if $\dim(V)$ is finite; otherwise, V is called infinite dimensional.

(c) $\dim(\{\mathbf{0}\})$ is defined to be 0, because the space $\{\mathbf{0}\}$ contains no linearly independent vector.

Example 6.5.5.

(a) $\dim \mathbb{R}^m = m$.

(b) for math major only. $\dim M_{mn} = mn$.

(c) for math major only. $\dim P_n = n + 1$.

(d) for math major only. Let P be the set of all real polynomials. The infinite set $\{1, x, x^2, x^3, \ldots\}$ is the basis, so $\dim P = \infty$.

(e) for math major only. Let S_2 be the set of 2×2 symmetric matrices. For $A \in S_2$,

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

We can show that

$$T = \{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \}$$

is a basis for S_2. Hence $\dim S_2 = 3$.

Geometric Interpretation:

If $\mathbf{x}$ is a nonzero vector in $\mathbb{R}^3$,

$$\text{Span}\{\mathbf{x}\} = \{\alpha \mathbf{x} \,:\, \alpha \in \mathbb{R}\}$$

is an one-dimensional subspace, which represents a line through the origin and parallel to $\mathbf{x}$.

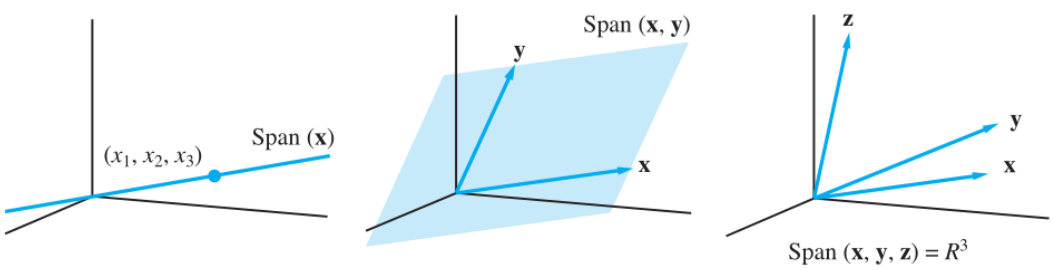
If $\mathbf{x}, \mathbf{y}$ are two linearly independent vectors in $\mathbb{R}^3$,

$$\text{Span}\{\mathbf{x}, \mathbf{y}\} = \{\alpha \mathbf{x} + \beta y \,:\, \alpha, \beta \in \mathbb{R}\}$$

is an two-dimensional subspace, which represents a plane through the origin and spanned by $\mathbf{x}$ and $\mathbf{y}$.

If $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are three linearly independent vectors in $\mathbb{R}^3$, they must form a basis and span the whole three-dimensional space $\mathbb{R}^3$ (why?)

$$\text{Span}\{\mathbf{x}, \mathbf{y}, \mathbf{z}\} = \mathbb{R}^3.$$



Theorem 6.5.3. *If a vector space V has dimension $m > 0$, then*

(a) any set of m linearly independent vectors spans V, so they form a basis of V.

(b) any m vectors that span V are linearly independent, so they form a basis of V.

Proof.

(a) Suppose $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ are linearly independent. Let $\mathbf{v}$ be any vector in V. By Theorem 6.5.1, $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m, \mathbf{v}$ must be linearly dependent. So there exist scalars $c_1, c_2, \ldots, c_m, c_{m+1}$, not all zero, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_m\mathbf{v}_m + c_{m+1}\mathbf{v} = \mathbf{0}.$$

The scalar c_{m+1} cannot be zero, otherwise the above relation would imply that $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ are linearly dependent. Hence,

$$\mathbf{v} = -\frac{c_1}{c_{m+1}}\mathbf{v}_1 - \frac{c_2}{c_{m+1}}\mathbf{v}_2 - \cdots - \frac{c_m}{c_{m+1}}\mathbf{v}_m.$$

Since $\mathbf{v}$ is an arbitrary vector in V, it follows that $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ span V.

(b) Suppose $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ span V. If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ are linearly dependent, then one of the $\mathbf{v}_i$'s, say, $\mathbf{v}_m$, can be written as a linear combination of the others. Then $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{m-1}$ still span V. If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{m-1}$ are linearly dependent, we can eliminate another vector and still have a spanning set. We can continue eliminating vectors in this way until we arrive at a linearly independent spanning set with fewer than m vectors. This contradicts $\dim(V) = m$. Therefore $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ must be linearly independent.

□

Example 6.5.6. Show that $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$, $\mathbf{u}_3 = \begin{bmatrix} 3 \\ 8 \\ 7 \end{bmatrix}$ form a basis of $\mathbb{R}^3$.

Proof. Step 1: To show linear independence.

Suppose $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 = \mathbf{0}$, then

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 8 \\ 2 & 4 & 7 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 3 & 8 & 0 \\ 2 & 4 & 7 & 0 \end{array}\right] \xrightarrow{\text{row-reduced}} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right].$$

It has only trivial solution $c_1 = c_2 = c_3 = 0$. So $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$ are linearly independent.

Step 2: To show the span.

Method 1: Since $\dim(\mathbb{R}^3) = 3$, and $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$ are 3 linearly independent vectors, by Theorem 6.5.3, $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$ must span $\mathbb{R}^3$.

Method 2: use definition. Let $\mathbf{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ be any vector. Solve $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 = \mathbf{v}$, i.e.,

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 8 \\ 2 & 4 & 7 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

It has been shown that the coefficient matrix is nonsingular, so this system must have a solution. Hence, $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$ span $\mathbb{R}^3$. □

Theorem 6.5.4. *If a vector space V has dimension $m > 0$, then*

(a) no set of fewer than m vectors can span V;

(b) any subset of fewer than m linearly independent vectors can be extended to a basis for V;

(c) any spanning set containing more than m vectors can be reduced to a basis for V.

Proof.

(a) If a set of fewer than m vectors can span V, using the approach in part (b) of Theorem 6.5.3, we can continue to eliminate those unnecessary vectors (which can be expressed as a linear combination of others) to obtain a basis containing fewer than m vectors. This contradicts $\dim(V) = m$.

(b) Suppose that $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ are linearly independent and $k < m$, then $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is a proper subspace of V, and there exists a vector $\mathbf{v}_{k+1}$ that is in V but not in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$. It follows from Theorem 6.4.6 that $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k, \mathbf{v}_{k+1}$ are linearly independent. If $k + 1 < m$, then, in the same manner, $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{k+1}\}$ can be extended to a set of $k+2$ linearly independent vectors. This extension process may be continued until a set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\}$ of n linearly independent vectors (which is a basis of V) is obtained.

(c) Suppose that $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ span V and $n > m$. Then, by Theorem 6.5.1, $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ must be linearly dependent. It follows that one of the vectors, say, $\mathbf{v}_n$, can be written as a linear combination of the others. Hence, if $\mathbf{v}_n$ is eliminated from the set, the remaining $n-1$ vectors will still span V. If $n-1 > m$, we can continue to eliminate vectors in this manner until we arrive at a spanning set containing m elements.

□

Theorem 6.5.5. *If W is a subspace of V, then*

(a) $\dim(W) \leq \dim(V)$; *and*

(b) $\dim(W) = \dim(V)$ *holds if and only if* $W = V$.

Proof.

(a) Suppose $\dim(W) = k$ and $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ is a basis of W.
Suppose $\dim(V) = m$ and $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\}$ is a basis of V.

Since $W \subseteq V$, each of $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k$ is also in V. By Theorem 6.5.1, any set of more than n vectors in V are linearly dependent, so we must have $k \leq m$, because otherwise, $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m$ are linearly dependent, contradicting the assumption that they form a basis of W.

(b) If $W = V$, then obviously $\dim(W) = \dim(V)$.

Now suppose $\dim(W) = \dim(V) = m$, and $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ is a basis of W. Then $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ is a set of m linearly independent vectors, so it follows from Theorem 6.5.3 that $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m$ span V. Therefore, $W = V$.

$\square$

6.6 Null Space, Column Space, Row Space

Let A be an $m \times n$ matrix. We will study the dimension and basis of several spaces related to A.

6.6.1 Null Space of a Matrix

Example 6.6.1. Let

$$A = \begin{bmatrix} 1 & 3 & 3 & -1 & -5 \\ 2 & 5 & 7 & 1 & 1 \\ 1 & 1 & 5 & 1 & 5 \\ -1 & -4 & -2 & 0 & 4 \end{bmatrix}$$

Solution. A can be row-reduced to

$$\begin{bmatrix} \boxed{1} & 0 & 6 & 0 & 4 \\ 0 & \boxed{1} & -1 & 0 & -2 \\ 0 & 0 & 0 & \boxed{1} & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

So the homogeneous system $A\mathbf{x} = \mathbf{0}$ has x_3, x_5 as free variables, and x_1, x_2, x_4 as lead variables. The solution can be expressed as

$$\mathbf{x} = \begin{bmatrix} -6x_3 - 4x_5 \\ x_3 + 2x_5 \\ x_3 \\ -3x_5 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} -6 \\ 1 \\ \boxed{1} \\ 0 \\ \boxed{0} \end{bmatrix} + x_5 \begin{bmatrix} -4 \\ 2 \\ \boxed{0} \\ -3 \\ \boxed{1} \end{bmatrix}, x_3, x_5 \in \mathbb{R}.$$

Let $\mathbf{z}_1 = \begin{bmatrix} -6 \\ 1 \\ \boxed{1} \\ 0 \\ \boxed{0} \end{bmatrix}$, $\mathbf{z}_2 = \begin{bmatrix} -4 \\ 2 \\ \boxed{0} \\ -3 \\ \boxed{1} \end{bmatrix}$, then $\mathcal{N}(A) = \text{Span}\{\mathbf{z}_1, \mathbf{z}_2\}$.

Next, we show $\mathbf{z}_1, \mathbf{z}_2$ are linearly independent. Suppose $\alpha \begin{bmatrix} -6 \\ 1 \\ \boxed{1} \\ 0 \\ \boxed{0} \end{bmatrix} + \beta \begin{bmatrix} -4 \\ 2 \\ \boxed{0} \\ -3 \\ \boxed{1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$.

Only considering the 3-rd and 5-th entries, we have

$$\begin{cases} 1\alpha + 0\beta = 0 \\ 0\alpha + 1\beta = 0. \end{cases}$$

So there is only trivial solution $\alpha = \beta = 0$, which means $\mathbf{z}_1, \mathbf{z}_2$ are linearly independent. Hence, $\mathbf{z}_1, \mathbf{z}_2$ form the basis of $\mathcal{N}(A)$ and $\dim(\mathcal{N}(A)) = 2$. $\square$

Remark 6.6.1. Another equivalent method to find the basis $\{\mathbf{z}_1, \mathbf{z}_2\}$:

(1) Let the free variables $x_3 = 1, x_5 = 0$, and solve the lead variables $x_1 = -6, x_2 = 1, x_4 = 0$, then we find one solution $\mathbf{z}_1 = \begin{bmatrix} -6 \\ 1 \\ \boxed{1} \\ 0 \\ \boxed{0} \end{bmatrix}$.

(2) Let the free variables $x_3 = 0, x_5 = 1$, and solve the lead variables $x_1 = -4, x_2 = 2, x_4 = -3$, then we find another solution $\mathbf{z}_2 = \begin{bmatrix} -4 \\ 2 \\ \boxed{0} \\ -3 \\ \boxed{1} \end{bmatrix}$.

The pattern of zeros and ones in free variables forces $\mathbf{z}_1, \mathbf{z}_2$ to be linearly independent!!!

Procedure to find a basis of $\mathcal{N}(A)$ for an $m \times n$ matrix A:

(i) Row-reduce A to REF or RREF with r pivot columns.

(ii) Determine the r lead variables, and $n-r$ free variables for the homogeneous system $A\mathbf{x} = \mathbf{0}$.

(iii) Each time, set one free variable to be 1 and other free variables to be 0, and solve all the lead variables, then we can get one solution. In total, we will obtain $n-r$ solutions $\mathbf{z}_1, \mathbf{z}_2, \cdots, \mathbf{z}_{n-r}$, which form the basis of $\mathcal{N}(A)$, and

$$\dim(\mathcal{N}(A)) = n - r.$$

The next two theorems give the explicit formulas of solutions to homogeneous system $A\mathbf{x} = 0$ and nonhomogeneous system $A\mathbf{x} = \mathbf{b}$. You can skip them if the notations are too complicated for you. Focus on examples!

Theorem 6.6.1 (Solutions to $A\mathbf{x} = \mathbf{0}$). *Suppose A is an $m \times n$ matrix and B is the RREF of A. Suppose B has r pivot columns, and their indices are*

$$D = \{d_1,\, d_2,\, \ldots,\, d_r\},$$

while the indices of $n-r$ non-pivot columns are

$$F = \{f_1,\, f_2,\, \ldots,\, f_{n-r}\}.$$

Construct the $n-r$ vectors $\mathbf{z}_j$, $1 \le j \le n-r$ of size n,

$$[\mathbf{z}_j]_i = \begin{cases} 1 & \text{if } i \in F,\ i = f_j \\ 0 & \text{if } i \in F,\ i \neq f_j \\ -[B]_{k,f_j} & \text{if } i \in D,\ i = d_k \end{cases}$$

Then the null space of A is given by

$$\mathcal{N}(A) = Span\{\mathbf{z}_1,\, \mathbf{z}_2,\, \ldots,\, \mathbf{z}_{n-r}\}.$$

Theorem 6.6.2 (**Solutions to $A\mathbf{x} = \mathbf{b}$**). *Suppose $A\mathbf{x} = \mathbf{b}$ is a consistent $m \times n$ linear system. Let B be the RREF of the the augmented matrix $[A|\mathbf{b}]$. Suppose B has r pivot columns, with indices $D = \{d_1, d_2, \ldots, d_r\}$, while the $n-r$ non-pivot columns have indices $F = \{f_1, f_2, \ldots, f_{n-r}\}$. Define vectors $\mathbf{c}$, $\mathbf{z}_j$, $1 \leq j \leq n-r$ of size n by*

$$[\mathbf{c}]_i = \begin{cases} 0 & \text{if } i \in F \\ [B]_{k,n+1} & \text{if } i \in D,\ i = d_k \end{cases}$$

$$[\mathbf{z}_j]_i = \begin{cases} 1 & \text{if } i \in F,\ i = f_j \\ 0 & \text{if } i \in F,\ i \neq f_j \\ -[B]_{k,f_j} & \text{if } i \in D,\ i = d_k \end{cases}$$

Then the set of solutions to $A\mathbf{x} = \mathbf{b}$ is

$$\{\mathbf{c} + \alpha_1\mathbf{z}_1 + \alpha_2\mathbf{z}_2 + \alpha_3\mathbf{u}_3 + \cdots + \alpha_{n-r}\mathbf{z}_{n-r} \,|\, \alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_{n-r} \in \mathbb{R}.\}$$

Here $\mathbf{c}$ is a particular solution to $A\mathbf{x} = \mathbf{b}$, and $\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_{n-r}$ are the basis of $\mathcal{N}(A)$.

Example 6.6.2. Let

$$A = \begin{bmatrix} -2 & -1 & -2 & -4 & 4 \\ -6 & -5 & -4 & -4 & 6 \\ 10 & 7 & 7 & 10 & -13 \\ -7 & -5 & -6 & -9 & 10 \\ -4 & -3 & -4 & -6 & 6 \end{bmatrix}.$$

The RREF of A is

$$B = \begin{bmatrix} \boxed{1} & 0 & 0 & 1 & -2 \\ 0 & \boxed{1} & 0 & -2 & 2 \\ 0 & 0 & \boxed{1} & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The free variables are x_4 and x_5.

Setting $x_4 = 1$ and $x_5 = 0$ gives $\mathbf{z}_1 = \begin{bmatrix} -1 \\ 2 \\ -2 \\ \boxed{1} \\ \boxed{0} \end{bmatrix}$.

Setting $x_4 = 0$ and $x_5 = 1$ gives $\mathbf{z}_2 = \begin{bmatrix} 2 \\ -2 \\ 1 \\ \boxed{0} \\ \boxed{1} \end{bmatrix}$. Then $\mathbf{z}_1, \mathbf{z}_2$ form the basis of $\mathcal{N}(A)$.

Example 6.6.3.

$$A = \begin{bmatrix} 2 & 1 & 5 & 1 & 5 & 1 \\ 1 & 1 & 3 & 1 & 6 & -1 \\ -1 & 1 & -1 & 0 & 4 & -3 \\ -3 & 2 & -4 & -4 & -7 & 0 \\ 3 & -1 & 5 & 2 & 2 & 3 \end{bmatrix} \xrightarrow{\text{row-reduced}} \begin{bmatrix} \boxed{1} & 0 & 2 & 0 & -1 & 2 \\ 0 & \boxed{1} & 1 & 0 & 3 & -1 \\ 0 & 0 & 0 & \boxed{1} & 4 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

x_1, x_2, x_4 are lead variables, and x_3, x_5, x_6 are free variables. Following the pattern of zeros and ones in free variables, set

$$\mathbf{z}_1 = \begin{bmatrix} -2 \\ -1 \\ \boxed{1} \\ 0 \\ \boxed{0} \\ \boxed{0} \end{bmatrix}, \ \mathbf{z}_2 = \begin{bmatrix} 1 \\ -3 \\ \boxed{0} \\ -4 \\ \boxed{1} \\ \boxed{0} \end{bmatrix}, \ \mathbf{z}_3 = \begin{bmatrix} -2 \\ 1 \\ \boxed{0} \\ 2 \\ \boxed{0} \\ \boxed{1} \end{bmatrix},$$

which form the basis of $\mathcal{N}(A)$.

Remark 6.6.2. Try to show that $\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3$ are linearly independent by definition.

Example 6.6.4 (Plane in $\mathbb{R}^3$: $ax + by + cz = d$). Suppose a, b, c are not all zero.

(i)

$$ax + by + cz = 0$$

has one lead variable and two free variables. The solution set can be spanned by two vectors, i.e.,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2, \ \ c_1, c_2 \in \mathbb{R}.$$

So $ax + by + cz = 0$ represents a plane $\mathcal{P}_1$ through the origin and spanned by $\mathbf{u}_1, \mathbf{u}_2$.

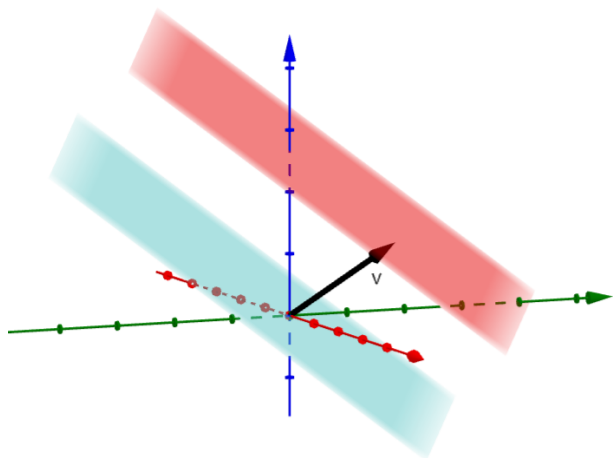
(ii) The solution set to

$$ax + by + cz = b$$

can be written as

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{v} + c_1\mathbf{u}_1 + c_2\mathbf{u}_2, \ \ c_1, c_2 \in \mathbb{R}.$$

So $ax + by + cz = d$ represents a plane $\mathcal{P}_2$ which is parallel to $\mathcal{P}_1$ and can be obtained by translating (or shifting) $\mathcal{P}_1$ by $\mathbf{v}$.



6.6.2 Column Space

Definition 6.6.1 (Column Space of a Matrix). Let A be an $m \times n$ matrix. The **column space** of A, written $\mathcal{C}(A)$, is the subspace of $\mathbb{R}^m$ spanned by the columns of A. If the columns of A are denoted by $\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_n$, then

$$\mathcal{C}(A) = \text{Span}\{\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_n\}.$$

Example 6.6.5. Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Then the column space of A is the set of all vectors of the form

$$c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

Thus $\mathcal{C}(A) = \mathbb{R}^2$.

Theorem 6.6.3 (Column Spaces and Consistent Systems). *Suppose A is an $m \times n$ matrix and $\mathbf{b}$ is a vector in $\mathbb{R}^m$. Then $\mathbf{b} \in \mathcal{C}(A)$ if and only if $A\mathbf{x} = \mathbf{b}$ is consistent.*

Proof. $\mathbf{b} \in \mathcal{C}(A)$ if and only if $\mathbf{b}$ can be expressed as a linear combination of the columns of A, i.e.,

$$\mathbf{b} = c_1\mathbf{A}_1 + c_2\mathbf{A}_2 + \cdots + c_n\mathbf{A}_n$$

for some scalars $c_1, c_2, \ldots, c_n$, which is just equivalent to that $A\mathbf{x} = \mathbf{b}$ is consistent. □

Remark 6.6.3. An alternative definition of the column space of an $m \times n$ matrix A is

$$\mathcal{C}(A) = \{\mathbf{y} \in \mathbb{R}^m \,|\, \mathbf{y} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\} = \{A\mathbf{x} \,|\, \mathbf{x} \in \mathbb{R}^n\}$$

Example 6.6.6. Let

$$A = \begin{bmatrix} 3 & 2 & 1 & -4 \\ -1 & 1 & -2 & 3 \\ 2 & -4 & 6 & -8 \end{bmatrix}$$

Determine if $\mathbf{v} = \begin{bmatrix} 18 \\ -6 \\ 12 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$ are in the column space of A.

Solution.

(i) The augmented matrix of $A\mathbf{x} = \mathbf{v}$ is

$$\begin{bmatrix} 3 & 2 & 1 & -4 & 18 \\ -1 & 1 & -2 & 3 & -6 \\ 2 & -4 & 6 & -8 & 12 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \boxed{1} & 0 & 1 & -2 & 6 \\ 0 & \boxed{1} & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The system is consistent and hence $v \in \mathcal{C}(A)$. In fact, $\mathbf{x} = \begin{bmatrix} 6 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ is one solution, so we have

$$\mathbf{v} = 6\mathbf{A}_1.$$

(ii) The augmented matrix of $A\mathbf{x} = \mathbf{w}$ is

$$\begin{bmatrix} 3 & 2 & 1 & -4 & 2 \\ -1 & 1 & -2 & 3 & 1 \\ 2 & -4 & 6 & -8 & -3 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \boxed{1} & 0 & 1 & -2 & 0 \\ 0 & \boxed{1} & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

The system is inconsistent and therefore $\mathbf{w} \notin \mathcal{C}(A)$. $\square$

I. Describe the column space of A as a null space of B

Question: Can we give explicit conditions on a vector $\mathbf{v}$ such that $\mathbf{v} \in \mathcal{C}(A)$?

Example 6.6.7. Let

$$A = \begin{bmatrix} 1 & 2 & 7 & 1 & -1 \\ 1 & 1 & 3 & 1 & 0 \\ 3 & 2 & 5 & -1 & 9 \\ 1 & -1 & -5 & 2 & 0 \end{bmatrix}.$$

$\mathbf{v} \in \mathcal{C}(A)$ if and only if $A\mathbf{x} = \mathbf{v}$ is consistent.

The augmented matrix

$$[A|\mathbf{v}] = \left[\begin{array}{ccccc|c} 1 & 2 & 7 & 1 & -1 & v_1 \\ 1 & 1 & 3 & 1 & 0 & v_2 \\ 3 & 2 & 5 & -1 & 9 & v_3 \\ 1 & -1 & -5 & 2 & 0 & v_4 \end{array}\right],$$

$$\xrightarrow[-1R_1+R_4]{-1R_1+R_2,-3R_1+R_3} \left[\begin{array}{ccccc|c} 1 & 2 & 7 & 1 & -1 & v_1 \\ 0 & -1 & -4 & 0 & 1 & v_2 - v_1 \\ 0 & -4 & -16 & -4 & 12 & v_3 - 3v_1 \\ 0 & -3 & -12 & 1 & 1 & v_4 - v_1 \end{array}\right]$$

$$\xrightarrow{-1R_2} \left[\begin{array}{ccccc|c} 1 & 2 & 7 & 1 & -1 & v_1 \\ 0 & 1 & 4 & 0 & -1 & v1 - v_2 \\ 0 & -4 & -16 & -4 & 12 & v_3 - 3v_1 \\ 0 & -3 & -12 & 1 & 1 & v_4 - v_1 \end{array}\right]$$

$$\xrightarrow[3R_2+R_4]{-2R_2+R_1,4R_2+R_3} \left[\begin{array}{ccccc|c} 1 & 0 & -1 & 1 & -1 & -v_1 + 2v_2 \\ 0 & 1 & 4 & 0 & -1 & v_1 - v_2 \\ 0 & 0 & 0 & -4 & 8 & v_1 - 4v_2 + v_3 \\ 0 & 0 & 0 & 1 & -2 & 2v_1 - 3v_2 + v_4 \end{array}\right]$$

$$\xrightarrow{R_3 \leftrightarrow R_4} \left[\begin{array}{ccccc|c} 1 & 0 & -1 & 1 & -1 & -v_1 + 2v_2 \\ 0 & 1 & 4 & 0 & -1 & v_1 - v_2 \\ 0 & 0 & 0 & 1 & -2 & 2v_1 - 3v_2 + v_4 \\ 0 & 0 & 0 & -4 & 8 & v_1 - 4v_2 + v_3 \end{array}\right]$$

$$\xrightarrow[4R_3+R_4]{-1R_3+R_1} \left[\begin{array}{ccccc|c} 1 & 0 & -1 & 0 & 3 & -3v_1 + 5v_2 - v_4 \\ 0 & 1 & 4 & 0 & -1 & v_1 - v_2 \\ 0 & 0 & 0 & 1 & -2 & 2v_1 - 3v_2 + v_4 \\ 0 & 0 & 0 & 0 & 0 & 9v_1 - 16v_2 + v_3 + 4v_4 \end{array}\right].$$

The system is consistent if and only if

$$9v_1 - 16v_2 + v_3 + 4v_4 = 0.$$

We therefore conclude that $\mathbf{v} \in \mathcal{C}(A)$ if and only if $9v_1 - 16v_2 + v_3 + 4v_4 = 0$. Let $B = \begin{bmatrix} 9 & -16 & 1 & 4 \end{bmatrix}$, then

$$\mathcal{C}(A) = \mathcal{N}(B)\,.$$

Example 6.6.8.

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 & 3 & 2 & 4 \\ 0 & 0 & 0 & 1 & 0 & -1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 5 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 2 & 4 & 1 & 0 & 5 & 3 & 7 \end{bmatrix}.$$

The augmented matrix of $A\mathbf{x} = \mathbf{v}$ is

$$[A|\mathbf{v}] = \left[\begin{array}{ccccccc|c} 1 & 2 & 1 & 0 & 3 & 2 & 4 & v_1 \\ 0 & 0 & 0 & 1 & 0 & -1 & 2 & v_2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 5 & v_3 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & v_4 \\ 2 & 4 & 1 & 0 & 5 & 3 & 7 & v_5 \end{array}\right]$$

$$\xrightarrow[-2R_1+R_5]{-1R_1+R_3} \left[\begin{array}{ccccccc|c} 1 & 2 & 1 & 0 & 3 & 2 & 4 & v_1 \\ 0 & 0 & 0 & 1 & 0 & -1 & 2 & v_2 \\ 0 & 0 & -1 & 1 & -1 & -2 & 1 & v_3 - v_1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & v_4 \\ 0 & 0 & -1 & 0 & -1 & -1 & -1 & v_5 - 2v_1 \end{array}\right]$$

$$\xrightarrow{R_3 \leftrightarrow R_4} \left[\begin{array}{ccccccc|c} 1 & 2 & 1 & 0 & 3 & 2 & 4 & v_1 \\ 0 & 0 & 0 & 1 & 0 & -1 & 2 & v_2 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & v_4 \\ 0 & 0 & -1 & 1 & -1 & -2 & 1 & v_3 - v_1 \\ 0 & 0 & -1 & 0 & -1 & -1 & -1 & v_5 - 2v_1 \end{array}\right]$$

$$\xrightarrow[1R_3+R_5]{-1R_3+R_1, 1R_3+R_4} \left[\begin{array}{ccccccc|c} 1 & 2 & 0 & 0 & 2 & 1 & 3 & v_1 - v_4 \\ 0 & 0 & 0 & 1 & 0 & -1 & 2 & v_2 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & v_4 \\ 0 & 0 & 0 & 1 & 0 & -1 & 2 & -v_1 + v_3 + v_4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2v_1 + v_4 + v_5 \end{array}\right]$$

$$\xrightarrow[R_2 \leftrightarrow R_4]{-1R_2+R_4} \left[\begin{array}{ccccccc|c} 1 & 2 & 0 & 0 & 2 & 1 & 3 & v_1 - v_4 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & v_4 \\ 0 & 0 & 0 & 1 & 0 & -1 & 2 & v_2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -v_1 - v_2 + v_3 + v_4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2v_1 + v_4 + v_5 \end{array}\right].$$

The system is consistent if and only if

$$-v_1 - v_2 + v_3 + v_4 = 0$$
$$-2v_1 + v_4 + v_5 = 0$$

Let B be the coefficient matrix

$$B = \begin{bmatrix} -1 & -1 & 1 & 1 & 0 \\ -2 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

Then $\mathbf{v} \in \mathcal{C}(A)$ if and only if $\mathbf{v} \in \mathcal{N}(B)$, i.e., $\mathcal{C}(A) = \mathcal{N}(B)$.

$\boxed{\textbf{II. Basis and dimension of } \mathcal{C}(A)}$

Example 6.6.9. Let

$$A = \begin{bmatrix} 1 & 4 & 0 & -1 & 0 & 7 \\ 2 & 8 & -1 & 3 & 9 & -13 \\ 0 & 0 & 2 & -3 & -4 & 12 \\ -1 & -4 & 2 & 4 & 8 & -31 \end{bmatrix},$$

A has 6 columns $\{\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_6\}$. How to select a minimal spanning set (basis) of $\mathcal{C}(A)$ from $\{\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_6\}$?

$$A \xrightarrow{\text{RREF}} B = \begin{bmatrix} \boxed{1} & 4 & 0 & 0 & 2 & 1 \\ 0 & 0 & \boxed{1} & 0 & 1 & -3 \\ 0 & 0 & 0 & \boxed{1} & 2 & -6 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The indexes of the pivot columns of B are $\{1, 3, 4\}$. It is easy to check that

(i) the pivot columns $\mathbf{B}_1, \mathbf{B}_3, \mathbf{B}_4$ are linearly independent (why?);

(ii) the non-pivot columns $\mathbf{B}_2, \mathbf{B}_5, \mathbf{B}_6$ can be spanned by $\mathbf{B}_1, \mathbf{B}_3, \mathbf{B}_4$ with

$$\mathbf{B}_2 = 4\mathbf{B}_1$$

$$\mathbf{B}_5 = 2\mathbf{B}_1 + \mathbf{B}_3 + 2\mathbf{B}_4$$

$$\mathbf{B}_6 = \mathbf{B}_1 - 3\mathbf{B}_3 + 6\mathbf{B}_4$$

We claim that:

(i) the corresponding columns $\mathbf{A}_1, \mathbf{A}_3, \mathbf{A}_4$ are also linearly independent;

(ii) the other columns $\mathbf{A}_2, \mathbf{A}_5, \mathbf{A}_6$ can also be spanned by $\mathbf{A}_1, \mathbf{A}_3, \mathbf{A}_4$ with

$$\mathbf{A}_2 = 4\mathbf{A}_1$$

$$\mathbf{A}_5 = 2\mathbf{A}_1 + \mathbf{A}_3 + 2\mathbf{A}_4$$

$$\mathbf{A}_6 = \mathbf{A}_1 - 3\mathbf{A}_3 + 6\mathbf{A}_4$$

Why? Why does the linear relations of columns in A and the linear relations of columns in B are preserved under elementary row operations? Let's explain it using two approaches.

Approach I: from the point of linear system. We know that $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solutions, i.e.,

$$x_1\mathbf{A}_1 + x_2\mathbf{A}_2 + \cdots + x_6\mathbf{A}_6 = \mathbf{0}$$

if and only if

$$x_1\mathbf{B}_1 + x_2\mathbf{B}_2 + \cdots + x_6\mathbf{B}_6 = \mathbf{0}.$$

We first show $\mathbf{A}_1, \mathbf{A}_3, \mathbf{A}_4$ are linearly independent. Suppose

$$c_1\mathbf{A}_1 + c_3\mathbf{A}_3 + c_4\mathbf{A}_4 = \mathbf{0},$$

for some scalars c_1, c_3, c_4, then $(c_1, 0, c_3, c_4, 0, 0)$ is a solution to $A\mathbf{x} = \mathbf{0}$, i.e.,

$$c_1\mathbf{A}_1 + 0\mathbf{A}_2 + c_3\mathbf{A}_3 + c_4\mathbf{A}_4 + 0\mathbf{A}_5 + 0\mathbf{A}_6 = \mathbf{0},$$

Therefore, $(c_1, 0, c_3, c_4, 0, 0)$ is also a solution to $B\mathbf{x} = \mathbf{0}$, i.e.,

$$c_1\mathbf{B}_1 + 0\mathbf{B}_2 + c_3\mathbf{B}_3 + c_4\mathbf{B}_4 + 0\mathbf{B}_5 + 0\mathbf{B}_6 = \mathbf{0},$$

so we obtain

$$c_1\mathbf{B}_1 + c_3\mathbf{B}_3 + c_4\mathbf{B}_4 = \mathbf{0}.$$

It follows from the linear independence of $\mathbf{B}_1, \mathbf{B}_3, \mathbf{B}_4$ that c_1, c_3, c_4 must be all zero. Hence, $\mathbf{A}_1, \mathbf{A}_3, \mathbf{A}_4$ are linearly independent.

Next we show $\mathbf{A}_2, \mathbf{A}_5, \mathbf{A}_6$ can be spanned by $\mathbf{A}_1, \mathbf{A}_3, \mathbf{A}_4$, in the same way as $\mathbf{B}_2, \mathbf{B}_5, \mathbf{B}_6$ are spanned by $\mathbf{B}_1, \mathbf{B}_3, \mathbf{B}_4$.

For example, since

$$\mathbf{B}_5 = 2\mathbf{B}_1 + \mathbf{B}_3 + 2\mathbf{B}_4,$$

$(2, 0, 1, 2, -1, 0)$ is a solution to $B\mathbf{x} = \mathbf{0}$ as well as $A\mathbf{x} = \mathbf{0}$, then we have the same relation

$$\mathbf{A}_5 = 2\mathbf{A}_1 + \mathbf{A}_3 + 2\mathbf{A}_4.$$

Approach II: using elementary matrix. We know that there exists a sequence of elementary matrices $E_1, E_2, \cdots, E_k$ such that

$$A = E_k E_{k-1} \cdots E_1 B.$$

Denote $P = E_k E_{k-1} \cdots E_1$. Then P is nonsingular, and $A = PB$, $\mathbf{A}_i = P\mathbf{B}_i$ for all i.

We first show $\mathbf{A}_1, \mathbf{A}_3, \mathbf{A}_4$ are linearly independent. Suppose

$$c_1\mathbf{A}_1 + c_3\mathbf{A}_3 + c_4\mathbf{A}_4 = \mathbf{0},$$

for some scalars c_1, c_3, c_4, then

$$\begin{aligned}
c_1 P\mathbf{B}_1 + c_3 P\mathbf{B}_3 + c_4 P\mathbf{B}_4 &= \mathbf{0} \\
P(c_1\mathbf{B}_1 + c_3\mathbf{B}_3 + c_4\mathbf{B}_4) &= \mathbf{0} \\
c_1\mathbf{B}_1 + c_3\mathbf{B}_3 + c_4\mathbf{B}_4 &= \mathbf{0} \quad (\text{ since } P \text{ is invertible })\\
c_1 = c_3 = c_4 &= 0 \quad (\text{ since } \mathbf{B}_1, \mathbf{B}_3, \mathbf{B}_4 \text{ are linearly independent })
\end{aligned}$$

Therefore, $\mathbf{A}_1, \mathbf{A}_3, \mathbf{A}_4$ are linearly independent.

Next, by multiplying P on the left to

$$\begin{aligned}
\mathbf{B}_2 &= 4\mathbf{B}_1 \\
\mathbf{B}_5 &= 2\mathbf{B}_1 + \mathbf{B}_3 + 2\mathbf{B}_4 \\
\mathbf{B}_6 &= \mathbf{B}_1 - 3\mathbf{B}_3 + 6\mathbf{B}_4
\end{aligned}$$

we immediately obtain the same linear relations for columns of A.

Hence, we have proved our claim. As a consequence, a basis of $\mathcal{C}(A)$ can be chosen as

$$\{\mathbf{A}_1, \mathbf{A}_3, \mathbf{A}_4\} = \{\begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ -3 \\ 4 \end{bmatrix}\}$$

The second approach can be generalized to the following theorem:

Theorem 6.6.4. *Let* $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k$ *be vectors in* $\mathbb{R}^m$*, and* P *be an* $m \times m$ *nonsingular matrix.*

(a) If $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k$ *are linear independent, then* $P\mathbf{u}_1, P\mathbf{u}_2, \cdots, P\mathbf{u}_k$ *are also linear independent.*

(b) If $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k$ *are linear dependent with a linear relation, say*

$$\mathbf{u}_k = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_k\mathbf{u}_{k-1},$$

then $P\mathbf{u}_1, P\mathbf{u}_2, \cdots, P\mathbf{u}_k$ *are also linear dependent with the same linear relation*

$$P\mathbf{u}_k = c_1P\mathbf{u}_1 + c_2P\mathbf{u}_2 + \cdots + c_kP\mathbf{u}_{k-1}.$$

Proof. Exercise. □

The above example motivates the next fundamental theorem.

Theorem 6.6.5 (Basis and Dimension of Column Space). *Let* A *be an* $m \times n$ *matrix with columns* $\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_n$*, and* B *be the RREF of* A*. Suppose* B *has* r *pivot columns with indices* $\{d_1, d_2, \ldots, d_r\}$*. Then*

(a) $\{\mathbf{A}_{d_1}, \mathbf{A}_{d_2}, \ldots, \mathbf{A}_{d_r}\}$ *is a basis of* $\mathcal{C}(A)$*.*

(b) $\dim \mathcal{C}(A) = r$*.*

Remark 6.6.4. For a matrix A,

the number of pivot columns of its RREF is called the rank of A.

The dimension of column space $\mathcal{C}(A)$ is called the column rank of A.

As we will define later, the dimension of row space $\mathcal{R}(A)$ is called the row rank of A.

They are all equal, and are just called "rank of A" in some textbooks.

Example 6.6.10.

$$A = \begin{bmatrix} 1 & 2 & 2 & 7 & 0 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 4 & -1 & -1 & 5 \\ 1 & 2 & 1 & 4 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \boxed{1} & 2 & 0 & 1 & 2 \\ 0 & 0 & \boxed{1} & 3 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 1 and 3 are the pivot columns. So a basis of $\mathcal{C}(A)$ can be chosen as

$$\{\mathbf{A}_1, \mathbf{A}_3\} = \{ \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \end{bmatrix} \}.$$

Example 6.6.11. Find the dimension and basis of the subspace spanned by

$$\mathbf{u}_1 = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 2 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} -8 \\ -1 \\ -9 \\ -4 \end{bmatrix}, \mathbf{u}_4 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ -2 \end{bmatrix}, \mathbf{u}_5 = \begin{bmatrix} -10 \\ -1 \\ -1 \\ 4 \end{bmatrix}.$$

Solution. The subspace $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_5\}$ is the same as the column space of

$$A = \begin{bmatrix} 2 & -1 & -8 & 3 & -10 \\ 1 & 1 & -1 & 1 & -1 \\ 3 & 0 & -9 & -1 & -1 \\ 2 & 1 & -4 & -2 & 4 \end{bmatrix}$$

The RREF of A is

$$\begin{bmatrix} \boxed{1} & 0 & -3 & 0 & -1 \\ 0 & \boxed{1} & 2 & 0 & 2 \\ 0 & 0 & 0 & \boxed{1} & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 1, 2 and 4 are the pivot columns. So

$$\{ \begin{bmatrix} 2 \\ 1 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -1 \\ -2 \end{bmatrix} \}$$

is the basis of $\mathcal{C}(A)$, and the dimension is 3. $\square$

6.6.3 Row Space of a Matrix

Definition 6.6.2 (Row Space of a Matrix). Let A be an $m \times n$ matrix. The **row space** of A, written $\mathcal{R}(A)$, is the subspace of $\mathbb{R}^n$ spanned by the rows of A. If the rows of A are denoted by $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$, then

$$\mathcal{R}(A) = \text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}.$$

Example 6.6.12. Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Then the row space of A is the set of all vectors of the form

$$c_1 \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & 0 \end{bmatrix}.$$

Thus $\mathcal{R}(A)$ is the x_1x_2-plane in $\mathbb{R}^3$.

Theorem 6.6.6 (Row-Equivalent Matrices Have the Same Row Spaces). *Suppose A and B are row-equivalent matrices. Then $\mathcal{R}(A) = \mathcal{R}(B)$.*

Proof.

If B is row equivalent to A, then B can be formed from A by a finite sequence of row operations. Thus, each row vector of B must be a linear combinations of the row vectors of A. Consequently, the row space of B must be a subspace of the row space of A (why?).

Since A is row equivalent to B, by the same reasoning, the row space of A is a subspace of the row space of B. Therefore, $\mathcal{R}(A) = \mathcal{R}(B)$. □

Suppose a matrix A is row-equivalent to its RREF B, then A and B has the same row spaces. As shown in the next example, it is very easy to pick the basis of row space from B.

Example 6.6.13. Same matrix A as in Example 6.6.9

$$A = \begin{bmatrix} 1 & 4 & 0 & -1 & 0 & 7 \\ 2 & 8 & -1 & 3 & 9 & -13 \\ 0 & 0 & 2 & -3 & -4 & 12 \\ -1 & -4 & 2 & 4 & 8 & -31 \end{bmatrix} \xrightarrow{\text{RREF}} B = \begin{bmatrix} \boxed{1} & 4 & 0 & 0 & 2 & 1 \\ 0 & 0 & \boxed{1} & 0 & 1 & -3 \\ 0 & 0 & 0 & \boxed{1} & 2 & -6 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The nonzero rows of B, including row 1, 2, 3, are linearly independent (why?), and can be chosen as the basis of $\mathcal{R}(B)$. Therefore, the three row vectors

$$\begin{bmatrix} 1 & 4 & 0 & 0 & 2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 & 1 & -3 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 2 & -6 \end{bmatrix}$$

form a basis of $\mathcal{R}(A)$.

Theorem 6.6.7 (Basis and Dimension of Row Space). *Let A be an $m \times n$ matrix, and B be the RREF of A with rows $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m$. Suppose B has r leading 1's. Then*

(a) $\{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_r\}$ *is a basis of* $\mathcal{R}(A)$.

(b) $\dim \mathcal{R}(A) = r$.

Example 6.6.14. Same matrix A as in Example 6.6.10

$$A = \begin{bmatrix} 1 & 2 & 2 & 7 & 0 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 4 & -1 & -1 & 5 \\ 1 & 2 & 1 & 4 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} B = \begin{bmatrix} \boxed{1} & 2 & 0 & 1 & 2 \\ 0 & 0 & \boxed{1} & 3 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Since B has 2 nonzero rows, so

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 3 & -1 \end{bmatrix}$$

form a basis of $\mathcal{R}(A)$.

I. Find basis of $\mathcal{C}(A)$ and $\mathcal{R}(A)$ by row operations on A:

Step 1. Row reduce A to its RREF B. Suppose the columns $d_1, d_2, \ldots, d_r$ of B are

pivot columns.

Step 2. Pick the corresponding columns $d_1, d_2, \ldots, d_r$ of A to form the basis of $\mathcal{C}(A)$.

Step 3. Pick the r nonzero rows of B to form the basis of $\mathcal{R}(A)$.

Remark 6.6.5. Note the difference: the basis of $\mathcal{C}(A)$ is chosen from the original columns of A, while the basis of $\mathcal{R}(A)$ is chosen from rows of B with pattern of zeros and ones. Elementary row operations preserve the row space and linear relation of columns, but it does not preserve column space or linear relation of rows.

Can we also find a basis of $\mathcal{R}(A)$ from its original rows, and find a basis of $\mathcal{C}(A)$ from its RREF with pattern of zeros and ones? Yes, consider A^t and note that

$$\boxed{\mathcal{C}(A)=\mathcal{R}(A^t),\ \mathcal{R}(A)=\mathcal{C}(A^t)}$$

if we do not distinguish row vector and column vector in Euclidean space.

Example 6.6.15. Compare this example with Example 6.6.9 and 6.6.13.

$$A = \begin{bmatrix} 1 & 4 & 0 & -1 & 0 & 7 \\ 2 & 8 & -1 & 3 & 9 & -13 \\ 0 & 0 & 2 & -3 & -4 & 12 \\ -1 & -4 & 2 & 4 & 8 & -31 \end{bmatrix}.$$

$$A^t = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 4 & 8 & 0 & -4 \\ 0 & -1 & 2 & 2 \\ -1 & 3 & -3 & 4 \\ 0 & 9 & -4 & 8 \\ 7 & -13 & 12 & -31 \end{bmatrix} \xrightarrow{\text{RREF}} C = \begin{bmatrix} \boxed{1} & 0 & 0 & -\frac{31}{7} \\ 0 & \boxed{1} & 0 & \frac{12}{7} \\ 0 & 0 & \boxed{1} & \frac{13}{7} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(i) Basis of $\mathcal{C}(A^t)$:

$$\begin{bmatrix} 1 \\ 4 \\ 0 \\ -1 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} 2 \\ 8 \\ -1 \\ 3 \\ 9 \\ -13 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 2 \\ -3 \\ -4 \\ 12 \end{bmatrix}$$

Therefore, the basis of $\mathcal{R}(A)$ is the same as the above. If we prefer to express them into row vectors, then just transpose them.

Basis of $\mathcal{R}(A)$:

$$\begin{bmatrix} 1 & 4 & 0 & -1 & 0 & 7 \end{bmatrix}, \begin{bmatrix} 2 & 8 & -1 & 3 & 9 & -13 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 2 & -3 & -4 & 12 \end{bmatrix}$$

(ii) Basis of $\mathcal{R}(A^t)$:

$$\begin{bmatrix} 1 & 0 & 0 & -\frac{31}{7} \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & \frac{12}{7} \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & \frac{13}{7} \end{bmatrix}$$

Therefore, the basis of $\mathcal{C}(A)$ is the same as the above. If we prefer to express them into column vectors, then just transpose them.

Basis of $\mathcal{C}(A)$:

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ -\frac{31}{7} \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ \frac{12}{7} \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ \frac{13}{7} \end{bmatrix}$$

II. Find basis of $\mathcal{C}(A)$ and $\mathcal{R}(A)$ by row operations on A^t:

Step 1. Row reduce A^t to its RREF C. Suppose the columns $d_1, d_2, \ldots, d_r$ of C are pivot columns.

Step 2. Pick the corresponding columns $d_1, d_2, \ldots, d_r$ of A^t to form the basis of $\mathcal{C}(A^t) = \mathcal{R}(A)$.

Step 3. Pick the r nonzero rows of C to form the basis of $\mathcal{R}(A^t) = \mathcal{C}(A)$.

Example 6.6.16. Let $A = \begin{bmatrix} 1 & 2 & 2 & 7 & 0 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 4 & -1 & -1 & 5 \\ 1 & 2 & 1 & 4 & 1 \end{bmatrix}$.

(a) Find the basis of $\mathcal{C}(A)$ using two methods.

(b) Determine if $\mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}$ is in $\mathcal{C}(A)$.

Solution.

(a) **Method 1**:

$$A = \begin{bmatrix} 1 & 2 & 2 & 7 & 0 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 4 & -1 & -1 & 5 \\ 1 & 2 & 1 & 4 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} B = \begin{bmatrix} \boxed{1} & 2 & 0 & 1 & 2 \\ 0 & 0 & \boxed{1} & 3 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 1 and 3 of B are the pivot columns. So a basis of $\mathcal{C}(A)$ can be chosen as

$$\{\mathbf{A}_1, \mathbf{A}_3\} = \{ \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \end{bmatrix} \}.$$

Method 2:

$$A^t = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 2 & 2 & 4 & 2 \\ 2 & 0 & -1 & 1 \\ 7 & 1 & -1 & 4 \\ 0 & 2 & 5 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} C = \begin{bmatrix} \boxed{1} & 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & \boxed{1} & \frac{5}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Rows 1 and 3 of C form a basis of $\mathcal{R}(A^t)$. Therefore a basis of $\mathcal{C}(A)$ can be chosen as

$$\{\begin{bmatrix} 1 \\ 0 \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ \frac{5}{2} \\ \frac{1}{2} \end{bmatrix}\}.$$

(b) It is easy to check it using the basis obtained in Method 2. Suppose

$$\mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ \frac{5}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ -\frac{1}{2}c_1 + \frac{5}{2}c_2 \\ \frac{1}{2}c_1 + \frac{1}{2}c_2 \end{bmatrix}.$$

From the first two equations, we get $c_1 = 2, c_2 = 3$. However, the last two equations do not hold. So $\mathbf{b} \notin \mathcal{C}(A)$.

$\square$

6.6.4 Rank and Nullity

Definition 6.6.3 (Nullity of a matrix). Let A be an $m \times n$ matrix. Then the **nullity** of A is the dimension of the null space of A,

$$n(A) = \dim(\mathcal{N}(A)).$$

If the RREF of A has r pivot columns, then

$$n(A) = n - r.$$

Definition 6.6.4 (Rank of a matrix). Let A be an $m \times n$ matrix. Then the **rank** of A is the dimension of the column/row space of A,

$$r(A) = \dim(\mathcal{C}(A)) = \dim(\mathcal{R}(A)).$$

If the RREF of A has r pivot columns, then

$$r(A) = r.$$

Theorem 6.6.8 (Dimension formula). *Let A be an $m \times n$ matrix. Then*

$$r(A) + n(A) = n.$$

Let us take a look at two extreme cases:

Case 1: $r(A) = 0, n(A) = n$, then $\mathcal{C}(A) = \{\mathbf{0}\}$ and $\mathcal{N}(A) = \mathbb{R}^n$, so

$$A = \mathcal{O}_{m \times n}.$$

Case 2: $r(A) = n, n(A) = 0$, then $\mathcal{C}(A) = \mathbb{R}^n$ and $\mathcal{N}(A) = \{\mathbf{0}\}$. In this case, if A is an $n \times n$ square matrix, then

$$A \text{ is nonsingular.}$$

We can update our list of equivalences about nonsingular matrices.

Theorem 6.6.9 (Nonsingular Matrix Equivalences). *Let A be an $n \times n$ square matrix. The following statements are equivalent.*

1. *A is nonsingular.*
2. *A is row equivalent to I_n.*
3. *The null space of A contains only the zero vector, $\mathcal{N}(A) = \{\mathbf{0}\}$.*
4. *The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every choice of $\mathbf{b}$.*
5. *A is invertible.*
6. *There exists B such that $AB = I_n$.*
7. *There exists B such that $BA = I_n$.*
8. *The n columns of A are linearly independent.*
9. *The n columns of A span $\mathbb{R}^n$, $\mathcal{C}(A) = \mathbb{R}^n$.*
10. *The n columns of A form a basis of $\mathbb{R}^n$.*
11. *The rank of A is n, $r(A) = n$.*
12. *The nullity of A is zero, $n(A) = 0$.*

Proof. All the equivalences can be proved by definition. Exercise. □

7 Determinant

Question: For an $n \times n$ square matrix A, is it possible to associate A with a real number which can tell us whether A is invertible/nonsingular? Yes, that number will be called the **"determinant"** of A, denoted by $\det(A)$ or $|A|$.

7.1 Determinant of a Square Matrix

1×1 **Matrices:** If $A = (a)$ is a 1×1 matrix, then A is invertible if and only if $a \neq 0$. So we define

$$\det(A) = a,$$

then A is invertible if and only if $\det(A) \neq 0$.

2×2 **Matrices:** Let

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

Then A is invertible if and only if A is row equivalent to I_2.

Case (1): if $a_{11} \neq 0$, then

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \xrightarrow[a_{11}R_2]{-\frac{a_{21}}{a_{11}}R_1+R_2} \begin{bmatrix} a_{11} & a_{12} \\ 0 & a_{11}a_{22} - a_{21}a_{12} \end{bmatrix}.$$

Then A is invertible if and only if

$$a_{11}a_{22} - a_{21}a_{12} \neq 0.$$

Case (2): if $a_{11} = 0$, then

$$\begin{bmatrix} 0 & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} a_{21} & a_{12} \\ 0 & a_{12} \end{bmatrix}.$$

Then A is invertible if and only if $a_{21}a_{12} \neq 0$, which is equivalent to $a_{11}a_{22} - a_{21}a_{12} \neq 0$ when $a_{11} = 0$. Therefore we define

$$\det(A) = a_{11}a_{22} - a_{21}a_{12},$$

then A is invertible if and only if $\det(A) \neq 0$.

Before we go to the general case, let's introduce two notations: "minor" and "cofactor". Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix},$$

(i) the minor of the a_{ij}, denoted by M_{ij}, is the determinant of the $(n-1) \times (n-1)$ matrix obtained from A by deleting the row and column containing a_{ij};

(ii) the cofactor of a_{ij}, denoted by A_{ij}, is defined by

$$A_{ij} = (-1)^{i+j} M_{ij}.$$

Using these notations, the determinant of 2×2 matrix A, can be written as

$$\det(A) = a_{11}A_{11} + a_{12}A_{12}.$$

Definition 7.1.1 (Determinant: cofactor expansion along the first row). Let A be an $n \times n$ matrix. The **determinant** of A, denoted by $\det(A)$ or $|A|$, is defined inductively as follows:

(a) If $n = 1$,

$$\det(A) = a_{11}.$$

(b) If $n > 1$,

$$\begin{aligned} \det(A) =& a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1n}A_{1n} \\ =& a_{11}M_{11} - a_{12}M_{12} + \cdots + (-1)^{1+n} a_{1n}M_{1n}. \end{aligned}$$

Remark 7.1.1. The determinant of A can be also denoted by

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

Example 7.1.1. 2×2 **determinant.**

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}A_{11} + a_{12}A_{12} = a_{11}M_{11} - a_{12}M_{12} = a_{11}a_{22} - a_{21}a_{12}.$$

Example 7.1.2. 3×3 **determinant.**

$$\begin{aligned}
\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} &= a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} \\
&= a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\
&= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \\
&= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}
\end{aligned}$$

For example,

$$\begin{aligned}
\begin{vmatrix} 3 & 2 & -1 \\ 4 & 1 & 6 \\ -3 & -1 & 2 \end{vmatrix} &= 3\begin{vmatrix} 1 & 6 \\ -1 & 2 \end{vmatrix} - 2\begin{vmatrix} 4 & 6 \\ -3 & 2 \end{vmatrix} + (-1)\begin{vmatrix} 4 & 1 \\ -3 & -1 \end{vmatrix} \\
&= 3\,(1 \times 2 - 6 \times (-1)) - 2\,(4 \times 2 - 6 \times (-3)) - (4 \times (-1) - 1 \times (-3)) \\
&= -27.
\end{aligned}$$

Example 7.1.3.

$$\begin{aligned}
\begin{vmatrix} 1 & 0 & 0 & 2 \\ 1 & 1 & 0 & 1 \\ 0 & 2 & 3 & 0 \\ 1 & 0 & 1 & 4 \end{vmatrix} &= 1\begin{vmatrix} 1 & 0 & 1 \\ 2 & 3 & 0 \\ 0 & 1 & 4 \end{vmatrix} - 0\begin{vmatrix} 1 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 1 & 4 \end{vmatrix} + 0\begin{vmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 4 \end{vmatrix} - 2\begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 3 \\ 1 & 0 & 1 \end{vmatrix} \\
&= 4.
\end{aligned}$$

7.2 Properties of Determinants

Theorem 7.2.1 (**Expansion along Any Row**). *Suppose that A is an $n \times n$ square matrix. Then for each $1 \leq i \leq n$,*

$$\begin{aligned}\det(A) =& a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in} \\ =& (-1)^{i+1}a_{i1}M_{i1} + (-1)^{i+2}a_{i2}M_{i2} + \cdots + (-1)^{i+n}a_{in}M_{in}\end{aligned}$$

which is known as **expansion** *along row i.*

Proof. Skip the proof. See Beezer, p.266. □

Theorem 7.2.2 (**Expansion along Any Column**). *Suppose that A is a an $n \times n$ square matrix. Then for each $1 \leq j \leq n$,*

$$\begin{aligned}\det(A) =& a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj} \\ =& (-1)^{1+j}a_{1j}M_{1j} + (-1)^{2+j}a_{2j}M_{2j} + \cdots + (-1)^{n+j}a_{nj}M_{nj}\end{aligned}$$

which is known as **expansion** *along column j.*

Proof. Skip the proof. See Beezer, p.268. □

Theorem 7.2.3 (**Determinant of Transpose**). *Suppose that A is an $n \times n$ square matrix. Then*

$$\det\left(A^t\right) = \det(A).$$

Proof. Skip the proof. See Beezer, p.267. □

Example 7.2.1.

$$A = \begin{bmatrix} -2 & 3 & 0 & 1 \\ 9 & -2 & 0 & 1 \\ 1 & 3 & -2 & -1 \\ 4 & 1 & 2 & 6 \end{bmatrix}$$

Expanding along column 3 yields

$$|A| = (0)(-1)^{1+3}\begin{vmatrix} 9 & -2 & 1 \\ 1 & 3 & -1 \\ 4 & 1 & 6 \end{vmatrix} + (0)(-1)^{2+3}\begin{vmatrix} -2 & 3 & 1 \\ 1 & 3 & -1 \\ 4 & 1 & 6 \end{vmatrix} +$$

$$(-2)(-1)^{3+3}\begin{vmatrix} -2 & 3 & 1 \\ 9 & -2 & 1 \\ 4 & 1 & 6 \end{vmatrix} + (2)(-1)^{4+3}\begin{vmatrix} -2 & 3 & 1 \\ 9 & -2 & 1 \\ 1 & 3 & -1 \end{vmatrix}$$

$$= 0 + 0 + (-2)(-107) + (-2)(61) = 92$$

Remark 7.2.1. Expand along a suitable row/column with many zeros to simplify computations.

Theorem 7.2.4 (Upper/Lower Triangular Determinants).

(a)

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{vmatrix} = a_{11}a_{22}\cdots a_{nn}.$$

(b)

$$\begin{vmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & 0 \\ a_{31} & a_{32} & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix} = a_{11}a_{22}\cdots a_{nn}.$$

Proof. We only prove (a).

$$\begin{aligned}
\begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{vmatrix} &= a_{11} \begin{vmatrix} a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix} && \text{expand along column 1} \\
&= a_{11}a_{22} \begin{vmatrix} a_{33} & \cdots & a_{3n} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & a_{nn} \end{vmatrix} && \text{expand along column 1} \\
&\cdots \\
&= a_{11}a_{22}\cdots a_{nn}
\end{aligned}$$

□

Theorem 7.2.5 (Determinant with Zero Row or Column). *Suppose that A is a square matrix with a zero row, or a zero column. Then*

$$\det(A) = 0.$$

Proof. Just expand along that zero row or zero column. □

Theorem 7.2.6 (Determinant with Equal Rows or Columns). *Suppose that A is a square matrix with two equal rows, or two equal columns. Then*

$$\det(A) = 0.$$

Proof. Skip the proof. See Beezer p.274. □

Theorem 7.2.7 (Determinant under Row/Column Operation). *Suppose that A and B are both $n \times n$ square matrices.*

(i) **Type I**: *If $A \xrightarrow{R_i \leftrightarrow R_j} B$, or $A \xrightarrow{C_i \leftrightarrow C_j} B$, then*

$$\det(B) = -\det(A).$$

(ii) **Type II**: *If $A \xrightarrow{\alpha R_i} B$, or $A \xrightarrow{\alpha C_i} B$ where $\alpha \in \mathbb{R}$, then*

$$\det(B) = \alpha \det(A).$$

(iii) **Type III**: *If $A \xrightarrow{\alpha R_i + R_j} B$, or $A \xrightarrow{\alpha C_i + C_j} B$, then*

$$\det(B) = \det(A).$$

Proof. Skip the proof. See Beezer p.273-275. □

The next example shows how to use row operations to convert a determinant to upper triangular form, whose value is given by the product of its diagonal entries.

Example 7.2.2.

$$\begin{aligned}
&\begin{vmatrix} 2 & 0 & 2 & 3 \\ 1 & 3 & -1 & 1 \\ -1 & 1 & -1 & 2 \\ 3 & 5 & 4 & 0 \end{vmatrix} \\
= -&\begin{vmatrix} 1 & 3 & -1 & 1 \\ 2 & 0 & 2 & 3 \\ -1 & 1 & -1 & 2 \\ 3 & 5 & 4 & 0 \end{vmatrix} \qquad (R_1 \leftrightarrow R_2) \\
= -&\begin{vmatrix} 1 & 3 & -1 & 1 \\ 0 & -6 & 4 & 1 \\ 0 & 4 & -2 & 3 \\ 0 & -4 & 7 & -3 \end{vmatrix} \qquad (-2R_1 + R_2, 1R_1 + R_3, -3R_1 + R_4)
\end{aligned}$$

$$= 2\begin{vmatrix} 1 & 3 & -1 & 1 \\ 0 & 1 & -1 & -2 \\ 0 & 4 & -2 & 3 \\ 0 & -4 & 7 & -3 \end{vmatrix} \qquad (1R_3 + R_2, -\tfrac{1}{2}R_2)$$

$$= 2\begin{vmatrix} 1 & 3 & -1 & 1 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 2 & 11 \\ 0 & 0 & 1 & -22 \end{vmatrix} \qquad (-4R_2 + R_3, -4R_2 + R_4, -1R_3 + R_4)$$

$$= -2\begin{vmatrix} 1 & 3 & -1 & 1 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & -22 \\ 0 & 0 & 0 & 55 \end{vmatrix} \qquad (-2R_4 + R_3, R_3 \leftrightarrow R_4)$$

$$= -2 \times 1 \times 1 \times 1 \times 55 = -110.$$

Theorem 7.2.8. *Suppose that A is an $n \times n$ matrix and E is an $n \times n$ elementary matrix. Then*

$$\det(EA) = \det(E)\det(A).$$

Proof.

If E is of Type I, i.e., E is obtained from I by swapping two rows, then by the previous theorem, $\det(E) = -\det(I) = -1$. Moreover, considering $B = EA$ in the previous theorem, we have

$$\det(EA) = -\det(A) = \det(E)\det(A).$$

Similarly, if E is of Type II, then $\det(E) = \alpha\det(I) = \alpha$, and

$$\det(EA) = \alpha\det(A) = \det(E)\det(A).$$

If E is of Type III, then $\det(E) = \det(I) = 1$, and

$$\det(EA) = \det(A) = \det(E)\det(A).$$

□

Theorem 7.2.9 (Use Determinant to Determine Invertibility). *Let a be an $n \times n$ matrix. Then A is invertible if and only if*

$$\det(A) \neq 0.$$

Proof. A can be reduced to its RREF B by a finite sequence of row operations. Thus,

$$B = E_k E_{k-1} \cdots E_1 = A$$

where E_i' s are all elementary matrices. It follows from Theorem 7.2.8 that

$$\det(B) = \det(E_k E_{k-1} \cdots E_1 A) = \det(E_k)\det(E_{k-1}) \cdots \det(E_1)\det(A).$$

Since the determinants of the E_i' s are all nonzero, so $\det(A) \neq 0$ if and only if $\det(B) \neq 0$.

If A is invertible, then $B = I$ and $\det(B) = 1 \neq 0$, so $\det(A) \neq 0$. If A is not invertible, then B has a zero row and $\det(B) = 0$, so $\det(A) = 0$. Therefore A is invertible if and only if $\det(A) \neq 0$. □

Theorem 7.2.10 (Determinant of Matrices Product). *Let A and B be $n \times n$ square matrices. Then*

$$\det(AB) = \det(A)\det(B).$$

Proof.

Case 1: if A is singular, then it follows from Theorem 5.2.3 that AB is also singular. So $\det(A) = 0 = \det(AB)$, and $\det(AB) = \det(A)\det(B)$ is obvious.

Case 2: if A is nonsingular, then A can be written as a product of elementary matrices $A = E_1 E_2 \cdots E_k$. Therefore

$$\begin{aligned}
\det(AB) &= \det(E_1 E_2 \cdots E_k B) \\
&= \det(E_1)\det(E_2) \cdots \det(E_k)\det(B) \\
&= \det(E_1 E_2 \cdots E_k)\det(B) \\
&= \det(A)\det(B).
\end{aligned}$$

□

Example 7.2.3. Find x such that

$$A = \begin{bmatrix} 2 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & x \\ 0 & 2 & 3 & 1 \end{bmatrix}$$

is singular.

Expand along the first column

$$\det(A) = 2 \begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & x \\ 2 & 3 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & 3 & 1 \end{vmatrix}.$$

Expand along the second row

$$\begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & x \\ 2 & 3 & 1 \end{vmatrix} = -x \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} = -x.$$

Finally

$$\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & 3 & 1 \end{vmatrix} = -1.$$

Hence

$$\det(A) = -2x - 1.$$

It is singular if and only if $\det(A) = 0$ if and only if $x = -\frac{1}{2}$.

Theorem 7.2.11. *If* $\det(A) \neq 0$*, then* A *is invertible and*

$$\det\left(A^{-1}\right) = \frac{1}{\det(A)}.$$

Proof. It follows from $A^{-1}A = I$ and

$$\det\left(A^{-1}\right)\det(A) = \det(I) = 1.$$

□

We can update our list of equivalences about nonsingular matrices.

Theorem 7.2.12 (Nonsingular Matrix Equivalences). *Let A be an $n \times n$ square matrix. The following statements are equivalent.*

1. *A is nonsingular.*
2. *A is row equivalent to I_n.*
3. *The null space of A contains only the zero vector, $\mathcal{N}(A) = \{\mathbf{0}\}$.*
4. *The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every choice of $\mathbf{b}$.*
5. *A is invertible.*
6. *There exists B such that $AB = I_n$.*
7. *There exists B such that $BA = I_n$.*
8. *The n columns of A are linearly independent.*
9. *The n columns of A span $\mathbb{R}^n$, $\mathcal{C}(A) = \mathbb{R}^n$.*
10. *The n columns of A form a basis of $\mathbb{R}^n$.*
11. *The rank of A is n, $r(A) = n$.*
12. *The nullity of A is zero, $n(A) = 0$.*
13. $\det(A) \neq 0$.

Theorem 7.2.13.

(a)

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{i-1,1} & a_{i-1,2} & \cdots & a_{i-1,n} \\ b_1 + c_1 & b_2 + c_2 & \cdots & b_n + c_n \\ a_{i+1,1} & a_{i+1,2} & \cdots & a_{i+1,n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

$$= \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{i-1,1} & a_{i-1,2} & \cdots & a_{i-1,n} \\ b_1 & b_2 & \cdots & b_n \\ a_{i+1,1} & a_{i+1,2} & \cdots & a_{i+1,n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{i-1,1} & a_{i-1,2} & \cdots & a_{i-1,n} \\ c_1 & c_2 & \cdots & c_n \\ a_{i+1,1} & a_{i+1,2} & \cdots & a_{i+1,n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

(b)

$$\begin{vmatrix} a_{11} & \cdots & a_{1,i-1} & b_1 + c_1 & a_{1,i+1} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2,i-1} & b_2 + c_2 & a_{2,i+1} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & \cdots & a_{n,i-1} & b_n + c_n & a_{n,i+1} & \cdots & a_{nn} \end{vmatrix}$$

$$= \begin{vmatrix} a_{11} & \cdots & a_{1,i-1} & b_1 & a_{1,i+1} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2,i-1} & b_2 & a_{2,i+1} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & \cdots & a_{n,i-1} & b_n & a_{n,i+1} & \cdots & a_{nn} \end{vmatrix} + \begin{vmatrix} a_{11} & \cdots & a_{1,i-1} & c_1 & a_{1,i+1} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2,i-1} & c_2 & a_{2,i+1} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & \cdots & a_{n,i-1} & c_n & a_{n,i+1} & \cdots & a_{nn} \end{vmatrix}$$

Proof. Expand along row i (or column i). $\square$

7.3 More Examples

Example 7.3.1.

$$\begin{vmatrix} 1 & a_1 & a_2 & a_3 \\ 1 & a_1+b_1 & a_2 & a_3 \\ 1 & a_1 & a_2+b_2 & a_3 \\ 1 & a_1 & a_2 & a_3+b_3 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a_1 & a_2 & a_3 \\ 0 & b_1 & 0 & 0 \\ 0 & 0 & b_2 & 0 \\ 0 & 0 & 0 & b_3 \end{vmatrix} \qquad (-1R_1+R_2, -1R_1+R_3, -1R_1+R_4)$$

$$= b_1b_2b_3.$$

Example 7.3.2.

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ 0 & b(b-a) & c(c-a) \end{vmatrix} \qquad (-aR_2+R_3)$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 0 & b-a & c-a \\ 0 & b(b-a) & c(c-a) \end{vmatrix} \qquad (-aR_1+R_2)$$

$$= \begin{vmatrix} b-a & c-a \\ b(b-a) & c(c-a) \end{vmatrix} \qquad \text{(expand along the first column)}$$

$$= (b-a)\begin{vmatrix} 1 & c-a \\ b & c(c-a) \end{vmatrix} \qquad \text{(pull out } b-a \text{ from column 1)}$$

$$= (b-a)(c-a)\begin{vmatrix} 1 & 1 \\ b & c \end{vmatrix} \qquad \text{(pull out } c-a \text{ from column 2)}$$

$$= (b-a)(c-a)(c-b).$$

Example 7.3.3.

$$\begin{vmatrix} 0 & 0 & 0 & a_{14} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}$$

$$= -\begin{vmatrix} a_{41} & a_{42} & a_{43} & a_{44} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{14} \end{vmatrix} \qquad (R_1 \leftrightarrow R_4)$$

$$= \begin{vmatrix} a_{41} & a_{42} & a_{43} & a_{44} \\ 0 & a_{32} & a_{33} & a_{34} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & 0 & 0 & a_{14} \end{vmatrix} \qquad (R_2 \leftrightarrow R_3)$$

$$= a_{41}a_{32}a_{23}a_{14}.$$

Example 7.3.4.

Given $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 1$, then

$$\begin{vmatrix} 2a_{11} & 3a_{12} & 4a_{13} \\ 2a_{21} & 3a_{22} & 4a_{23} \\ 2a_{31} & 3a_{32} & 4a_{33} \end{vmatrix} = 2 \times 3 \times 4 \times \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 24.$$

Example 7.3.5. Compute $\begin{vmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & 1 & 2 & 2 \\ 1 & 2 & 1 & 2 & 3 \\ 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 4 \end{vmatrix}$.

By applying operations $-R_1 + R_2$, $-R_1 + R_3$, $-R_1 + R_4$, $-R_1 + R_5$, it is

$$\begin{vmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{vmatrix} = (-1)^{1+3} \times 1 \times \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{vmatrix} \quad \text{(by expansion along column 3)}$$
$= 1 \times 1 \times 2 \times 3 = 6.$

Example 7.3.6. Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}.$$

$$C = \begin{bmatrix} A & \mathcal{O}_{32} \\ \mathcal{O}_{23} & B \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ 0 & 0 & 0 & b_{11} & b_{12} \\ 0 & 0 & 0 & b_{21} & b_{22} \end{bmatrix}$$

Show that

$$\det(C) = \det(A)\det(B).$$

Proof. Expand along the last row,

$$\det(C) = (-1)^{5+4} b_{21} \begin{vmatrix} a_{11} & a_{12} & a_{13} & 0 \\ a_{21} & a_{22} & a_{23} & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ 0 & 0 & 0 & b_{12} \end{vmatrix} + (-1)^{5+5} b_{22} \begin{vmatrix} a_{11} & a_{12} & a_{13} & 0 \\ a_{21} & a_{22} & a_{23} & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ 0 & 0 & 0 & b_{11} \end{vmatrix}$$

$$= (-1)^{5+4} b_{21} (-1)^{4+4} b_{12} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + (-1)^{5+5} b_{22} (-1)^{4+4} b_{11} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$= (b_{11}b_{22} - b_{21}b_{12}) \det(A) = \det(A)\det(B).$ □

Remark 7.3.1. The result is also valid when A is a square matrix of size n and B is a square matrix of size m.

Example 7.3.7. Compute $\begin{vmatrix} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{vmatrix}$.

By $-R_1 + R_2$, $-R_1 + R_3$, $-R_1 + R_4$, the above is $\begin{vmatrix} a & 1 & 1 & 1 \\ 1-a & a-1 & 0 & 0 \\ 1-a & 0 & a-1 & 0 \\ 1-a & 0 & 0 & a-1 \end{vmatrix}$.

Take out the common factor $a-1$ of row 2, row 3 and row 4, the above is

$$(a-1)^3 \begin{vmatrix} a & 1 & 1 & 1 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{vmatrix}$$

$$= (a-1)^3 \begin{vmatrix} a+3 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{vmatrix} \qquad (-1R_4 + R_1, -1R_3 + R_1, -1R_2 + R_1)$$

$$= (a+3)(a-1)^3.$$

Example 7.3.8. Compute $\begin{vmatrix} x & 1 & 1 & \cdots & 1 \\ 1 & x & 1 & \cdots & 1 \\ 1 & 1 & x & \cdots & 1 \\ \vdots & \vdots & \vdots & \vdots & 1 \\ 1 & 1 & 1 & \cdots & x \end{vmatrix}_{n\times n}$

Add $C_2, C_3, \ldots, C_n$ to C_1,

$$\begin{vmatrix} x+(n-1) & 1 & 1 & \cdots & 1 \\ x+(n-1) & x & 1 & \cdots & 1 \\ x+(n-1) & 1 & x & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & 1 \\ x+(n-1) & 1 & 1 & \cdots & x \end{vmatrix}_{n\times n} = (x+(n-1)) \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & x & 1 & \cdots & 1 \\ 1 & 1 & x & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & 1 \\ 1 & 1 & 1 & \cdots & x \end{vmatrix}_{n\times n}$$

$-C_1 + C_2$, $C_1 + C_3, \ldots, -C_1 + C_n$. The determinant is

$$(x + (n-1)) \begin{vmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & x-1 & 0 & \cdots & 0 \\ 1 & 0 & x-1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 \\ 1 & 0 & 0 & \cdots & x-1 \end{vmatrix}_{n \times n} = (x+n-1)(x-1)^{n-1}.$$

Example 7.3.9. Let B_n be a $n \times n$ matrix in the form

$$\begin{bmatrix} 1-a_1 & a_2 & 0 & \cdots & 0 & 0 \\ -1 & 1-a_2 & a_3 & \cdots & 0 & 0 \\ 0 & -1 & 1-a_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1-a_{n-1} & a_n \\ 0 & 0 & 0 & \cdots & -1 & 1-a_n \end{bmatrix}$$

1. Show that $\det(B_n) = \det(B_{n-1}) + (-1)^n (a_1 a_2 \cdots a_n)$.

2. Hence show $\det(B_n) = 1 + \sum_{i=1}^{n} (-1)^i (a_1 a_2 \cdots a_i)$.

Solution.

1. $\det(B_n) =$

$$\begin{vmatrix} 1-a_1 & a_2 & 0 & \cdots & 0 & 0 \\ -1 & 1-a_2 & a_3 & \cdots & 0 & 0 \\ 0 & -1 & 1-a_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1-a_{n-1} & a_n \\ -a_1 & 0 & 0 & \cdots & 0 & 1 \end{vmatrix} \quad (\text{add } R_1, \ldots, R_{n-1} \text{ to } R_n)$$

Expand along the last row, the above is

$$(-1)^{n+1}(-a_1)\begin{vmatrix} a_2 & 0 & \cdots & 0 & 0 \\ 1-a_2 & a_3 & \cdots & 0 & 0 \\ -1 & 1-a_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1-a_{n-1} & a_n \end{vmatrix} +$$

$$+(-1)^{n+n}\begin{vmatrix} 1-a_1 & a_2 & 0 & \cdots & 0 \\ -1 & 1-a_2 & a_3 & \cdots & 0 \\ 0 & -1 & 1-a_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1-a_{n-1} \end{vmatrix}$$

The first matrix is an lower triangular matrix, so the determinant is the product of the diagonal entries, the second matrix is B_{n-1}.

$$= (-1)^n(a_1 \cdots a_n) + \det(B_{n-1}).$$

2. We prove the result by mathematical induction:

   Step 1: The formula is valid for $n = 1$: $\det(B_1) = 1 - a_1$.

   Step 2: Suppose the formula is true for $n = k$, we want to show that the formula is true for $n = k + 1$:

$$B_{k+1} = (-1)^{k+1}(a_1 \cdots a_{k+1}) + \det(B_k)$$

$$= 1 + \sum_{i=1}^{k}(-1)^i(a_1 a_2 \cdots a_i) + (-1)^{k+1}(a_1 \cdots a_{k+1})$$

$$= 1 + \sum_{i=1}^{k+1}(-1)^i(a_1 \cdots a_i)$$

$$= 1 + \sum_{i=1}^{n}(-1)^i(a_1 \cdots a_i).$$

   The formulat is true for $n = k + 1$.

   Step 3: By mathematical induction, the formula is valid for all positive integer.

Explanation: the formula is true for $k = 1$, then it is true for $k + 1 = 2$, so true for $k + 1 = 3$, etc. Hence true for all integers. This process is called **mathematical induction**.

$\square$

Example 7.3.10. Let C_n be a $n \times n$ matrix given by

$$C_n = \left.\underbrace{\begin{bmatrix} x & a & a & \cdots & a & a \\ -a & x & a & \cdots & a & a \\ -a & -a & x & \cdots & a & a \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -a & -a & -a & \cdots & -a & x \end{bmatrix}}_{n}\right\} n$$

1. Show that $\det(C_n) = a(x+a)^{n-1} + (x-a)\det(C_{n-1})$.

2. Show that $\det(C_n) = \frac{1}{2}((x+a)^n + (x-a)^n)$.

Solution.

1. The last column can be written as

$$\begin{bmatrix} a \\ a \\ a \\ \vdots \\ x \end{bmatrix} = \begin{bmatrix} a \\ a \\ a \\ \vdots \\ a \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ x-a \end{bmatrix}.$$

Then $\det(C_n) =$

$$\left.\underbrace{\begin{vmatrix} x & a & a & \cdots & a & a \\ -a & x & a & \cdots & a & a \\ -a & -a & x & \cdots & a & a \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -a & -a & -a & \cdots & -a & a \end{vmatrix}}_{n}\right\} n + \left.\underbrace{\begin{vmatrix} x & a & a & \cdots & a & 0 \\ -a & x & a & \cdots & a & 0 \\ -a & -a & x & \cdots & a & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -a & -a & -a & \cdots & -a & x-a \end{vmatrix}}_{n}\right\} n$$

For the first determinant, pulling out a from the last column, it is equal to

$$a \left.\underbrace{\begin{vmatrix} x & a & a & \cdots & a & 1 \\ -a & x & a & \cdots & a & 1 \\ -a & -a & x & \cdots & a & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -a & -a & -a & \cdots & -a & 1 \end{vmatrix}}_{n}\right\} n$$

$$= a \left.\underbrace{\begin{vmatrix} x+a & 2a & 2a & \cdots & 2a & 0 \\ 0 & x+a & 2a & \cdots & 2a & 0 \\ 0 & 0 & x+a & \cdots & 2a & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -a & -a & -a & \cdots & -a & 1 \end{vmatrix}}_{n}\right\} n \ (-1R_n + R_1, \ldots, -1R_n + R_{n-1})$$

$$= (-1)^{n+n} a \left.\underbrace{\begin{vmatrix} x+a & 2a & 2a & \cdots & 2a \\ 0 & x+a & 2a & \cdots & 2a \\ 0 & 0 & x+a & \cdots & 2a \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x+a \end{vmatrix}}_{n-1}\right\} n-1 \ \text{(expand along the last column)}$$

$= a(x+a)^{n-1}$ (determinant of upper triangular matrix)

For the second determinant,

$$\left.\underbrace{\begin{vmatrix} x & a & a & \cdots & a & 0 \\ -a & x & a & \cdots & a & 0 \\ -a & -a & x & \cdots & a & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -a & -a & -a & \cdots & -a & x-a \end{vmatrix}}_{n}\right\} n$$

$$= (x-a)\underbrace{\begin{vmatrix} x & a & a & \cdots & a \\ -a & x & a & \cdots & a \\ -a & -a & x & \cdots & a \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -a & -a & -a & \cdots & x \end{vmatrix}}_{n-1} \Bigg\} n-1 \text{ (expand along the last column)}$$

$= (x-a)\det(C_{n-1})$.

Adding the results

$$\det(C_n) = a(x+a)^{n-1} + (x-a)\det(C_{n-1}).$$

2. We will prove the formula by induction.

   Step 1: when $n = 1$, $C_1 = [x]$, $\det(C_1) = x = \frac{1}{2}((x+a)+(x-a))$. So the formula is valid for $n = 1$.

   Step 2: Suppose the formula is valid for $n = k$, i.e.

$$\det(C_k) = \frac{1}{2}((x+a)^k + (x-a)^k).$$

   Then for $n = k+1$,

$$\det(C_{k+1}) = a(x+a)^k + (x-a)\det(C_k)$$

$$= a(x+a)^k + (x-a)\frac{1}{2}((x+a)^k + (x-a)^k)$$

$$= \frac{1}{2}(x+a)^k(2a+x-a) + \frac{1}{2}(x-a)^{k+1}$$

$$= \frac{1}{2}((x+a)^{k+1} + (x-a)^{k+1})$$

$$= \frac{1}{2}((x+a)^n + (x-a)^n).$$

   So the formula is valid for $n = k+1$.

   Step 3 By mathematical induction, the formula is valid for all integers $n \geq 1$.

$\square$

Example 7.3.11 (Vandermonde Determinant). This is the most important example of determinant. Let

$$V_n = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ a_1 & a_2 & a_3 & \cdots & a_n \\ a_1^2 & a_2^2 & a_3^2 & \cdots & a_n^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_1^{n-2} & a_2^{n-2} & a_3^{n-2} & \cdots & a_n^{n-2} \\ a_1^{n-1} & a_2^{n-1} & a_3^{n-1} & \cdots & a_n^{n-1} \end{bmatrix}$$

1. $\det(V_n) = \det(V_{n-1}) \prod_{i=1}^{n-1}(a_n - a_i)$.

2. $\det(V_n) = \prod_{1 \le i < j \le n}(a_j - a_i)$ for $n \ge 2$.

Solution.

1. $\det V_n =$

   (by $-a_nR_{n-1} + R_n$, $-a_nR_{n-2} + R_{n-1}$, $\ldots$, $-a_nR_1 + R_2$)

$$= \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ a_1 - a_n & a_2 - a_n & a_3 - a_n & \cdots & 0 \\ a_1^2 - a_1a_n & a_2^2 - a_2a_n & a_3^2 - a_3a_n & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_1^{n-2} - a_1^{n-3}a_n & a_2^{n-2} - a_2^{n-3}a_n & a_3^{n-2} - a_3^{n-3}a_n & \cdots & 0 \\ a_1^{n-1} - a_1^{n-2}a_n & a_2^{n-1} - a_2^{n-2}a_n & a_3^{n-1} - a_3^{n-2}a_n & \cdots & 0 \end{vmatrix}$$

   (expanding along the last column)

$$= (-1)^{1+n} \begin{vmatrix} a_1 - a_n & a_2 - a_n & a_3 - a_n & \cdots & a_{n-1} - a_n \\ a_1(a_1 - a_n) & a_2(a_2 - a_n) & a_3(a_3 - a_n) & \cdots & a_{n-1}(a_{n-1} - a_n) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_1^{n-3}(a_1 - a_n) & a_2^{n-3}(a_2 - a_n) & a_3^{n-3}(a_3 - a_n) & \cdots & a_{n-1}^{n-3}(a_{n-1} - a_n) \\ a_1^{n-2}(a_1 - a_n) & a_2^{n-2}(a_2 - a_n) & a_3^{n-2}(a_3 - a_n) & \cdots & a_{n-1}^{n-2}(a_{n-1} - a_n) \end{vmatrix}$$

(pull out factor $a_1 - a_n$ from column 1, $a_2 - a_n$ from column 2, ..., $a_{n-1} - a_n$ from column $n-1$)

$$= (-1)^{n-1}(a_1 - a_n)(a_2 - a_n)\cdots(a_{n-1} - a_n)\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ a_1 & a_2 & a_3 & \cdots & a_{n-1} \\ a_1^2 & a_2^2 & a_3^2 & \cdots & a_{n-1}^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_1^{n-2} & a_2^{n-2} & a_3^{n-2} & \cdots & a_{n-1}^{n-2} \end{vmatrix}$$

$$= (a_n - a_1)\cdots(a_n - a_{n-1})\det(V_{n-1}) = \det(V_{n-1})\prod_{i=1}^{n-1}(a_n - a_i).$$

2. Again by mathematical induction:

   Step 1 When $n = 2$,

$$\begin{vmatrix} 1 & 1 \\ a_1 & a_2 \end{vmatrix} = a_2 - a_1.$$

   So the formula is valid for $n = 2$.

   Step 2 Suppose the statement is true for $n = k$, i.e.

$$\det(V_k) = \prod_{1\le i<j\le k}(a_j - a_i).$$

   Then for $n = k + 1$

$$\det(V_{k+1}) = \det(V_k)\prod_{i=1}^{k}(a_{k+1} - a_i)$$

$$= \prod_{1\le i<j\le k}(a_j - a_i)\prod_{i=1}^{k}(a_{k+1} - a_i)$$

$$= \prod_{1\le i<j\le k+1}(a_j - a_i)$$

$$= \prod_{1\le 1<j\le n}(a_j - a_i).$$

   The formula is valid for $n = k + 1$.

   Step 3 By mathematical induction, the formula is valid for all $n \ge 2$.

   Or without mathematical induction, you can simple repeat the steps again and again until $n = 2$.

$\square$

7.4 Adjoint and Cramer's Rule

For an $n \times n$ matrix $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$, we define the "adjoint" of A by

$$\operatorname{adj}(A) = \begin{bmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \vdots & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{bmatrix}$$

where A_{ij} is the cofactor of a_{ij}. Thus, to form the adjoint, we must replace each term by its cofactor and then transpose the resulting matrix.

Theorem 7.4.1 (Matrix Inverse and Adjoint). *Suppose that A is an $n \times n$ square matrix. Then*

$$A \operatorname{adj}(A) = \det(A)\, I.$$

Moreover, if $\det(A) \neq 0$, then A is invertible with

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

Proof. The general ij-th entry of $A \operatorname{adj}(A)$ is

$$\begin{aligned} [A \operatorname{adj}(A)]_{ij} &= \sum_{k=1}^{n} a_{ik} [\operatorname{adj}(A)]_{kj} \\ &= \sum_{k=1}^{n} a_{ik} A_{jk} \\ &= \begin{cases} \det(A), & \text{if } i = j, \\ 0. & \text{if } i \neq j, \end{cases} \end{aligned}$$

Therefore, $A \operatorname{adj}(A) = \det(A)\, I$.

If $\det(A) \neq 0$, then $A \dfrac{1}{\det(A)} \operatorname{adj}(A) = I$, which implies $A^{-1} = \dfrac{1}{\det(A)} \operatorname{adj}(A)$. $\square$

Example 7.4.1. Let

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

and suppose

$$\det(A) = a_{11}a_{22} - a_{21}a_{12} \neq 0.$$

Then

$$\begin{aligned} \operatorname{adj}(A) &= \begin{bmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{bmatrix} \\ &= \begin{bmatrix} (-1)^{1+1}M_{11} & (-1)^{2+1}M_{21} \\ (-1)^{1+2}M_{12} & (-1)^{2+2}M_{22} \end{bmatrix} \\ &= \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}. \end{aligned}$$

Therefore,

$$A^{-1} = \frac{1}{a_{11}a_{22} - a_{21}a_{12}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}.$$

Example 7.4.2. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 1 & 1 & -1 \end{bmatrix}, \ \det(A) = 6.$$

$$M_{11} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1, \quad M_{12} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2,$$

$$M_{13} = \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1, \quad M_{21} = \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} = -5,$$

$$M_{22} = \begin{vmatrix} 1 & 3 \\ 1 & -1 \end{vmatrix} = -4, \quad M_{23} = \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = -1,$$

$$M_{31} = \begin{vmatrix} 2 & 3 \\ 0 & 1 \end{vmatrix} = 2, \quad M_{32} = \begin{vmatrix} 1 & 3 \\ 1 & 1 \end{vmatrix} = -2, \quad M_{33} = \begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} = -2.$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} M_{11} & -M_{21} & M_{31} \\ -M_{12} & M_{22} & -M_{32} \\ M_{13} & -M_{23} & M_{33} \end{bmatrix} = \begin{bmatrix} -\frac{1}{6} & \frac{5}{6} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{6} & -\frac{1}{3} \end{bmatrix}$$

Theorem 7.4.2 (Cramer's Rule). *Let A be an $n \times n$ invertible matrix and let $\mathbf{b} \in \mathbb{R}^n$. Let M_j be the matrix obtained by replacing the j-th column of A by $\mathbf{b}$. If $\mathbf{x}$ is the unique solution to $A\mathbf{x} = \mathbf{b}$, then*

$$x_j = \frac{\det(M_j)}{\det(A)}, \quad \textit{for } j = 1, \ldots, n.$$

Proof. Since

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{\det(A)} \mathrm{adj}(A)\mathbf{b},$$

we have

$$x_j = \frac{A_{1j}b_1 + A_{2j}b_2 + \cdots + A_{nj}b_n}{\det(A)} = \frac{\det(M_j)}{\det(A)}.$$

□

Example 7.4.3. Using Cramer's rule to solve the

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 2 \\ x_1 \qquad\quad + x_3 &= 3 \\ x_1 + x_2 - x_3 &= 1 \end{aligned}$$

Solution. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 1 & 1 & -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}.$$

Then

$$\det(A) = 6 \neq 0.$$

$$x_1 = \frac{1}{\det(A)} \begin{vmatrix} 2 & 2 & 3 \\ 3 & 0 & 1 \\ 1 & 1 & -1 \end{vmatrix} = \frac{15}{6} = \frac{5}{2}.$$

$$x_2 = \frac{1}{\det(A)} \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 1 \\ 1 & 1 & -1 \end{vmatrix} = \frac{-6}{6} = -1.$$

$$x_3 = \frac{1}{\det(A)} \begin{vmatrix} 1 & 2 & 2 \\ 1 & 0 & 3 \\ 1 & 1 & 1 \end{vmatrix} = \frac{3}{6} = \frac{1}{2}.$$

Thus the solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{5}{2} \\ -1 \\ \frac{1}{2} \end{bmatrix}.$$

□

8 Eigenvalues and Eigenvectors

This chapter is about eigenvalues, eigenvectors and diagonalization of square matrices. The most natural motivation comes from the study of linear transformation, which will be discussed in the second course: Linear Algebra II.

Simply speaking, given an $n \times n$ matrix A, multiplying any vector $x \in \mathbb{R}^n$ by A on the left gives another vector $A\mathbf{x}$. For example, let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix},$$

then $A\mathbf{e}_1 = 2\mathbf{e}_1$, $A\mathbf{e}_2 = 3\mathbf{e}_2$. Thus, multiplying $\mathbf{e}_1$ by A on the left results in $2\mathbf{e}_1$, and multiplying $\mathbf{e}_2$ by A on the left results in $3\mathbf{e}_2$.

Let

$$B = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix},$$

can we find some vector $\mathbf{x} \neq \mathbf{0}$, such that multiplying $\mathbf{x}$ by B on the left is equivalent to a scalar multiplication on $\mathbf{x}$, that is,

$$B\mathbf{x} = \lambda\mathbf{x}, \quad \text{for some scalar } \lambda.$$

This pair of λ and $\mathbf{x}$ are called the eigenvalue and eigenvector of B.

8.1 Eigenvalues and Eigenvectors

Definition 8.1.1 (Eigenvalues and Eigenvectors of a Matrix). Let A be an $n \times n$ matrix. A scalar λ is called an **eigenvalue** of A if there exists a nonzero vector $\mathbf{x}$ such that

$$A\mathbf{x} = \lambda\mathbf{x}.$$

This vector $\mathbf{x}$ is called an **eigenvector** of A associated with the eigenvalue λ.

Example 8.1.1.

(a) Let $A = \begin{bmatrix} 5 & -6 \\ 2 & -2 \end{bmatrix}$. We can check

$$A \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 & -6 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

So $\lambda = 2$ is an eigenvalue of A, and $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is an eigenvector associated with $\lambda = 2$.

(b) Let

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

and let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \ \mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \ \mathbf{w} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

Then

$$A\mathbf{u} = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} = 4\mathbf{u}, \ A\mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = 1\mathbf{v}, \ A\mathbf{w} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = 1\mathbf{w}.$$

$\mathbf{u}$ is an eigenvector of A associated with eigenvalue 4,

$\mathbf{v}$ is an eigenvector of A associated with eigenvalue 1,

$\mathbf{w}$ is an eigenvector A associated with eigenvalue 1.

Question: How to find the eigenvalues and corresponding eigenvectors?

Note that the equation $A\mathbf{x} = \lambda\mathbf{x}$ can be written as

$$(A - \lambda I_n)\,\mathbf{x} = \mathbf{0}. \tag{8.10}$$

Thus, λ is an eigenvalue of A if and only if (8.10) has a nontrivial solution. This happens if and only if $A - \lambda I_n$ is singular, i.e.,

$$\det(A - \lambda I_n) = 0.$$

If the above determinant is expanded, we obtain an nth degree polynomial about λ:

$$\begin{vmatrix} a_{11}-\lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22}-\lambda & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn}-\lambda \end{vmatrix} = (-1)^n\lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0.$$

Definition 8.1.2 (Characteristic Polynomial). Let A be an $n \times n$ matrix. The polynomial

$$p_A(\lambda) = \det(A - \lambda I_n)$$

is called the **characteristic polynomial** of A.

The equation $\det(A - \lambda I_n) = 0$ is called the **characteristic equation** of A.

The roots of the characteristic polynomial are the eigenvalues of A. By fundamental theorem of algebra, each nth degree polynomial has n complex roots if multiplicity is counted. That means, a characteristic polynomial may have no real roots. Therefore, it is necessary to expand the field of scalars to the complex numbers, and to allow complex entries for our vectors and matrices.

However, in this course we will focus on examples with real eigenvalues and real eigenvectors.

Theorem 8.1.1 (Equivalences on Eigenvalues). *Let A be an $n \times n$ matrix and λ be a scalar. The following statements are equivalent:*

(a) λ *is an eigenvalue of* A.

(b) $(A - \lambda I_n)\mathbf{x} = \mathbf{0}$ *has a nontrivial solution.*

(c) $\mathcal{N}(A - \lambda I_n) \neq \{\mathbf{0}\}$.

(d) $A - \lambda I_n$ *is singular.*

(e) $\det(A - \lambda I_n) = 0$.

If λ is an eigenvalue of A, the set of solutions to $(A - \lambda I_n)\mathbf{x} = \mathbf{0}$ is $\mathcal{N}(A - I_n)$, which is a subspace and called the eigenspace of A corresponding to the eigenvalue λ.

Definition 8.1.3 (Eigenspace of a Matrix). Let A be an $n \times n$ matrix and λ be an eigenvalue of A. The **eigenspace** of A for λ, denoted by $\mathcal{E}_A(\lambda)$, is the set of all eigenvectors of A with eigenvalue λ, together with the zero vector.

$$\mathcal{E}_A(\lambda) = \mathcal{N}(A - \lambda I_n).$$

Example 8.1.2. Let $A = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}$. Find the eigenvalues and the bases of the eigenspaces.

Solution. The characteristic equation is

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 2 \\ 3 & -2-\lambda \end{vmatrix} = \lambda^2 - \lambda - 12 = 0.$$

Thus, A has two eigenvalues $\lambda_1 = 4$ and $\lambda_2 = -3$.

For $\lambda_1 = 4$, solve $(A - 4I)\mathbf{x} = \mathbf{0}$.

$$\begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix} \to \begin{bmatrix} -1 & 2 \\ 0 & 0 \end{bmatrix}.$$

The eigenspace for λ_1 is of dimension 1, and the basis can be chosen as $\mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

For $\lambda_2 = -3$, solve $(A + 3I)\mathbf{x} = \mathbf{0}$.

$$\begin{bmatrix} 6 & 2 \\ 3 & 1 \end{bmatrix} \to \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix}.$$

The eigenspace for λ_2 is of dimension 1, and the basis can be chosen as $\mathbf{x}_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$. $\square$

Example 8.1.3. Not required in this course. Skip.

Let $A = \begin{bmatrix} 0 & 8 \\ -2 & 0 \end{bmatrix}$, then

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 8 \\ -2 & -\lambda \end{vmatrix} = \lambda^2 + 16 = 0.$$

Thus, A has two complex eigenvalues $\lambda_1 = 4i$ and $\lambda_2 = -4i$.

For $\lambda_1 = 4i$, solve $(A - 4iI)\mathbf{x} = \mathbf{0}$.

$$\begin{bmatrix} -4i & 8 \\ -2 & -4i \end{bmatrix} \to \begin{bmatrix} 1 & 2i \\ 0 & 0 \end{bmatrix}.$$

The eigenspace for λ_1 is of dimension 1, and the basis can be chosen as $\mathbf{x}_1 = \begin{bmatrix} -2i \\ 1 \end{bmatrix}$.

For $\lambda_2 = -4i$, solve $(A + 4iI)\mathbf{x} = \mathbf{0}$.

$$\begin{bmatrix} 4i & 8 \\ -2 & 4i \end{bmatrix} \to \begin{bmatrix} -1 & 2i \\ 0 & 0 \end{bmatrix}.$$

The eigenspace for λ_2 is of dimension 1, and the basis can be chosen as $\mathbf{x}_2 = \begin{bmatrix} 2i \\ 1 \end{bmatrix}$.

Theorem 8.1.2 (Properties of Eigenvalues).

(a) A is invertible if and only if 0 is not an eigenvalue of A.

(b) If λ is an eigenvalue of A, then $k\lambda$ is en eigenvalue of kA for any scalar k.

(c) If λ is an eigenvalue of A, then λ^k is en eigenvalue of A^k for any scalar non-negative integer k.

(d) If λ is an eigenvalue of A, then λ is an eigenvalue of A^t.

(e) If λ is an eigenvalue of A and A is invertible, then λ^{-1} is an eigenvalue of A^{-1}.

Proof. Exercise. □

Definition 8.1.4 (Algebraic Multiplicity and Geometric Multiplicity). Suppose the characteristic polynomial of A can be factorized as

$$\det(A - \lambda I_n) = (-1)^n(\lambda - \lambda_1)^{n_1}(\lambda - \lambda_2)^{n_2} \cdots (\lambda - \lambda_k)^{n_k},$$

where $\lambda_1, \cdots, \lambda_k$ are distinct roots. Then for each eigenvalue λ_i,

(a) the integer n_i is called the **algebraic multiplicity** of λ_i, denoted by $m_a(\lambda_i)$;

(b) the dimension of the eigenspace associated with λ_i is called the **geometric multiplicity** of λ_i, denoted by $m_g(\lambda_i)$.

Remark 8.1.1. In Example 8.1.2, A are two eigenvalues $\lambda_1 = 4$ and $\lambda_2 = -3$ with

$$m_a(\lambda_1) = m_g(\lambda_1) = 1, \ m_a(\lambda_2) = m_g(\lambda_2) = 1.$$

Example 8.1.4. Let

$$A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}.$$

The characteristic equation is

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 3 \\ 0 & 2-\lambda \end{vmatrix} = (\lambda - 2)^2 = 0.$$

A has a repeated eigenvalue $\lambda = 2$ with algebraic multiplicity 2.

Solving $(A - 2I)\mathbf{x} = \mathbf{0}$.

$$\begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

The eigenspace for λ is of dimension 1, and the basis can be chosen as $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, so the geometric multiplicity of λ is 1.

Example 8.1.5. Let

$$A = \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix}.$$

$$\det(A-\lambda I) = \begin{vmatrix} 2-\lambda & -3 & 1 \\ 1 & -2-\lambda & 1 \\ 1 & -3 & 2-\lambda \end{vmatrix} = -\lambda(\lambda-1)^2 = 0.$$

Thus, A has two distinct eigenvalues: $\lambda_1 = 0$ with algebraic multiplicity 1, and $\lambda_2 = 1$ with algebraic multiplicity 2.

For $\lambda_1 = 0$,

$$A - 0I = \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The basis of the eigenspace can be chosen as $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and the geometric multiplicity of λ_1 is 1.

For $\lambda_2 = 1$,

$$A - I = \begin{bmatrix} 1 & -3 & 1 \\ 1 & -3 & 1 \\ 1 & -3 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & -3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The basis of the eigenspace can be chosen as $\mathbf{x}_2 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, and the geometric multiplicity of λ_2 is 2.

Theorem 8.1.3. *Let A be an $n \times n$ matrix and λ be an eigenvalue of A, then*

$$1 \le m_g(\lambda) \le m_a(\lambda) \le n.$$

Proof. Skip. □

Example 8.1.6. Let

$$A = \begin{bmatrix} -1 & 1 & 0 \\ -4 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}.$$

$$\det(A-\lambda I) = \begin{vmatrix} -1-\lambda & 1 & 0 \\ -4 & 3-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda)(\lambda-1)^2 = 0.$$

Thus, A has two distinct eigenvalues: $\lambda_1 = 2$ with algebraic multiplicity 1, and $\lambda_2 = 1$ with algebraic multiplicity 2.

For $\lambda_1 = 2$,

$$A - 2I = \begin{bmatrix} -3 & 1 & 0 \\ -4 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The basis of the eigenspace can be chosen as $\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, and the geometric multiplicity of λ_1 is 1.

For $\lambda_2 = 1$,

$$A - I = \begin{bmatrix} -2 & 1 & 0 \\ -4 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

The basis of the eigenspace can be chosen as $\mathbf{x}_2 = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$, and the geometric multiplicity of λ_2 is 1.

8.2 Similar Matrices (may be skipped if no enough time)

Definition 8.2.1 (Similar Matrices). Let A and B be $n \times n$ matrices. A is said to be similar to B if there exists a nonsingular matrix S such that

$$S^{-1}AS = B.$$

Example 8.2.1. Let

$$A = \begin{bmatrix} 89 & 145 \\ -54 & -88 \end{bmatrix}, \quad B = \begin{bmatrix} -5 & -7 \\ 4 & 6 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}.$$

Check that S is nonsingular and

$$S^{-1}AS = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 89 & 145 \\ -54 & -88 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} = B.$$

It follows that A is similar to B.

Example 8.2.2.

$$A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} -13 & -8 & -4 \\ 12 & 7 & 4 \\ 24 & 16 & 7 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 1 & 2 \\ -2 & -1 & -3 \\ 1 & -2 & 0 \end{bmatrix}.$$

Check that S is nonsingular and

$$S^{-1}AS = \begin{bmatrix} -6 & -4 & -1 \\ -3 & -2 & -1 \\ 5 & 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ -2 & -1 & -3 \\ 1 & -2 & 0 \end{bmatrix} = B.$$

It follows that A is similar to B.

Theorem 8.2.1 (Similarity is an Equivalence Relation). *Let A, B and C be $n \times n$ matrices. Then*

(a) (Reflexive) A is similar to A.

(b) (Symmetric) If A is similar to B, then B is similar to A.

(c) (Transitive) If A is similar to B and B is similar to C, then A is similar to C.

Proof.

(a) $I_n^{-1} A I_n = A$, so by definition A is similar to A.

(b) If A is similar to B, then there exists a nonsingular matrix S such that

$$S^{-1}AS = B.$$

It follows that

$$A = SBS^{-1} = (S^{-1})^{-1}B(S^{-1}).$$

So by definition B is similar to A.

(c) If A is similar to B, then there exists a nonsingular matrix S such that

$$P^{-1}AP = B.$$

If B is similar to C, then there exists a nonsingular matrix Q such that

$$Q^{-1}BQ = C.$$

Then

$$C = Q^{-1}P^{-1}APQ = (PQ)^{-1}A(PQ).$$

So by definition A is similar to C.

□

Remark 8.2.1. By property (b), we can say A and B are similar.

Theorem 8.2.2 (Similar Matrices have Equal Eigenvalues). *Suppose A and B are similar matrices. Then the characteristic polynomials of A and B are equal, i.e., $p_A(\lambda) = p_B(\lambda)$.*

Proof. Since A and B are similar, there exists a nonsingular matrix S, such that $A = S^{-1}BS$. Then

$$\begin{aligned}
p_A(\lambda) &= \det(A - \lambda I) \\
&= \det\left(S^{-1}BS - \lambda I\right) \\
&= \det\left(S^{-1}(B - \lambda I)S\right) \\
&= \det\left(S^{-1}\right)\det(B - \lambda I)\det(S) \\
&= \det(B - \lambda I) \\
&= p_B(\lambda).
\end{aligned}$$

□

Example 8.2.3. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix}$$

Then

$$p_A(\lambda) = \begin{vmatrix} 1-\lambda & 2 \\ 3 & 4-\lambda \end{vmatrix} = (1-\lambda)(4-\lambda) - 6 = \lambda^2 - 5\lambda - 2$$

and

$$p_B(\lambda) = \begin{vmatrix} 1-\lambda & 2 \\ 0 & 4-\lambda \end{vmatrix} = (1-\lambda)(4-\lambda) = \lambda^2 - 5\lambda + 4.$$

Because $p_A(\lambda) \neq p_B(\lambda)$, we conclude that A and B are not similar.

Example 8.2.4. Same characteristic polynomial does not imply similarity. Let

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

We have

$$p_A(\lambda) = p_B(\lambda) = \lambda^2,$$

so A and B have equal characteristic polynomials. But A and B cannot be similar because for any nonsingular matrix P, $P^{-1}BP = \mathcal{O} \neq A$.

Therefore, the converse of the above theorem is false.

8.3 Diagonalization

Definition 8.3.1 (Diagonalizable Matrix). Let A be an $n \times n$ matrix. A is called **diagonalizable** if A is similar to a diagonal matrix, i.e., there exists a nonsingular matrix S and scalars $\lambda_1, \lambda_2, \ldots, \lambda_n$ such that

$$S^{-1}AS = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$$

Example 8.3.1. Let

$$B = \begin{bmatrix} -7 & -6 & -12 \\ 5 & 5 & 7 \\ 1 & 0 & 4 \end{bmatrix}, \quad S = \begin{bmatrix} -5 & -3 & -2 \\ 3 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

We can check that

$$S^{-1}BS == \begin{bmatrix} -1 & -1 & -1 \\ 2 & 3 & 1 \\ -1 & -2 & 1 \end{bmatrix} \begin{bmatrix} -7 & -6 & -12 \\ 5 & 5 & 7 \\ 1 & 0 & 4 \end{bmatrix} \begin{bmatrix} -5 & -3 & -2 \\ 3 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Hence, B is diagonalizable.

Theorem 8.3.1 (Diagonalization Characterization). *Let A be an $n \times n$ matrix. Then A is diagonalizable if and only if A has n linearly independent eigenvectors.*

Proof.

($\Rightarrow$) Suppose that A is diagonalizable. Then there exists an invertible matrix S and scalars $\lambda_1, \lambda_2, \ldots, \lambda_n$ such that

$$S^{-1}AS = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$$

Denote $S = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix}$, where $\mathbf{v}_i$ is the ith column of S. Then

$$\begin{aligned} AS =& S \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) \\ A \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix} =& \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix} \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) \\ \begin{bmatrix} A\mathbf{v}_1 & A\mathbf{v}_2 & \cdots & A\mathbf{v}_n \end{bmatrix} =& \begin{bmatrix} \lambda_1\mathbf{v}_1 & \lambda_2\mathbf{v}_2 & \cdots & \lambda_n\mathbf{v}_n \end{bmatrix} \\ A\mathbf{v}_i =& \lambda_i\mathbf{v}_i, \quad \text{for } i = 1, 2, \cdots, n. \end{aligned}$$

Noting that $\mathbf{v}_i \neq \mathbf{0}$ since S is nonsingular, the above equation implies that λ_i's are eigenvalues of A, and $\mathbf{v}_i$'s are eigenvectors associated with λ_i's. Moreover, $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$ are linearly independent since S is nonsingular. Therefore, A has n linearly independent eigenvectors.

($\Leftarrow$). Suppose A has n linearly independent eigenvectors associated with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, i.e. $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$ for $i = 1, 2, \ldots, n$. Let

$$S = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix},$$

then S is nonsingular. Similarly to the above computation, we have

$$\begin{aligned} AS &= \begin{bmatrix} A\mathbf{v}_1 & A\mathbf{v}_2 & \cdots & A\mathbf{v}_n \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1\mathbf{v}_1 & \lambda_2\mathbf{v}_2 & \cdots & \lambda_n\mathbf{v}_n \end{bmatrix} \\ &= S\,\mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n). \end{aligned}$$

So

$$S^{-1}AS = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$$

Therefore A is diagonalizable. □

Remark 8.3.1.

(a) If A is diagonalizable, i.e.,

$$S^{-1}AS = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n).$$

Then $\lambda_1, \lambda_2, \ldots, \lambda_n$ are just eigenvalues of A, and the ith column of S is just an eigenvector associated with λ_i.

(b) The diagonalizing matrix S is not unique. Reordering the columns of S or multiplying them by nonzero scalars will produce a new diagonalizing matrix.

Example 8.3.2. Diagonalize the matrix $A = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}$ in Example 8.1.2.

Solution. We have shown that A has 2 eigenvalues:

$\lambda_1 = 4$, with corresponding eigenvector $\mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$,

$\lambda_2 = -3$, with corresponding eigenvector $\mathbf{x}_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$.

It is guaranteed by next theorem that $\mathbf{x}_1$ and $\mathbf{x}_2$ must be linearly independent. Putting $\mathbf{x}_1$, $\mathbf{x}_2$ as columns of S:

$$S = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix},$$

then

$$S^{-1}AS = \begin{bmatrix} 4 & 0 \\ 0 & -3 \end{bmatrix}.$$

□

Example 8.3.3. Show that

$$A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}$$

in Example 8.3.3 is not diagonalizable.

Proof. We have shown A has a repeated eigenvalue $\lambda = 2$ with only one linearly independent eigenvector $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, so A is not diagonalizable. □

Theorem 8.3.2 (**Eigenvectors for distinct eigenvalues are linearly independent**). *If $\lambda_1, \lambda_2, \cdots, \lambda_k$ are distinct eigenvalues of an $n \times n$ matrix A with corresponding eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k$, then $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k$ are linearly independent.*

Proof. By induction for k. Skip. □

Remark 8.3.2. Suppose A has distinct eigenvalues $\lambda_1, \lambda_2, \cdots, \lambda_k$ with algebraic multiplicities $n_1, n_2, \cdots, n_k$, i.e.,

$$p_A(\lambda) = (-1)^n(\lambda - \lambda_1)^{n_1}(\lambda - \lambda_2)^{n_2} \cdots (\lambda - \lambda_k)^{n_k}$$

with $n_1 + n_2 + \cdots + n_k = n$.

Suppose the geometric multiplicities are $m_1, m_2, \cdots, m_k$, i.e., there are

$$
\begin{gathered}
m_1 \text{ linearly independent eigenvectors for } \lambda_1 \\
m_2 \text{ linearly independent eigenvectors for } \lambda_2 \\
\vdots \\
m_k \text{ linearly independent eigenvectors for } \lambda_k
\end{gathered}
$$

In total, there are $m_1 + m_2 + \cdots + m_k$ linearly independent eigenvectors.

A is diagonalizable if and only if $m_1 + m_2 + \cdots + m_k = n$.

Noting that $m_i \leq n_i$, and $m_1 + m_2 + \cdots + m_k \leq n_1 + n_2 + \cdots + n_k = n$, so A is diagonalizable if and only if $m_1 = n_1, m_2 = n_2, \cdots, m_k = n_k$.

Theorem 8.3.3. *Let A be an $n \times n$ matrix. The following statements are equivalent.*

(a) A is diagonalizable.

(b) A has n linearly independent eigenvectors.

(c) The algebraic multiplicity equals geometric multiplicity for each eigenvalue of A.

Corollary 8.3.1 ($n \times n$ **matrix with** n **distinct eigenvalues).** If an $n \times n$ matrix has n distinct eigenvalues, then it is diagonalizable.

Proof. In this case, the algebraic multiplicity and geometric multiplicity for each eigenvalue is 1, and A has n linealy independent eigenvectors. □

Example 8.3.4. Determine if $A = \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix}$ in Example 8.1.5 is diagonalizable.

Solution. A has two distinct eigenvalues: $\lambda_1 = 0$ with algebraic multiplicity 1, and $\lambda_2 = 1$ with algebraic multiplicity 2.

For $\lambda_1 = 0$, one eigenvector is $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

For $\lambda_2 = 1$, two linearly independent eigenvectors are $\mathbf{x}_2 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.

Therefore, A has 3 linearly independent eigenvectors, and A is diagonalizable. $\square$

Example 8.3.5. Determine if $A = \begin{bmatrix} -1 & 1 & 0 \\ -4 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ in Example 8.1.6 is diagonalizable.

Solution. A has two distinct eigenvalues: $\lambda_1 = 2$ with algebraic multiplicity 1, and $\lambda_2 = 1$ with algebraic multiplicity 2.

For $\lambda_1 = 2$, one eigenvector is $\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

For $\lambda_2 = 1$, there is only one linearly independent eigenvector $\mathbf{x}_2 = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$.

Therefore, A has only 2 linearly independent eigenvectors (geometric multiplicity $<$ algebraic multiplicity for λ_2), and A is not diagonalizable. $\square$

8.4 Powers of Matrices

Suppose A is diagonalizable, i.e.,

$$S^{-1}AS = D = \operatorname{diag}(\lambda_1, \lambda_2, \cdots, \lambda_n).$$

Then

$$\begin{aligned}
A =& SDS^{-1}, \\
A^k =& \underbrace{SDS^{-1}SDS^{-1}\cdots SDS^{-1}}_{k} \\
=& S\underbrace{D\cdots D}_{k}S^{-1} \\
=& SD^kS^{-1} \\
=& S\operatorname{diag}(\lambda_1^k, \lambda_2^k, \cdots, \lambda_n^k)S^{-1}.
\end{aligned}$$

Example 8.4.1. Let

$$A = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}.$$

Find A^5 and a general formula of A^k.

Solution. It follows from Example 8.3.2 that A is diagonalizable with

$$S^{-1}AS = D = \begin{bmatrix} 4 & 0 \\ 0 & -3 \end{bmatrix}$$

where $S = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$. Thus,

$$\begin{aligned}
A^5 =& (SDS^{-1})^5 \\
=& SD^5S^{-1} \\
=& \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 4^5 & 0 \\ 0 & (-3)^5 \end{bmatrix} \frac{1}{7} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \\
=& \begin{bmatrix} 843 & 362 \\ 543 & -62 \end{bmatrix}.
\end{aligned}$$

$$
\begin{aligned}
A^k =& \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 4^k & 0 \\ 0 & (-3)^k \end{bmatrix} \frac{1}{7} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \\
=& \frac{1}{7} \begin{bmatrix} 6 \cdot 4^k + (-3)^k & 2 \cdot 4^k - 2 \cdot (-3)^k \\ 3 \cdot 4^k - 3 \cdot (-3)^k & 4^k + 6 \cdot (-3)^k \end{bmatrix}.
\end{aligned}
$$

□

Example 8.4.2. Compute A^5 for

$$
A = \begin{bmatrix} 4 & -2 & 1 \\ 2 & 0 & 1 \\ 2 & -2 & 3 \end{bmatrix}.
$$

Solution. Following the procedure of diagonalization, we can find

$$
S = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix},
$$

and

$$
S^{-1}AS = D = \begin{bmatrix} 3 & & \\ & 2 & \\ & & 2 \end{bmatrix}
$$

$$
\begin{aligned}
A^5 =& SD^5S^{-1} \\
=& \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 3^5 & & \\ & 2^5 & \\ & & 2^5 \end{bmatrix} \begin{bmatrix} 2 & -2 & 1 \\ -2 & 3 & -1 \\ -1 & 1 & 0 \end{bmatrix} \\
=& \begin{bmatrix} 454 & -422 & 211 \\ 422 & -390 & 211 \\ 422 & -422 & 243 \end{bmatrix}.
\end{aligned}
$$

□

9 Inner Product Space

9.1 Basic Properties of Inner Product

Definition 9.1.1 (Inner Product). For two vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^m$, we define

$$\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{i=1}^{m} [\mathbf{v}]_i [\mathbf{w}]_i = [\mathbf{v}]_1 [\mathbf{w}]_1 + \cdots + [\mathbf{v}]_m [\mathbf{w}]_m. \tag{9.11}$$

It is called an **inner product (dot product)** of $\mathbb{R}^m$. The vector space $\mathbb{R}^m$ together with this inner product is called an inner product space. If we regard $\mathbf{v}$ and $\mathbf{w}$ as $m \times 1$ matrice, then

$$\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^t \mathbf{w}. \tag{9.12}$$

Example 9.1.1.

$$\left\langle \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\rangle = 1 \times 3 + 2 \times 4 = 11.$$

$$\left\langle \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \right\rangle = 1 \times 4 + 2 \times 5 + 3 \times 6 = 32.$$

Theorem 9.1.1. *For any* $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^m$ *and* $\alpha \in \mathbb{R}$*, we have*

(a) $\langle \mathbf{v} + \mathbf{w}, \mathbf{u} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle$.

(b) $\langle \alpha \mathbf{v}, \mathbf{w} \rangle = \alpha \langle \mathbf{v}, \mathbf{w} \rangle$.

(c) $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$.

(d) $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$. *Moreover,* $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ *if and only if* $\mathbf{v} = \mathbf{0}$.

Proof. (a)

$$\begin{aligned}\langle \mathbf{v}+\mathbf{w}, \mathbf{u}\rangle &= [\mathbf{v}+\mathbf{w}]_1[\mathbf{u}]_1 + [\mathbf{v}+\mathbf{w}]_2[\mathbf{u}]_2 + \cdots + [\mathbf{v}+\mathbf{w}]_m[\mathbf{u}]_m \\ &= ([\mathbf{v}]_1 + [\mathbf{w}]_1)[\mathbf{u}]_1 + ([\mathbf{v}]_2 + [\mathbf{w}]_2)[\mathbf{u}]_2 + \cdots + ([\mathbf{v}]_n + [\mathbf{w}]_m)[\mathbf{u}]_m \\ &= [\mathbf{v}]_1[\mathbf{u}]_1 + [\mathbf{w}]_1[\mathbf{u}]_1 + [\mathbf{v}]_2[\mathbf{u}]_2 + [\mathbf{w}]_2[\mathbf{u}]_2 + \cdots + [\mathbf{v}]_m[\mathbf{u}]_m + [\mathbf{w}]_m[\mathbf{u}]_m \\ &= [\mathbf{v}]_1[\mathbf{u}]_1 + \cdots + [\mathbf{v}]_m[\mathbf{u}]_m + [\mathbf{w}]_1[\mathbf{u}]_1 + \cdots + [\mathbf{w}]_m[\mathbf{u}]_m \\ &= \langle \mathbf{v}, \mathbf{u}\rangle + \langle \mathbf{w}, \mathbf{u}\rangle .\end{aligned}$$

Or we can use (9.12):

$$\begin{aligned}\langle \mathbf{v}+\mathbf{w}, \mathbf{u}\rangle &= (\mathbf{v}+\mathbf{w})^t\mathbf{u} = (\mathbf{v}^t + \mathbf{w}^t)\mathbf{u} \\ &= \mathbf{v}^t\mathbf{u} + \mathbf{w}^t\mathbf{u} = \langle \mathbf{v}, \mathbf{u}\rangle + \langle \mathbf{w}, \mathbf{u}\rangle .\end{aligned}$$

(b)

$$\begin{aligned}\langle \alpha\mathbf{v}, \mathbf{w}\rangle &= [\alpha\mathbf{v}]_1[\mathbf{w}]_1 + [\alpha\mathbf{v}]_2[\mathbf{w}]_2 + \cdots + [\alpha\mathbf{v}]_m[\mathbf{w}]_m \\ &= \alpha[\mathbf{v}]_1[\mathbf{w}]_1 + \alpha[\mathbf{v}]_2[\mathbf{w}]_2 + \cdots + \alpha[\mathbf{v}]_m[\mathbf{w}]_m \\ &= \alpha([\mathbf{v}]_1[\mathbf{w}]_1 + [\mathbf{v}]_2[\mathbf{w}]_2 + \cdots + [\mathbf{v}]_m[\mathbf{w}]_m) \\ &= \alpha \langle \mathbf{v}, \mathbf{w}\rangle .\end{aligned}$$

Or we can use (9.12):

$$\langle \alpha\mathbf{v}, \mathbf{w}\rangle = (\alpha\mathbf{v})^t\mathbf{w} = \alpha\mathbf{v}^t\mathbf{w} = \alpha \langle \mathbf{v}, \mathbf{w}\rangle .$$

(c)

$$\begin{aligned}\langle \mathbf{v}, \mathbf{w}\rangle &= [\mathbf{v}]_1[\mathbf{w}]_1 + [\mathbf{v}]_2[\mathbf{w}]_2 + \cdots + [\mathbf{v}]_m[\mathbf{w}]_m. \\ &= [\mathbf{w}]_1[\mathbf{v}]_1 + [\mathbf{w}]_2[\mathbf{v}]_2 + \cdots + [\mathbf{w}]_m[\mathbf{v}]_m. \\ &= \langle \mathbf{w}, \mathbf{v}\rangle .\end{aligned}$$

(d)

$$\langle \mathbf{v}, \mathbf{v}\rangle = [\mathbf{v}]_1^2 + [\mathbf{v}]_2^2 + \cdots + [\mathbf{v}]_m^2 \geq 0.$$

Noting that $\langle \mathbf{v}, \mathbf{v}\rangle = 0$ if and only if $[\mathbf{v}]_i = 0$ for all $1 \leq i \leq n$, we see that $\mathbf{v} = \mathbf{0}$.

□

Remark 9.1.1. Property (a)-(b) is called linearity. (c) is called symmetry. (d) is called positive definiteness. In general, an operation $\langle\ ,\ \rangle$ defined on a vector space V satisfying the property (a)-(d) is called an inner product on V. Therefore, Definition 9.1.1 is a very special (and also called "standard") inner product on $\mathbb{R}^m$, and it is not the unique one. All the following properties and theorems hold for general inner product space, but we will only focus on $\mathbb{R}^m$ with standard inner product.

Proposition 9.1.1. For any $\alpha, \beta \in \mathbb{R}$ and $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^m$, we have

(a) $\langle \alpha\mathbf{v} + \beta\mathbf{w}, \mathbf{u}\rangle = \alpha \langle \mathbf{v}, \mathbf{u}\rangle + \beta \langle \mathbf{w}, \mathbf{u}\rangle$.

(b) $\langle \mathbf{u}, \alpha\mathbf{v} + \beta\mathbf{w}\rangle = \alpha \langle \mathbf{u}, \mathbf{v}\rangle + \beta \langle \mathbf{u}, \mathbf{w}\rangle$.

(c) $\langle \mathbf{0}, \mathbf{v}\rangle = \langle \mathbf{v}, \mathbf{0}\rangle = 0$.

(d) If $\langle \mathbf{v}, \mathbf{x}\rangle = 0$ for all $\mathbf{x} \in \mathbb{R}^m$, then $\mathbf{v} = \mathbf{0}$.

(e) If $\langle \mathbf{v}, \mathbf{x}\rangle = \langle \mathbf{w}, \mathbf{x}\rangle$ for all $x \in \mathbb{R}^m$, then $\mathbf{v} = \mathbf{w}$.

Proof. (a)

$$\langle \alpha\mathbf{v} + \beta\mathbf{w}, \mathbf{u}\rangle = \langle \alpha\mathbf{v}, \mathbf{u}\rangle + \langle \beta\mathbf{w}, \mathbf{u}\rangle$$

$$= \alpha \langle \mathbf{v}, \mathbf{u}\rangle + \beta \langle \mathbf{w}, \mathbf{u}\rangle .$$

Or we can also use (9.12):

$$\langle \alpha\mathbf{v} + \beta\mathbf{w}, \mathbf{u}\rangle = (\alpha\mathbf{v} + \beta\mathbf{w})^t\mathbf{u}$$

$$= \alpha(\mathbf{v})^t\mathbf{u} + \beta(\mathbf{w})^t\mathbf{u} = \alpha \langle \mathbf{v}, \mathbf{u}\rangle + \beta \langle \mathbf{w}, \mathbf{u}\rangle .$$

(b)

$$\langle \mathbf{u}, \alpha\mathbf{v} + \beta\mathbf{w}\rangle = \langle \alpha\mathbf{v} + \beta\mathbf{w}, \mathbf{u}\rangle$$

$$= \alpha \langle \mathbf{v}, \mathbf{u}\rangle + \beta \langle \mathbf{w}, \mathbf{u}\rangle = \alpha \langle \mathbf{u}, \mathbf{v}\rangle + \beta \langle \mathbf{u}, \mathbf{w}\rangle .$$

Or we can also use (9.12) (fill the detail).

(c)

$$\langle \mathbf{0}, \mathbf{v} \rangle = 0[\mathbf{v}]_1 + \cdots + 0[\mathbf{v}]_m = 0$$

and

$$\langle \mathbf{v}, \mathbf{0} \rangle = [\mathbf{v}]_1 0 + \cdots + [\mathbf{v}]_m 0 = 0.$$

(d) Suppose $\langle \mathbf{v}, \mathbf{x} \rangle = 0$ for all $\mathbf{x} \in \mathbb{R}^m$. Let $\mathbf{x} = \mathbf{v}$. Then $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, so $\mathbf{v} = \mathbf{0}$.

(e) Suppose $\langle \mathbf{v}, \mathbf{x} \rangle = \langle \mathbf{w}, \mathbf{x} \rangle$, then $0 = \langle \mathbf{v}, \mathbf{x} \rangle - \langle \mathbf{w}, \mathbf{x} \rangle = \langle \mathbf{v} - \mathbf{w}, \mathbf{x} \rangle$ for all $\mathbf{x} \in V$. By the previous part $\mathbf{v} - \mathbf{w} = \mathbf{0}$. So $\mathbf{v} = \mathbf{w}$.

□

Definition 9.1.2 (Norm). The **norm** (or **length**) of $\mathbf{v} \in \mathbb{R}^m$ is defined to be $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.
Note that $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$. So the symbol $\sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$ is meaningful.

Example 9.1.2. Let

$$\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \mathbf{w} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Then

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$$

and

$$\|\mathbf{w}\| = \sqrt{\langle \mathbf{w}, \mathbf{w} \rangle} = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}.$$

Proposition 9.1.2.

(a) $\|\mathbf{v}\| \geq 0$ for any $\mathbf{v} \in \mathbb{R}^m$. Moreover, $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

(b) $\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\|$ for any $\mathbf{v} \in \mathbb{R}^m$, $\alpha \in \mathbb{R}$.

(c) Suppose $\mathbf{v} \neq \mathbf{0}$ and let $\alpha = \dfrac{1}{\|\mathbf{v}\|}$. Then $\|\alpha \mathbf{v}\| = 1$.

Proof. (a) $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} \geq 0$.

$$\|\mathbf{v}\| = 0 \iff 0 = \|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle \iff \mathbf{v} = \mathbf{0}.$$

(b) $\|\alpha\mathbf{v}\| = \sqrt{\langle \alpha\mathbf{v}, \alpha\mathbf{v} \rangle} = \sqrt{\alpha \langle \mathbf{v}, \alpha\mathbf{v} \rangle} = \sqrt{\alpha^2 \langle \mathbf{v}, \mathbf{v} \rangle} = |\alpha|\sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = |\alpha|\|\mathbf{v}\|$.

(c) By the previous part

$$\|\alpha\mathbf{v}\| = |\alpha|\|\mathbf{v}\| = \frac{1}{\|\mathbf{v}\|}\|\mathbf{v}\| = 1.$$

□

Definition 9.1.3 (unit vector). A vector $\mathbf{v} \in \mathbb{R}^m$ is said to be a **unit vector** if $\|\mathbf{v}\| = 1$.
A non-zero vector $\mathbf{v}$ can be **normalized** to a unit vector $\dfrac{\mathbf{v}}{\|\mathbf{v}\|}$.

Example 9.1.3. In Example 9.1.2, the vectors $\mathbf{v}$ and $\mathbf{w}$ can be normalized to

$$\frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\mathbf{v}}{\sqrt{14}} = \begin{bmatrix} \frac{1}{\sqrt{14}} \\ \frac{2}{\sqrt{14}} \\ \frac{3}{\sqrt{14}} \end{bmatrix}$$

and

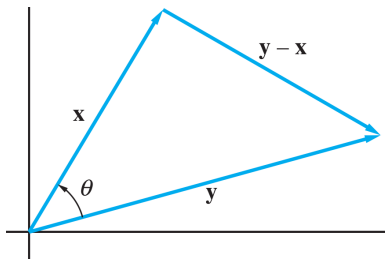
$$\frac{\mathbf{w}}{\|\mathbf{w}\|} = \frac{\mathbf{w}}{\sqrt{2}} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

respectively.

9.2 Orthogonal Sets

Theorem 9.2.1. *Let* $\mathbf{x}$ *and* $\mathbf{y}$ *be two nonzero vectors in either* $\mathbb{R}^2$ *or* $\mathbb{R}^3$ *and* θ *be the angle between them, then*

$$\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\| \|\mathbf{y}\| \cos\theta.$$



Proof. The vectors $\mathbf{x}$, $\mathbf{y}$, $\mathbf{y} - \mathbf{x}$ form a triangle. By the law of cosines,

$$\|\mathbf{y} - \mathbf{x}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\|\mathbf{x}\| \|\mathbf{y}\| \cos\theta.$$

and hence it follows that

$$\begin{aligned}
\|\mathbf{x}\| \|\mathbf{y}\| \cos\theta =& \frac{1}{2}(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - \|\mathbf{y} - \mathbf{x}\|^2) \\
=& \frac{1}{2}(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - \langle \mathbf{y} - \mathbf{x}, \mathbf{y} - \mathbf{x} \rangle) \\
=& \frac{1}{2}(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - (\langle \mathbf{y}, \mathbf{y} \rangle - \langle \mathbf{y}, \mathbf{x} \rangle - \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{x} \rangle)) \\
=& \langle \mathbf{x}, \mathbf{y} \rangle
\end{aligned}$$

□

Remark 9.2.1. For two nonzero vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^2$ or $\mathbb{R}^3$, the angle θ between them can be calculated as

$$\theta = \arccos\left(\frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}\right).$$

In particular, $\mathbf{x}$ and $\mathbf{y}$ are perpendicular, i.e., $\theta = 90°$, if and only if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$. This fact can be generalized to vectors in $\mathbb{R}^m$.

Definition 9.2.1. Two vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^m$ are said **orthogonal** or **perpendicular** if $\langle \mathbf{v}, \mathbf{w} \rangle = 0$. In this case it is denoted by $\mathbf{v} \perp \mathbf{w}$.

Example 9.2.1.

(a)
$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \perp \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix},$$
because
$$\left\langle \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\rangle = 1 \times (-1) + 2 \times (-1) + 3 \times 1 = 0.$$

(b) In $\mathbb{R}^m$, $\mathbf{e}_i \perp \mathbf{e}_j$ if $i \neq j$.

Definition 9.2.2 (Orthogonal set). A subset $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ of $\mathbb{R}^m$ is said to be **orthogonal** if the following conditions hold:

1. $\mathbf{0} \notin S$, i.e. $\mathbf{v}_i \neq \mathbf{0}$ for $i = 1, \ldots, k$.

2. $\mathbf{v}_i \perp \mathbf{v}_j$ for $i \neq j$, i.e., $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for $i \neq j$.

Example 9.2.2.

(a) $S = \{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \end{bmatrix} \}$ is orthogonal.

(b) $S = \{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \}$ is orthogonal.

(c) For any $k \leq m$, the set $S = \{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_k\} \subset \mathbb{R}^m$ is orthogonal.

Proposition 9.2.1. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k\}$ be an orthogonal subset of $\mathbb{R}^m$. Let

$$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k,$$

$$\mathbf{w} = \beta_1 \mathbf{v}_1 + \cdots + \beta_k \mathbf{v}_k.$$

Then

$$\langle \mathbf{v}, \mathbf{w} \rangle = \alpha_1 \beta_1 \|\mathbf{v}_1\|^2 + \cdots + \alpha_k \beta_k \|\mathbf{v}_k\|^2.$$

Proof. First for $1 \leq i \leq k$, we compute

$$\begin{aligned}
\langle \mathbf{v}, \mathbf{v}_i \rangle &= \langle \alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k, \mathbf{v}_i \rangle \\
&= \alpha_1 \langle \mathbf{v}_1, \mathbf{v}_i \rangle + \cdots + \alpha_k \langle \mathbf{v}_k, \mathbf{v}_i \rangle \\
&= \alpha_i \langle \mathbf{v}_i, \mathbf{v}_i \rangle = \alpha_i \|\mathbf{v}_i\|^2.
\end{aligned}$$

The last step follows from the fact that $\langle \mathbf{v}_j, \mathbf{v}_i \rangle = 0$ for $j \neq i$. Then

$$\begin{aligned}
\langle \mathbf{v}, \mathbf{w} \rangle &= \langle \mathbf{v}, \beta_1 \mathbf{v}_1 + \cdots + \beta_k \mathbf{v}_k \rangle \\
&= \beta_1 \langle \mathbf{v}, \mathbf{v}_1 \rangle + \cdots + \beta_k \langle \mathbf{v}, \mathbf{v}_k \rangle \\
&= \alpha_1 \beta_1 \|\mathbf{v}_1\|^2 + \cdots + \alpha_k \beta_k \|\mathbf{v}_k\|^2.
\end{aligned}$$

□

Theorem 9.2.2. *Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k\}$ be an orthogonal subset of $\mathbb{R}^m$. Then S is linearly independent.*

Proof. Suppose

$$\alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k = \mathbf{0},$$

then for $1 \leq i \leq k$ we have

$$\begin{aligned}
0 &= \langle \mathbf{0}, \mathbf{v}_i \rangle \\
&= \langle \alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k, \mathbf{v}_i \rangle \\
&= \alpha_1 \langle \mathbf{v}_1, \mathbf{v}_i \rangle + \cdots + \alpha_k \langle \mathbf{v}_k, \mathbf{v}_i \rangle \\
&= \alpha_i \|\mathbf{v}_i\|^2.
\end{aligned}$$

Noting that $\mathbf{v}_i \neq \mathbf{0}$, we get

$$\alpha_i = 0.$$

Therefore the linear relation is trivial. Hence S is linearly independent. $\square$

Theorem 9.2.3. *Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k\}$ be an orthogonal subset of $\mathbb{R}^m$. Suppose $\mathbf{v} \in \mathrm{Span}\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k\}$ with*

$$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k$$

for some $\alpha_i \in \mathbb{R}$, $i = 1, \ldots, k$. Then

$$\alpha_i = \frac{\langle \mathbf{v}, \mathbf{v}_i \rangle}{\|\mathbf{v}_i\|^2},$$

and

$$\mathbf{v} = \frac{\langle \mathbf{v}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 + \cdots + \frac{\langle \mathbf{v}, \mathbf{v}_k \rangle}{\|\mathbf{v}_k\|^2}\mathbf{v}_k.$$

Proof. For $1 \leq i \leq k$, we compute

$$\begin{aligned}
\langle \mathbf{v}, \mathbf{v}_i \rangle &= \langle \alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k, \mathbf{v}_i \rangle \\
&= \alpha_1 \langle \mathbf{v}_1, \mathbf{v}_i \rangle + \cdots + \alpha_k \langle \mathbf{v}_k, \mathbf{v}_i \rangle \\
&= \alpha_i \langle \mathbf{v}_i, \mathbf{v}_i \rangle = \alpha_i \|\mathbf{v}_i\|^2.
\end{aligned}$$

Hence

$$\alpha_i = \frac{\langle \mathbf{v}, \mathbf{v}_i \rangle}{\|\mathbf{v}_i\|^2}.$$

$\square$

Remark 9.2.2. We do not need to solve linear system to find the coefficients α_i in the linear combination.

Example 9.2.3. Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$. It is easy to check

$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is orthogonal. Let

$$\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \in \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}.$$

Then

$$\alpha_1 = \frac{\langle \mathbf{v}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} = \frac{6}{3} = 2.$$

$$\alpha_2 = \frac{\langle \mathbf{v}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} = \frac{-1}{2} = -\frac{1}{2}.$$

$$\alpha_3 = \frac{\langle \mathbf{v}, \mathbf{v}_3 \rangle}{\|\mathbf{v}_3\|^2} = \frac{-3}{6} = -\frac{1}{2}.$$

Hence

$$\mathbf{v} = 2\mathbf{v}_1 - \frac{1}{2}\mathbf{v}_2 - \frac{1}{2}\mathbf{v}_3.$$

Definition 9.2.3 (Orthogonal Basis). Let V be a subspace of $\mathbb{R}^m$. A subset S of V is said to be an **orthogonal basis** for V if S is a basis of V and S is orthogonal.

If S is an orthogonal subset of V, then by Theorem 9.2.2, it is automatically linearly independent. So in order to check if S is an orthogonal basis, we need only check that S spans V. So we have the following result.

Theorem 9.2.4. *Let V be a subspace of $\mathbb{R}^m$. Suppose that S is an orthogonal subset of V. Then S is an orthogonal basis if and only if S spans V.*

Corollary 9.2.1. Let V be a subspace of $\mathbb{R}^m$. Suppose that $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is an orthogonal basis of V. Then for any $\mathbf{v} \in V$, we have

$$\mathbf{v} = \frac{\langle \mathbf{v}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 + \cdots + \frac{\langle \mathbf{v}, \mathbf{v}_n \rangle}{\|\mathbf{v}_n\|^2}\mathbf{v}_n.$$

Proof. This follows from Theorem 9.2.3. □

Example 9.2.4.

(a) The set $S = \{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \end{bmatrix} \}$ an orthogonal basis of $\mathbb{R}^2$.

(b) The set $S = \{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \}$ is an orthogonal basis of $\mathbb{R}^3$.

(c) The set $S = \{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_m\}$ is an orthogonal basis of $\mathbb{R}^m$. It is called **the standard basis** for V.

Definition 9.2.4. A subset $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ of $\mathbb{R}^m$ is said to be **orthonormal** if it is orthogonal and every vector in S is a unit vector, i.e.

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Let V be a subspace of $\mathbb{R}^m$. The subset S is said to be an **orthonormal basis** for V if it is orthonormal and is a basis of V.

Because an orthonormal set S is orthogonal, the above theorems regarding orthogonal sets are also true for orthonormal sets. In particular we have the following result.

Theorem 9.2.5. *Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ be an orthonormal subset of $\mathbb{R}^m$ and let $\mathbf{v} \in \operatorname{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$. Then*

$$\mathbf{v} = \langle \mathbf{v}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \cdots + \langle \mathbf{v}, \mathbf{v}_k \rangle \mathbf{v}_k.$$

Proof. By Theorem 9.2.3 and $\|\mathbf{v}_i\| = 1$ for $i = 1, \ldots, k$. □

If $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is an orthogonal subset of $\mathbb{R}^m$, then $\{\frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}, \ldots, \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|}\}$ is an orthonormal subset. The process is called **normalization**.

Example 9.2.5.

(a) The set $S = \{\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}-2\\1\end{bmatrix}\}$ is an orthogonal basis of $\mathbb{R}^2$. Normalizing it, we obtain an orthonormal basis

$$S' = \{\frac{1}{\sqrt{5}}\begin{bmatrix}1\\2\end{bmatrix}, \frac{1}{\sqrt{5}}\begin{bmatrix}-2\\1\end{bmatrix}\}.$$

(b) The set $S = \{\begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\-1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\-2\end{bmatrix}\}$ is an orthogonal basis of $\mathbb{R}^3$. Normalizing it, we obtain an orthonormal basis

$$S' = \{\frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\1\end{bmatrix}, \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\\0\end{bmatrix}, \frac{1}{\sqrt{6}}\begin{bmatrix}1\\1\\-2\end{bmatrix}\}.$$

9.3 Gram-Schmidt Orthogonalization Process

Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ be an orthogonal subset of $\mathbb{R}^m$. If $\mathbf{w} \in \operatorname{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$, then

$$\mathbf{w} = \frac{\langle \mathbf{w}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 + \cdots + \frac{\langle \mathbf{w}, \mathbf{v}_k \rangle}{\|\mathbf{v}_k\|^2}\mathbf{v}_k.$$

But what if $\mathbf{w} \notin \operatorname{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$? Let's compare the difference. We have the following theorem.

Theorem 9.3.1. *Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ be an orthogonal subset of $\mathbb{R}^m$ and $\mathbf{w} \in \mathbb{R}^m$. Then the vector*

$$\mathbf{v} = \mathbf{w} - \frac{\langle \mathbf{w}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \cdots - \frac{\langle \mathbf{w}, \mathbf{v}_k \rangle}{\|\mathbf{v}_k\|^2}\mathbf{v}_k$$

is perpendicular to each $\mathbf{v}_i$, $i = 1, \ldots, k$.

Proof. For $1 \leq i \leq k$, we compute

$$\langle \mathbf{v}, \mathbf{v}_i \rangle = \langle \mathbf{w}, \mathbf{v}_i \rangle - \frac{\langle \mathbf{w}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\langle \mathbf{v}_1, \mathbf{v}_i \rangle - \cdots - \frac{\langle \mathbf{w}, \mathbf{v}_k \rangle}{\|\mathbf{v}_k\|^2}\langle \mathbf{v}_k, \mathbf{v}_i \rangle.$$

Because $\langle \mathbf{v}_j, \mathbf{v}_i \rangle$ is 0 unless $j = i$, the above becomes

$$\langle \mathbf{w}, \mathbf{v}_i \rangle - \frac{\langle \mathbf{w}, \mathbf{v}_i \rangle}{\|\mathbf{v}_i\|^2}\langle \mathbf{v}_i, \mathbf{v}_i \rangle = \langle \mathbf{w}, \mathbf{v}_i \rangle - \langle \mathbf{w}, \mathbf{v}_i \rangle = 0.$$

Hence $\mathbf{v} \perp \mathbf{v}_i$ for $i = 1, \ldots, k$. □

Theorem 9.3.2 (Gram-Schmidt Orthogonalization Process).
Let $S = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_k\}$ be a linearly independent subset of $\mathbb{R}^m$. Let

$$\begin{aligned}
\mathbf{v}_1 =& \mathbf{w}_1, \\
\mathbf{v}_\ell =& \mathbf{w}_\ell - \frac{\langle \mathbf{w}_\ell, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \cdots - \frac{\langle \mathbf{w}_\ell, \mathbf{v}_{\ell-1} \rangle}{\|\mathbf{v}_{\ell-1}\|^2}\mathbf{v}_{\ell-1}, \quad \textit{for } 2 \leq \ell \leq k.
\end{aligned}$$

Then $S' = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is an orthogonal set.

Moreover, $\operatorname{Span}\{\mathbf{w}_1, \ldots, \mathbf{w}_\ell\} = \operatorname{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_\ell\}$ *for $\ell = 1, \ldots, k$.*

In particular, $\operatorname{Span}\{\mathbf{w}_1, \ldots, \mathbf{w}_k\} = \operatorname{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$.

The process of obtaining S' by the above procedure is called the **Gram-Schmidt Orthogonalization process**.

Proof. We will prove it by induction for ℓ.

First, $\text{Span}\{\mathbf{v}_1\} = \text{Span}\{\mathbf{w}_1\}$.

Next, suppose that $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_{\ell-1}\} = \text{Span}\{\mathbf{w}_1, \ldots, \mathbf{w}_{\ell-1}\}$ and that the set $\{\mathbf{v}_1, \ldots, \mathbf{v}_{\ell-1}\}$ is orthogonal.

Then $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_{\ell-1}, \mathbf{v}_\ell\} = \text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_{\ell-1}, \mathbf{w}_\ell\} = \text{Span}\{\mathbf{w}_1, \ldots, \mathbf{w}_{\ell-1}, \mathbf{w}_\ell\}$.

By Theorem 9.3.1, $\mathbf{v}_\ell \perp \mathbf{v}_1, \ldots, \mathbf{v}_{\ell-1}$. Thus $\{\mathbf{v}_1, \ldots, \mathbf{v}_\ell\}$ is orthogonal. □

Corollary 9.3.1. Suppose that V is a subspace of $\mathbb{R}^m$. Then there exists an orthogonal (orthonormal basis) of V.

Proof. Suppose $S = \{\mathbf{w}_1, \ldots, \mathbf{w}_k\}$ is a basis of V. Applying Gram-Schmidt orthogonalization process to S, we obtain an orthogonal set $S' = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ with $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} = \text{Span}\{\mathbf{w}_1, \ldots, \mathbf{w}_k\} = V$. By Theorem 9.2.4, S' is an orthogonal basis.

Normalizing S', we can also obtain an orthonormal basis. □

The above proof actually describes a method to find an orthogonal (orthonormal) basis of V.

Example 9.3.1. Let V be a subspace of $\mathbb{R}^4$ with a basis:

$$\{\mathbf{w}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{w}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{w}_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix}\}$$

We can apply Gram-Schmidt orthogonalization process to find an orthogonal (orthonormal) basis of V.

Let

$$\mathbf{v}_1 = \mathbf{w}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

$$\mathbf{v}_2 = \mathbf{w}_2 - \frac{\langle \mathbf{w}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{2}\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

$$\mathbf{v}_3 = \mathbf{w}_3 - \frac{\langle \mathbf{w}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{\langle \mathbf{w}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2}\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix} - \frac{2}{2}\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} - \frac{2}{2}\begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

The set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthogonal basis of V.

Normalize the vectors

$$\mathbf{u}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{u}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{u}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|} = \frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

The set $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal basis of V.

Example 9.3.2. Let $V = \mathcal{N}\left(\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}\right)$. Find an orthonormal basis of V.

V has a basis:

$$\{\mathbf{w}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{w}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{w}_3 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}\}$$

Let

$$\mathbf{v}_1 = \mathbf{w}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

$$\mathbf{v}_2 = \mathbf{w}_2 - \frac{\langle \mathbf{w}_2, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/2 \\ -1/2 \\ 1 \\ 0 \end{bmatrix}.$$

$$\mathbf{v}_3 = \mathbf{w}_3 - \frac{\langle \mathbf{w}_3, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{\langle \mathbf{w}_3, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2}\mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{3}\begin{bmatrix} -1/2 \\ -1/2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/3 \\ -1/3 \\ -1/3 \\ 1 \end{bmatrix}.$$

The set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthogonal basis of V.

Normalizing it, we can obtain an orthonormal basis of V:

$$\{\frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{6}}\begin{bmatrix} -1 \\ -1 \\ 2 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{12}}\begin{bmatrix} -1 \\ -1 \\ -1 \\ 3 \end{bmatrix}\}.$$

9.4 Two Inequalities (Optional!)

Theorem 9.4.1 (Cauchy-Schwartz inequality). *For* $\mathbf{v}, \mathbf{w} \in \mathbb{R}^m$,

$$|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \|\mathbf{v}\| \|\mathbf{w}\|.$$

Moreover, the equality holds if and only if $\mathbf{v}$ *and* $\mathbf{w}$ *are linearly dependent.*

Proof. Let $t \in \mathbb{R}$ and consider the function about t:

$$\begin{aligned} F(t) =& \|t\mathbf{w} - \mathbf{v}\|^2 \\ =& \langle t\mathbf{w} - \mathbf{v}, t\mathbf{w} - \mathbf{v} \rangle \\ =& t^2 \langle \mathbf{w}, \mathbf{w} \rangle - 2t \langle \mathbf{v}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle \end{aligned}$$

Then $F(t) \geq 0$ for all $t \in \mathbb{R}$.

Case 1: if $\mathbf{w} = \mathbf{0}$, then the Cauchy-Schwartz inequality is trivial.

Case 2: if $\mathbf{w} \neq \mathbf{0}$, then $F(t)$ is a nonnegative quadratic polynomial, so the discriminant

$$\Delta = (2 \langle \mathbf{v}, \mathbf{w} \rangle)^2 - 4 \langle \mathbf{v}, \mathbf{v} \rangle \langle \mathbf{w}, \mathbf{w} \rangle \leq 0,$$

which can be simplified as

$$|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \|\mathbf{v}\| \|\mathbf{w}\|.$$

The equality holds if and only if $\mathbf{w} = \mathbf{0}$ or $F(t_0) = 0$ for some t_0, i.e., $\mathbf{v} = t_0\mathbf{w}$ for some t_0. More simply stated, equality will hold if and only if $\mathbf{v}$ and $\mathbf{w}$ are linearly dependent. □

Theorem 9.4.2 (Triangle inequality). *For* $\mathbf{v}, \mathbf{w} \in \mathbb{R}^m$,

$$\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|.$$

Moreover, the equality holds if and only if $\mathbf{w} = \mathbf{0}$ *or* $\mathbf{v} = \alpha\mathbf{w}$ *for some* $\alpha \geq 0$.

Proof.

$$\|\mathbf{v} + \mathbf{w}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle + \langle \mathbf{w}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle = \|\mathbf{v}\|^2 + 2 \langle \mathbf{v}, \mathbf{w} \rangle + \|\mathbf{w}\|^2.$$

By the Cauchy-Schwarz inequality

$$|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \|\mathbf{v}\| \|\mathbf{w}\|,$$

thus

$$\|\mathbf{v} + \mathbf{w}\|^2 \leq \|\mathbf{v}\|^2 + 2\langle \mathbf{v}, \mathbf{w} \rangle + \|\mathbf{w}\|^2 = (\|\mathbf{v}\| + \|\mathbf{w}\|)^2.$$

The triangle inequality follows by taking square roots on both sides.

Equality case happens if and only if $\langle \mathbf{v}, \mathbf{w} \rangle = \|\mathbf{v}\| \|\mathbf{w}\|$, so either $\mathbf{w} = \mathbf{0}$ or $\mathbf{v} = \alpha \mathbf{w}$ for some $\alpha \geq 0$.

□

Remark 9.4.1. The above two theorems hold for general inner product spaces. In particular, for $\mathbb{R}^m$ with the standard inner product, they can be written into explicit form. Let

$$\mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_m \end{bmatrix}, \ \mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_m \end{bmatrix}.$$

Cauchy-Schwarz inequality:

$$|v_1 w_1 + \cdots + v_m w_m| \leq \sqrt{v_1^2 + \cdots + v_m^2} \sqrt{w_1^2 + \cdots + w_m^2}.$$

Triangle inequality:

$$\sqrt{(v_1 + w_1)^2 + \cdots + (v_m + w_m)^2} \leq \sqrt{v_1^2 + \cdots + v_m^2} + \sqrt{w_1^2 + \cdots + w_m^2}.$$